KU LEUVEN

FACULTY OF SCIENCE

Integral Affine Structures

Classification of Integral Affine Structures on Compact Surfaces

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Preface

TODO: Add Preface

Summary

TODO: Add summary

List of symbols

$C^\infty(M, N)$	Smooth maps from M to N

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Introduction

TODO: Add introduction

CHAPTER ONE

Preliminaries

1.1 Geometric Structures on Manifolds

1.1.1 Basic definition and properties

A lot of the content of this project extends in a nice way to more general types of geometries, namely locally homogeneous geometric structures. The idea behind these structures is that we can locally describe our "geometry" on a manifold M by charts into a space X together with a group G that tells us how to transition between charts on the overlaps.

We will require that our group G be a Lie group acting transitively on M.

Definition 1.1. Let $U \subset X$ be an open set of our model space X. We say that a smooth map $f : U \mapsto X$ is *locally*-G if for each component $U_i \subset U$ there is a $g_i \in G$ such that $f|_{U_i} = g_i|_{U_i}$.

Remark 1.2. Locally-G maps satisfy the following unique extension property. If $U \subset X$ and $f : U \mapsto X$ is a locally-G map then there is a unique $g \in G$ restricting to f.

Definition 1.3. A (G, X)-atlas is a pair $(\mathcal{U}, \Phi)$ consisting of an open covering for M and a collection of charts such that the transition maps are locally-G on their respective overlaps.

Definition 1.4. A map $f : M \mapsto N$ between two (G, X)-manifolds is a (G, X)-map if for each pair of charts (ϕ, φ) then $\phi \circ f \circ \varphi^{-1}$ is locally-G on the relevant intersection.

TODO: Add some examples (euclidean/projective/hyperbolic) TODO: At some point include classification of 1-dim integral affine (This is easy in - textbook)

1.1.2 The developing map and holonomy representation

For (G, X)-manifolds, there is an inherently nice way to "globalize" the charts via a (G, X)-map called the developing map. The study of this map will be of importance to us throughout this thesis.

Proposition 1.5. Let M be a simply connected (G, X)-manifold. Then there exists a (G, X)-map

$$\mathrm{dev} : M \to X.$$

Proof. We pick a base point $x_0 \in M$ and a chart (U_0, φ_0) around x_0. Now for any $x \in M$ we define the map dev as follows. We pick a curve $x(t)$ on M satisfying $x(0) = x_0$ and $x(1) = x$. As the image of our curve is compact, we can cover it by finitely many coordinate charts (U_i, φ_i) with $i \in \{0, \ldots, n\}$ such that $x(t) \in U_i$ for $t \in (a_i, b_i)$ with

$$a_0 < 0 < a_1 < b_0 < a_2 < b_1 < a_3 \cdots < a_n < b_{n-1} < 1 < b_n$$

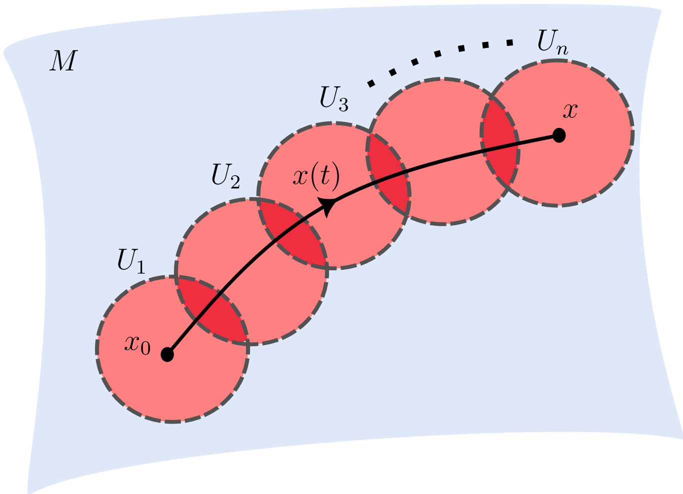

Figure 1.1: Developing along a curve

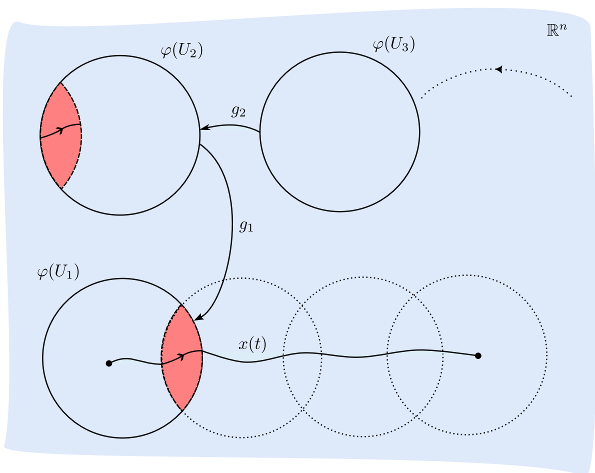

Figure 1.2: We successively glue our charts together using the transition maps

Now we consider where these charts overlap. Since our manifold is equipped with a (G, X)-structure our transition maps differ by an element of G. Let $g_i = \varphi_{i-1} \circ \varphi_i^{-1} \in G$ be the transition map between U_i and U_{i-1}. We then define

$$\mathrm{dev}(x) = g_1 g_2 \dots g_n \varphi_n(x).$$

We must show that this is a well-defined map. We first show that this map does not change if we refine the cover. Suppose we insert a chart (V, ϕ) between U_i and U_{i-1}. Let $h_i, h_{i-1} \in G$ be the unique group elements such that

$$\varphi_{i-1} = h_{i-1} \circ \phi \text{ on } V \cap U_{i-1} \text{ and}$$
$$\phi = h_i \varphi_i \text{ on } V \cap U_{i-1}.$$

Then on $V \cap U_i \cap U_{i-1}$ we have that $\varphi_{i-1} = h_{i-1} \circ h_i \circ \varphi_i$. This gives us that $g_i = \varphi_{i-1} \circ \varphi_i^{-1} = h_{i-1} \circ h_i$ as we require that our transition maps satisfy a unique extension property.

Now when we develop along our curve with respect to this new covering, we obtain that:

$$\mathrm{dev}(x) = g_1 \dots g_{i-1} h_{i-1} h_i g_{i+1} \dots g_n \varphi_n(x) = g_1 \dots g_{i-1} g_i g_{i+1} \dots g_n \varphi_n(x)$$

This defines dev along $x(t)$. We now need to show that it does not depend on our choice of curve. As M is simply connected, all curves with the same start point and end point will be homotopic. By compactness we can split our homotopy into smaller homotopies such that we can break up our curve into regions

$$a_1 = 0 < a_2 < \dots < a_n = 1$$

such that during each small homotopy the segment $x((a_i, a_{i+1}))$ lies entirely inside a coordinate patch. It then follows that dev is independent of our choice of paths completing the proof.

$\square$

This shows that we have a well defined developing map on any simply connected (G, X)-manifold. However, this map depended on the choice of base point and initial chart. We immediately get the following corollary from this fact.

Corollary 1.6. Let M be a (G, X)-manifold and let $\tilde{M}$ be its universal cover. Then there exists a pair $(\mathrm{dev}, \mathrm{hol})$ satisfying

$$\mathrm{dev} : \tilde{M} \to X$$
$$\mathrm{hol} : \pi_1(M) \to G$$

where $\mathrm{hol}(\gamma)$ is the element $g_1 \dots g_n$ of G obtained from developing around a loop $\gamma \in M$.

The development pair satisfy the following equivariance condition. Let $\gamma \in \pi_1(M)$ act by deck transformations on $\tilde{M}$. Then we have the following. (TODO: Fix this proof)

Lemma 1.7. Let $\gamma \in \pi_1(M)$. Then

$$\mathrm{dev}(\gamma \cdot x) = \mathrm{hol}(\gamma)\mathrm{dev}(x)$$

for $x \in \tilde{M}$.

Proof. Let $\{(V_i, \phi_i)\}$ be a cover for γ and lift this to a cover of a lift of γ which we denote by $\tilde{\gamma}$. Denote the transition maps on $V_i \cap V_{i+1}$ by g_i. Then by definition

$$\begin{aligned} \mathrm{dev}(\gamma \cdot x) &= g_1 \dots g_n \varphi_k(\gamma(1)) \\ &= \mathrm{hol}(\gamma)\mathrm{dev}(x) \end{aligned}$$

□

The role of the development pair is to "globalize" the coordinate chart of M. Indeed the following proposition makes this statement more clear.

Proposition 1.8. Let M be a (G, X)-manifold with development pair $(\mathrm{dev}, \mathrm{hol})$. Denote by Γ the image of $\pi_1(M)$ under the holonomy map. Then there exists a (G, X)-atlas such that the transition maps lie in Γ.

Proof. We pick $m \in M$ and fix a lift $\tilde{m} \in \tilde{M}$ of m. Then, as dev is a local diffeomorphism, there is an open set $\tilde{U}$ of $\tilde{m}$ such that dev is a local diffeomorphism onto X.

We pick $V \subset M$ an open neighborhood of m and take its inverse under the covering map $p^{-1}(V)$ which we may assume dev is a diffeomorphism on by intersecting with $\tilde{U}$.

(TODO: Make this proof more clear) Now, define coordinate charts on M by $(V_\alpha, \psi_\alpha = \mathrm{dev} \circ p_\alpha^{-1})$ at a point $m \in M$ where by p_α^{-1} we mean we have fixed a component of the preimage p^{-1} restricted to V_α. Now suppose that $V_i \cap V_j \neq \emptyset$ and let $\tilde{U}_i = p_i^{-1}(V_i)$ and $\tilde{U}_j = p_j^{-1}(V_j)$. Then the preimage of their intersection will differ by a deck transformation $\gamma \in \pi_1(M)$. Therefore, on the intersection $V_i \cap V_j$

$$\begin{aligned} \mathrm{dev} \circ p_i^{-1} &= \mathrm{dev} \circ \gamma \circ p_j^{-1} \\ \Rightarrow p_i^{-1} \circ p_j &= \gamma \end{aligned}$$

Now our by definition of our transition maps

$$\begin{aligned} \psi_i \circ \psi_j^{-1} &= \mathrm{dev} \circ p_{i-1} \circ p_j \circ dev^{-1} \\ &= \mathrm{dev} \circ \gamma \circ dev^{-1} \\ &= \mathrm{hol}(\gamma) dev \circ dev^{-1} \\ &= \mathrm{hol}(\gamma) \in \Gamma \end{aligned}$$

and so we have thus given our coordinate charts on M in terms of the development pair. □

This shows that once we have fixed a development pair, our (G, X)-structure is entirely determined by this. Conversely, the following lemma shows us that once we have fixed our (G, X)-structure, the development pair is determined up to certain factors/conjugations.

Lemma 1.9. Let M be a (G, X)-manifold with development pair $(\mathrm{dev}, \mathrm{hol})$. Suppose that $(\mathrm{dev}', \mathrm{hol}')$ is another development pair for M. Then $\mathrm{dev}' = g \circ \mathrm{dev}$ and $\mathrm{hol}' = g \circ \mathrm{hol} \circ g^{-1}$ for some $g \in G$.

Proof. Our development pair was determined once we picked a base point and a chart (U, ϕ). Suppose that we pick a different base point and chart (V, φ). Now we consider how we define $\operatorname{dev}(x)$ for $x \in \tilde{M}$. We choose a curve γ connecting x_0 and x and choose a cover $U_{i\, i\in I}$ of γ and develop around it. Now, with respect to our new initial chart, $V \cup \{U_i\}_{i\in I}$ is an open cover of γ. If we develop starting with this new chart, we can first glue all the U_i to U_1 as before and then glue this chain to V with an element of G by gluing U_1 to V on their overlap. Hence the entire development will have just changed by that element of G.

Now to show that $\operatorname{hol}' = g \circ \operatorname{hol} \circ g^{-1}$ we consider

$$\begin{aligned}
\operatorname{dev}'(\gamma \star \alpha) &= g \circ \operatorname{dev}(\gamma \star \alpha) \\
&= g \circ \operatorname{hol}(\gamma) \circ \operatorname{dev}(\alpha) \\
&= (g \circ \operatorname{hol}(\gamma) \circ g^{-1})(g \circ \operatorname{dev}(\alpha)) \\
&= (g \circ \operatorname{hol}(\gamma) \circ g^{-1})\operatorname{dev}'(\alpha)
\end{aligned}$$

and as the development pair satisfied the equivariance property it follows that $\operatorname{hol}' = g \circ \operatorname{hol} \circ g^{-1}$. □

This shows that once we have fixed a development pair, any other development pair is related by $(\operatorname{dev}', \operatorname{hol}') = (g \circ \operatorname{dev}, g \circ \operatorname{hol} \circ g^{-1})$. We should therefore note that if we want to classify (G, X)-structures on a manifold M by classifying the possible development pairs, it is important to consider them up to this relation.

1.2 Completeness of (G, X)-structures

An important property the developing map can have is that of *completeness*.

Definition 1.10. A (G, X)-manifold M is said to be complete if its developing map is a diffeomorphism onto X.

In the case that X is simply connected e.g. $X = \mathbb{R}^n$, we have that M is diffeomorphic to X/Γ where $\Gamma = \operatorname{hol}(\pi_1(M))$.

In the case that M is a Riemannian manifold, this notion of completeness is equivalent to completeness of the Levi-Civita connection of M.

TODO : Examples of complete/ incomplete manifolds e.g. Hopf

1.2.1 Affine Structures on the Torus

The standard affine structure on the torus is obtained from viewing the torus as $\mathbb{R}^n/\mathbb{Z}^n$ where $\mathbb{Z}^n$ acts by translations. This structure is easily seen to be a complete integral affine manifold. Indeed, the action is an action by the integral affine group and is both free and proper. Covering space theory tells us that $\pi : \mathbb{R}^n \to \mathbb{R}^n/\mathbb{Z}^n$ is a covering space with the group of deck transformations via our $\mathbb{Z}^n$-action. A theorem of Auslander and Markus (L. Auslander and L. Markus, Holonomy of flat affinely connected manifolds) tells us that affine structures obtained in this manner are complete. However, even for the 2-torus, not all affine structures are complete. We now give an example of an incomplete structure on $\mathbb{T}^2$.

Example 1.11. Let $\lambda > 1$ be a real number and consider the action of $\mathbb{Z}$ generated by multiplication by λ on $\mathbb{R}^2 - \{0\}$. This action is a free and proper action by the affine group and so the quotient $M = \mathbb{R}^2 - \{0\}/\langle\lambda\rangle$ is an affine manifold.

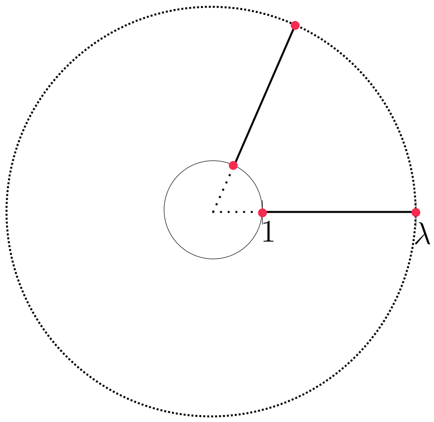

Figure 1.3: Visualisation of the Hopf manifold. The points 1 and λ are identified.

The map

$$\rho : \mathbb{R}^2 \to \mathbb{R}^2 - \{0\}$$
$$(x, y) \mapsto e^x(\cos y, \sin y)$$

is a covering map. Basing our fundamental group at $[(1, 0)] \in M$, our group is generated by the loops γ_1 and γ_2 where γ_1 is the projection of the line segment joining the point $(1, 0) \mapsto (\lambda, 0)$ and γ_2 is the loop going around the unit circle once. These lift to curves $\bar{\gamma}_1$ and $\bar{\gamma}_2$ in the universal cover satisfying

$$\bar{\gamma}_1(0) = (0, \ln(1)) = (0, 0), \quad \bar{\gamma}_1(1) = (0, \ln(\lambda)),$$
$$\bar{\gamma}_2(0) = (0, 0), \quad \bar{\gamma}_2(1) = (0, 2\pi),$$

from this we see that γ_1 and γ_2 act on (x, y) by

$$\gamma_1 \cdot (x, y) = (\ln(\gamma) + x, y)$$
$$\gamma_2(x, y) = (x, y + 2\pi)$$

Now, we define our developing pair by

$$\mathrm{dev}(x, y) = \rho(x, y),$$
$$\mathrm{hol}(\gamma_1) \cdot (x, y) = (\lambda x, \lambda y),$$
$$\mathrm{hol}(\gamma_2) = \mathrm{id}_{\mathbb{R}^2}$$

We check that these satisfy the equivariant property.

$$\begin{aligned}\mathrm{dev}(\gamma_1 \cdot (x, y)) &= \mathrm{dev}((\ln(\lambda) + x, y)) \\ &= e^{\ln(\lambda)} e^x(\cos y, \sin y) \\ &= \lambda \mathrm{dev}((x, y)) \\ &= \mathrm{hol}(\gamma_1)\mathrm{dev}((x, y))\end{aligned}$$

$$\begin{aligned}\mathrm{dev}(\gamma_2 \cdot (x, y)) &= \mathrm{dev}((x, y + 2\pi)) \\ &= e^x(\cos(y + 2\pi), \sin(y + 2\pi)) \\ &= e^x(\cos y, \sin y) \\ &= \mathrm{hol}(\gamma_2)\mathrm{dev}((x, y))\end{aligned}$$

It is now clear that this structure is incomplete as $0 \notin \mathrm{dev}(\mathbb{R}^2)$. Indeed, it is typical that the fixed point of a radiant structure is not in the developing image as shown in Theorem 6.5.2 [Gol22].

1.3 Representation Theory and Affine Manifolds

1.3.1 Affine representations

We now turn our attention to affine manifolds before specializing to the integral affine case. An affine manifold is a (G, X)-manifold where $X = \mathbb{R}^n$ and $G = \text{Aff}(\mathbb{R}^n)$. An important tool in classifying affine structures is the representation theory of groups. Often, by studying properties of the holonomy group of an affine manifold M, we are able to obtain information about the affine structure of M. Of fundamental importance is the case where the holonomy group is nilpotent which we shall see in Chapter 2. We will use these tools to determine conditions for the affine structure to be complete.

Definition 1.12. An affine transformation of $\mathbb{R}^n$ is a map

$$A : \mathbb{R}^n \to \mathbb{R}^n$$

such that $A(x) = Gx + b$ where $G \in GL(\mathbb{R}^n)$ and $b \in \mathbb{R}^n$.

The group $\text{Aff}(\mathbb{R}^n)$ is the group of affine transformations of $\mathbb{R}^n$ and is equal to $\text{Aff}(\mathbb{R}^n) = \text{Gl}(\mathbb{R}^n) \rtimes \mathbb{R}^n$.

Definition 1.13. An affine representation of a group G is a group homomorphism

$$\alpha : G \to \text{Aff}(\mathbb{R}^n).$$

Example 1.14. Let M be a (G, X)-manifold with $G = \text{Aff}(\mathbb{R}^n)$ and $X = -\mathbb{R}^n$. Then the holonomy representation is an affine representation of $\pi_1(M)$.

Definition 1.15. Let G be any group and let A be an abelian group. We will call A a G-module if there is a group homomorphism

$$\rho : G \to \text{Aut}(A).$$

Remark 1.16. A G-module A is a module in the usual sense over the group ring $\mathbb{Z}[G]$ which consists of formal integer sums of elements of G.

Remark 1.17. An affine representation α splits into a linear component and a translation component, i.e. $\alpha(g) = (\lambda(g), \mu(g))$ where $\lambda : G \to \text{Gl}(\mathbb{R}^n)$ and $\mu : G \to \mathbb{R}^n$. The linear component makes $\mathbb{R}^n$ into a G-module. We will denote this G-module by E when we want to emphasize that we are viewing $\mathbb{R}^n$ as a G-module.

Analogous to irreducible representations in the usual sense, there is a notion of irreducibility for affine representations.

Definition 1.18. An affine representation $\alpha : G \to \text{Aff}(E)$ is said to be *irreducible* if α leaves a proper affine subspace $F \subset E$ invariant.

Remark 1.19. Note that unlike standard representations, for affine representations we do not allow an origin to be fixed in the notion of irreducibility as there is no concept of an origin for affine spaces.

We now wish to illustrate that completeness of the affine structure has implications on the group theoretic properties of the affine holonomy group. We will use freely some of the properties of Eilenberg Mac-Lane spaces which can be found in [Hat02] although we will recall the definition.

Definition 1.20. Let G be a group and let $n \geq 1$ be an integer. A path-connected topological space X is called an *Eilenberg–MacLane space* of type $K(G, n)$ if

$$\pi_i(X) \cong \begin{cases} G & \text{if } i = n, \\ 0 & \text{if } i \neq n. \end{cases}$$

Equivalently, X has a single nontrivial homotopy group, which occurs in degree n and is isomorphic to G.

Theorem 1.21. Let M be a compact complete affine manifold. Then the affine holonomy representation is irreducible.

Proof. Since M is complete we have that $M = \mathbb{R}^n/\Gamma$ where Γ is the affine holonomy group. Now, if Γ is not irreducible, then there is an invariant affine subspace $V \subset \mathbb{R}^n$. Since V is invariant under Γ and Γ acts freely and properly on $\mathbb{R}^n$ it follows that V/Γ is an affine manifold with universal cover V. Since both $\mathbb{R}^n$ and V are contractible, it follows by Whiteheads theorem that V/Γ and M are Eilenberg Mac-Lane $K(\Gamma, 1)$ spaces.

Let $i : V \to \mathbb{R}^n$ be the inclusion. The induced map

$$i_* : \pi_1(V/\Gamma) \to \pi_1(M)$$

is an isomorphism of fundamental groups and hence since both spaces are $K(\Gamma, 1)$ this is a homotopy equivalence. Since $R^n\Gamma$ and V/Γ are both compact, it follows that $\dim V/\Gamma = \dim \mathbb{R}^n/\Gamma$ and hence $V = \mathbb{R}^n$. □

We will need some more machinary from the theory of group cohomology. This leads us to the following definitions.

Definition 1.22 (Crossed Homomorphism)**.** Let G be a group and E a G-module. A crossed homomorphism for $\lambda : G \to \mathrm{Aut}(E)$ is a group homomorphism

$$\mu : G \to E$$

that satisfies $\mu(gh) = \mu(g) + \lambda(g)\mu(h)$.

Proposition 1.23. Let $\alpha(g) = (\lambda(g), \mu(g))$ be an affine representation on E of a group G. Then μ is a crossed homomorphism for λ.

Proof. Let $g, h \in G$ and $\alpha = (\mu, \lambda)$ be an affine representation. Then

$$\begin{aligned} \alpha(gh) &= (\mu(gh), \lambda(gh)) \\ &= \alpha(g) \cdot \alpha(h) \\ &= (\mu(g), \lambda(g)) \cdot (\mu(h), \lambda(h)) \\ &= (\mu(g) + \lambda(g)\mu(h), \lambda(g)\lambda(h)) \\ \Rightarrow \mu(gh) &= \mu(g) + \lambda(g)\mu(h) \end{aligned}$$

and so μ is a crossed homomorphism for λ. □

Definition 1.24 (Principal Crossed Homomorphism)**.** A *principal crossed homomorphism* for λ is a crossed homomorphism of the form $\mu(g) = y - \lambda(g)y$ for some $y \in E$.

Remark 1.25. If μ is a principal crossed homomorphism for λ determined by some $y \in E$, we will often write $\mu = D_y$. This comes from the fact that principal crossed homomorphisms are sometimes also called *principal derivations*.

Lemma 1.26. The group cohomology $H^1(G, E)$ is isomorphic to

$$H^1(G, E) \cong \frac{\{\text{crossed homomorphisms for } \lambda\}}{\{\text{principal crossed homomorphisms for } \lambda\}}$$

Proof. This is a standard result in group cohomology. See for example [Tom23]. □

1.3.2 Radiant affine manifolds

Of importance to completeness of the affine structure is the case that our affine representation has a fixed point. It turns out that such structures will always be incomplete.

Definition 1.27. An affine representation with a fixed point is called *radiant*.

Proposition 1.28. Let $y \in E$. Then y is a *stationary point* of the action of α if and only if $\mu(g) = y - \lambda(g)y$ for all $g \in G$.

Proof. Suppose $y \in E$ is a stationary point of α. Then $\alpha(g)y = y$ for every $g \in G$. This gives that

$$\begin{aligned}(\lambda(g) + \mu(g))y &= \lambda(g)y + \mu(g)\\ &= y\\ \Rightarrow \mu(g) &= y - \lambda(g)y\end{aligned}$$

Conversely, suppose that $\mu(g) = y - \lambda(g)y$. Then $y = \mu(g) + \lambda(g)y = \alpha(g)y$ □

Definition 1.29 (Radiance Obstruction)**.** Let α be an affine representation. The radiance obstruction of α is the cohomology class

$$c_\alpha = [\mu] \in H^1(G, E)$$

where μ is the translational part of α.

Lemma 1.30. The radiance obstruction vanishes, e.g. $c_\alpha = 0$ if and only if α is conjugate to its linear part by a translation, if and only if α has a stationary point.

Proof. By Lemma 1.28, $y \in E$ is stationary if and only if $\mu(g) = D_y$ is a principal crossed homomorphism for λ if and only if $c_\alpha = [\mu] = 0$.

Now suppose that $\mu = D_y$ for some $y \in E$ so that $\mu(g) = y - \lambda(g)y$. We define the translation

$$\begin{aligned} T_y &: E \to E \\ x &\mapsto x - y. \end{aligned}$$

Now, conjugating α by this translation yields

$$\begin{aligned} T_y \circ \alpha(g) \circ T_y^{-1}(x) &= T_y \circ \alpha(g)(x + y) \\ &= T_y \circ (\lambda(g)x + \lambda(g)y + \mu(g)) \\ &= T_y \circ (\lambda(g)x + \lambda(g)y + y - \lambda(g)y) \\ &= T_y \circ (\lambda(g)x + y) \\ &= \lambda(g)x + y - y \\ &= \lambda(g)x. \end{aligned}$$

Conversely, suppose that $T_y \circ \alpha(g) \circ T_y^{-1} = \lambda(g)$ for all $g \in G$ and $x \in E$. Then

$$\begin{aligned} T_y \circ \alpha(g) \circ T_y(x) &= T_y \circ \alpha(g)(x + y) \\ &= T_y \circ (\lambda(g)(x + y) + \mu(g)) \\ &= \lambda(g)(x + y) + \mu(g) - y \\ &= \lambda(g)x \end{aligned}$$

and so

$$\begin{aligned} \mu(g) &= \lambda(g)(x) - \lambda(g)(x) - \lambda(g)(y) + y \\ &= y - \lambda(g)y \end{aligned}$$

as required. □

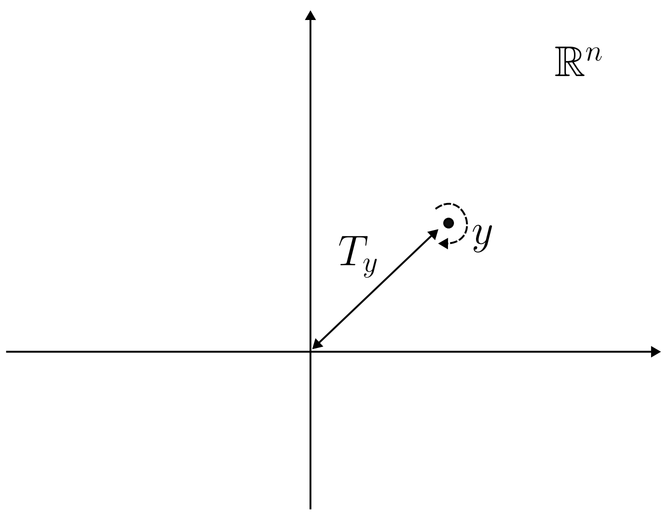

Figure 1.4: Conjugating a fixed point y under the affine action by a translation to the origin

Remark 1.31. This lemma will allow us to alter the development map by a translation (which alters the holonomy representation by a conjugation) to assume that our fixed point is at the origin.

Definition 1.32 (Contragradient Representation)**.** Let E be a G-module by $\lambda : G \to \mathrm{Aut}(V)$. We define the contragradient representation on E^* by

$$\begin{aligned} \lambda^* : G &\to GL(E^*) \\ g &\mapsto \lambda^* g \end{aligned}$$

such that $\lambda^* g(f)(x) = f(\lambda(g^{-1})x)$ for all $x \in E$ and $f \in E^*$.

Remark 1.33. Once a representation $\lambda : G \to \mathrm{Aut}(E)$ has been fixed, we will often write the action of g on an element x of E or E^* as $g \cdot x$. It is clear from whether x is an element of E or E^* whether this action should be taken as the action of $\lambda(g)$ or $\lambda^* g$.

Remark 1.34. This induces a representation on $\wedge^k E^*$ by
$(g \cdot \omega)(x_1, \ldots, x_n) = \omega(g^{-1} \cdot x_1, \ldots, g^{-1} \cdot x_n)$ for $\omega \in \wedge^k E^*$ and $x_i \in E$.

We will make use of the following lemma when looking at the relationship between the holonomy representation and parallel volume forms.

Lemma 1.35. Let $F : V \mapsto V$ be a linear map from a vector space V of dimension n to itself and let $\omega \in \wedge^n(V^*)$. Then

$$F^* \omega = \det(A)\omega$$

where A is any matrix representing the linear transformation F.

Proof. As $\dim(\wedge^n(V^*)) = 1$ and F is a linear map, it follows that our map corresponds to multiplication by some scalar $\lambda \in \mathbb{R}$. Let $\varphi : V \to \mathbb{R}^n$ be an isomorphism identifying our vector space V with $\mathbb{R}$. Then we have

$$\begin{aligned} F^* \varphi^* \det &= \lambda \varphi^* \det \\ \Rightarrow (\varphi^{-1})^* F^* \varphi^* \det &= \lambda \det \\ \Rightarrow (\varphi^{-1} F \varphi)^* \det &= \lambda \det . \end{aligned}$$

If we let $A = \varphi^{-1} F \varphi$ and $\{e_1, \ldots, e_n\}$ the standard basis of $\mathbb{R}^n$ then

$$A^* \det(e_1 . \ldots , e_n) = \lambda \det(e_1, \ldots, e_n) = \lambda = \det(Ae_1, \ldots, Ae_n)$$

which gives us that $\det(A) = \lambda$ as required. $\square$

This leads us to the following elementary but important property of the action of the holonomy representation on volume forms. (TODO: define the affine connection!!!!!)

Corollary 1.36. Let M be an affine manifold and $\lambda : G \to GL(\mathbb{R}^n)$ be the linear part of the holonomy representation. Then M has a parallel volume form with respect to the affine connection if and only if $\det(\lambda(g)) = 1$ for all $g \in G$.

Proof. Suppose ω is a parallel volume form on M. Then the parallel transport of ω around a loop in M leaves ω unchanged, e.g. the linear holonomy action preserves ω. This gives us that

$$\begin{aligned} g \cdot \omega &= \omega \\ \Rightarrow \omega(g^{-1} \cdot x_1, \dots, g^{-1} \cdot x_n) &= \omega(x_1, \dots, x_n) \end{aligned}$$

for all $x_i \in M$. But by Lemma 1.35

$$\omega(g^{-1} \cdot x_1, \dots, g^{-1} \cdot x_n) = \det(\lambda(g^{-1})) \cdot \omega(x_1, \dots, x_n)$$

and so the volume is preserved if $\det(\lambda(g)) = 1$

Conversely, suppose that the determinant of the linear holonomy is 1 for all $g \in G$. If we pick a point $m_0 \in M$ then locally there exists a volume form ω_0 on a chart U of M with respect to the affine connection, corresponding to a constant, top degree form. We now claim that we can uniquely extend such a form. Indeed, as parallel transport provides a linear isomorphism between tangent spaces we can attempt to define a volume form on M by parallel transporting ω_0.

Pick another point $m \in M$. We claim that the parallel transport of ω_0 along a curve γ from m_0 to m does not depend on the choice of curve. If γ' is another curve then the parallel transport $(P^{\gamma'}_{m_0 \to m})^{-1} \circ P^{\gamma}_{m_0 \to m} = g$ for some g in the linear holonomy group. It follows that

$$\begin{aligned} g^* \circ \omega_0 &= ((P^{\gamma'}_{m_0 \to m})^{-1} \circ P^{\gamma}_{m_0 \to m})^* \circ \omega_0 \\ &= \det(\lambda(g)) \cdot \omega_0 \\ &= \omega_0 \\ \Rightarrow (P^{\gamma}_{m_0 \to m})^* \circ \omega_0 &= (P^{\gamma'}_{m_0 \to m})^* \circ \omega_0 \end{aligned}$$

and so the parallel transport does from m to m_0 does not depend on the choice of path. We note that this defines a continuous maps as in each affine chart the parallel transport identifies the tangent spaces via the identity map and on over laps they differ by an element of $GL(\mathbb{R}^n)$ as the derivative of an affine map extracts the linear part. ☐

Definition 1.37 (Radiant Manifold)**.** An affine manifold is said to be radiant if its affine holonomy representation has a fixed point.

Remark 1.38. By Lemma 1.30 we may conjugate our developing map by a translation to assume that the fixed point is at the origin.

When M is radiant the above assumption shows that the radiant vector field $R = \sum_i x_i \frac{\partial}{\partial x_i}$ is fixed by the affine action. There is then a unique vector field $\tilde{X}$ on the universal cover $\tilde{M}$ that is related to R by the developing map. Since this is preserved by deck transformations (by equivariance of developing map and since the representation is radiant) it descends to a vector field X on M.

Definition 1.39. The vector field X on M is called the radiant vector field of M.

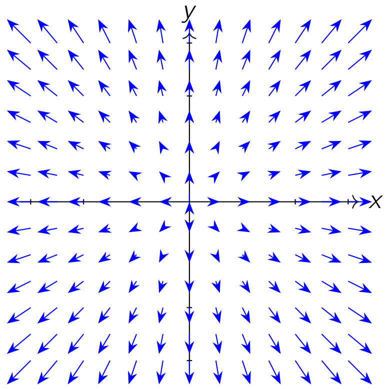

Figure 1.5: Radiant Vector Field on $\mathbb{R}^2$

1.4 Integral Affine Manifolds

One of the main objects of interest in this thesis is integral affine manifolds. These are (G, X)-structures whereby $G = \mathrm{Aff}_{\mathbb{Z}}(\mathbb{R}^n)$ and $X = \mathbb{R}^n$. They show up in symplectic geometry as the base space of a regular Lagrangian torus fibration inherits such a structure as we explain in Chapter 3. They are also important objects in the study of the SYZ conjecture in physics which relates a Calabi-Yau manifold X to its "mirror" $\tilde{X}$ by a Lagrangian fibration over a common base space B [Gro08]. In this chapter we give some basic properties of these manifolds as well as some examples.

Proposition 1.40. Let M be a smooth manifold. An integral affine structure on M is equivalent to a full-rank lattice $\Lambda \subset T^*M$ that is locally spanned by closed one-forms.

Proof. Given an integral affine structure on M, we can define a lattice in T^*M as follows. Let $(U, x_1, \dots, x_n)$ be a chart on M. We define $\Lambda|_U = \mathrm{span}_{\mathbb{Z}}(dx_1, \dots, dx_n)$. Carrying out this construction in each affine chart defines a full-rank lattice in T^*M. On overlaps, these lattices will differ by the differential of an integral affine map $x \mapsto Ax + b$ with $A \in \mathrm{Gl}(\mathbb{Z}^n)$ and $b \in \mathbb{R}^n$. But the differential of such a map is given by the linear part A and hence is an automorphism of the lattice.

Conversely, given a full-rank lattice $\Lambda \subset T^*M$ that is locally spanned by closed one-forms, by shrinking charts if necessary we can assume that our lattice is spanned by exact one-forms $(dx_1, \dots, dx_n)$ in an open neighborhood U. Since the $dx_i|_p$ span a full-rank lattice in T_p^*U, they are linearly independent and hence if we define coordinates by $(U, x_1, \dots, x_n)$ by the inverse function theorem the map $\bar{x} = (x_1, \dots, x_n)$ is a local diffeomorphism. On the overlap of two charts $U_i \cap U_j$, we know that the pullback of the lattice in U_i to U_j must be an automorphism of the lattice. But these maps are given exactly by elements of the integral affine group $\mathrm{Aff}_{\mathbb{Z}}(\mathbb{R}^n)$. $\square$

We now give some examples of integral affine manifolds.

Example 1.41. Let $\mathbb{Z}^n$ act on $\mathbb{R}^n$ by translations. Then the quotient $(\mathbb{R}/\mathbb{Z})^n$ inherits an integral affine structure and is the standard integral affine torus. We note that this integral affine structure can not be equivalent to any of the structures obtained from Example 1.11 as these are incomplete. In fact, it is also true that a closed integral affine manifold cannot be radiant as we shall see in (TODO: corollary).

Example 1.42. Let B be an integral affine manifold and let $f : M \to B$ be a local diffeomorphism where M is any smooth manifold. Then there is a unique integral affine structure on M such that f is an integral affine map as defined in Definition 1.1. The developing map is an example of such a map and in this sense the integral affine structure on any integral affine manifold can be realised as the pullback of the standard integral affine structure on $\mathbb{R}^n$.

Example 1.43. Let M be an integral affine manifold and let Γ be a discrete group acting freely and properly on M by integral affine diffeomorphisms. Then the quotient M/Γ inherits an integral affine structure and the projection map $\pi : M \to M/\Gamma$ is an integral affine map.

We can of course define an integral affine structure by specifying a full-rank lattice in T^*M that is locally spanned by closed one-forms. Importantly, the cotangent bundle of any manifold M inherits a symplectic structure which gives rise to a volume form ω on T^*M. If $\Lambda, \Lambda' \subset T^*M$ are two lattices in T^*M such that they give M an integral affine structure, then the volume form descends to the quotient and the induced volume of T^*M/Λ and T^*M/Λ' must be the same if Λ and Λ' determine equivalent integral affine structures. This will be discussed more in Chapter 3.

Example 1.44 (Radiant integral affine manifold)**.** If we drop the compactness assumption, it is possible to equip a manifold M with an integral affine radiant structure. Let $M = \mathbb{R}^2 - \{0\}$ and

$$i : M \hookrightarrow \mathbb{R}^2$$

be the inclusion. Then this is a local diffeomorphism and hence, equipping $\mathbb{R}^2$ with the standard integral affine structure, this induces an integral affine structure on M. There is a covering map

$$\pi : (0, \infty) \times \mathbb{R} = \tilde{M},$$

$$(r, \theta) \mapsto (r\cos\theta, r\sin\theta)$$

and the composition $\widetilde{\mathrm{dev}} = i \circ \pi : \tilde{M} \to \mathbb{R}^2$ is the developing map for the integral affine structure induced on M. This developing map fails to be injective and surjective, hence this affine structure is not complete and is therefore not affinely diffeomorphic to the standard integral affine structure on $\mathbb{R}^2$. By considering $\theta \mapsto \theta + 2\pi$ one notes that $\widetilde{dev}(r, \theta + 2\pi) = \mathrm{id}_{\mathbb{R}^2}\widetilde{\mathrm{dev}}(r, \theta)$ and so the holonomy representation is the trivial one in this case.

We can modify the affine structure so that it becomes radiant as follows. For any $A \in \mathrm{Gl}(\mathbb{Z}^n)$, there is a free action of A on M by integral affine diffeomorphisms. This lifts to the universal cover $\tilde{M}$ so that $\pi \circ \tilde{A} = A \circ \pi$ where $\tilde{A}$ is the lift to the cover which is unique after fixing a base point by covering space theory. If we take a "branch cut" V, i.e. remove a ray where θ is constant, then the sheets above V are $U_n = \{(r, \theta) \mid 2\pi n < \theta < 2\pi(n+1)\}$. The developing map is a diffeomorphism when restricted to a single sheet and so the action of A on one of these sheets is given by $A \cdot U_n = \widetilde{\mathrm{dev}}^{-1}(A \cdot \widetilde{\mathrm{dev}}(U_n))$ where we view A as acting on $\mathbb{R}^2$ via the inclusion i. We can define a new development pair on M by the condition

$$\mathrm{dev} = A \cdot \widetilde{\mathrm{dev}}$$

$$\mathrm{hol}(\gamma^k) = (A^k, 0)$$

where $\gamma \in \pi_1(M)$ is a generator of $\pi_1(M)$. On a sheet U_n we have

$$\begin{aligned}
\mathrm{dev}(A \cdot (r, \theta)) &= A \cdot \widetilde{\mathrm{dev}}(A \cdot (r, \theta)) \\
&= \mathrm{hol}(\gamma) \cdot \widetilde{\mathrm{dev}}(A \cdot (r, \theta)) \\
&= \mathrm{hol}(\gamma) \cdot A \cdot \widetilde{\mathrm{dev}}(r, \theta) \\
&= \mathrm{hol}(\gamma) \cdot \mathrm{dev}(r, \theta)
\end{aligned}$$

and so the equivariance property is satisfied since the union of the U_n form an open, dense set. It follows that $(\mathrm{dev}, \mathrm{hol})$ defines a radiant integral affine structure on M.

Example 1.45 (Mapping tori)**.** Let M be an integral affine manifold and let $f : M \to M$ be an integral affine diffeomorphism. Then, equipping $\mathbb{R}$ with its standard integral affine structure, we can form the quotient manifold

$$M_f = \frac{M \times \mathbb{R}}{\sim}$$

where $(m, t) \sim (m', t')$ if $(m, t) = (f^{-n}(m'), t' + n)$ for some $n \in \mathbb{Z}$ where f^n denotes composition of f or f^{-1} depending on if n is positive. We note that even if f is not free, the $\mathbb{Z}$-action defining the quotient is free as it is a translation in the second component. The resulting manifold is called the *mapping torus* of f. We note that this is a fiber bundle over S^1 through projection onto the second component.

Remark 1.46. If Σ is compact and admits a $\mathbb{R}P^n$-structure and f is a projective automorphism, then the mapping torus of f admits a radiant affine structure [Gol22].

(TODO: probably some of our affine structures can be realised as mapping toris of $S^1 = \mathbb{R}/\mathbb{Z}$. I think definitely the upper triangular guys)

CHAPTER TWO

Nilpotent Holonomy and Completeness

2.1 Nilpotent Holonomy

2.1.1 Nilpotent Groups

In this section, we will see that the condition that the holonomy group of an affine manifold M be nilpotent will be useful in determining whether the affine structure is complete. We will build up some useful results before determining some equivalent conditions that force completeness of the affine structure.

Definition 2.1 (Central Series)**.** Let G be a group. Then a central series for G is a finite collection of normal subgroups such that

$$1 = G_0 \leq G_1 \leq \cdots \leq G_s = G$$

and

$$\frac{G_{i+1}}{G_i} \leq Z(\frac{G}{G_i})$$

where $Z(\frac{G}{G_i})$ denotes the center.

Definition 2.2 (Nilpotent Group)**.** A group G is said to be nilpotent if it admits a central series.

Remark 2.3. The *nilpotent class* of G is the length of the shortest central series.

Example 2.4. An important example of nilpotent groups are the upper triangular matrices. For example, the group

$$H(R) = \left\{ \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} \right\}$$

is called the Heisenberg group and is nilpotent of class 2. This group notably shows up in quantum mechanics.

For nilpotent groups, we have the following results regarding their group cohomology.

Lemma 2.5. Let G be a nilpotent group and E a G-module. If $H^0(G, E) = 0$ then also $H^i(G, E) = 0$ for all $i \geq 0$.

Proof. The proof of this can be found in [Hir78]. □

A consequence of this is the following lemma. (TODO: WTH is group homology? Should say some stuff about these things.)

Lemma 2.6. If G is a nilpotent group and E is a G-module, then the following statements are equivalent.

1. $H^0(G, E) = 0$
2. $H^0(G, E^*) = 0$
3. $H_0(G, E) = 0$
4. $H_0(G, E^*) = 0$

Proof. See [FGH81]. □

2.1.2 Unipotence and the Fitting submodule

In the case that the affine holonomy group is nilpotent, we will be able to show that completeness is equivalent to a property called unipotence of the linear holonomy. We will now outline some consequences of unipotence.

Definition 2.7 (Unipotent)**.** A G-module E is called unipotent if $(g - I)^n = 0$ for every $g \in G$ where $n = \dim(E)$.

Definition 2.8. An affine representation is unipotent if its linear part defines a unipotent G-module.

Although E may not be unipotent itself, we can consider the smallest submodule (possibly the zero module) of E which is unipotent. We shall denote this submodule by E_U.

Remark 2.9. The submodule E_U is called the *Fitting submodule* of E.

Often, it will be useful to quotient out by the unipotent part of a G-module. The following lemma gives some insight on why this is a useful idea. (TODO: check this lemma)

Lemma 2.10. Let E_U be the Fitting submodule of E. Then

$$H^0(G, E/E_U) = 0$$

Proof. Take E_U to be the Fitting submodule and assume $E_U \neq E$ otherwise the result is trivial. As $H^0(G, E/E_U)$ is the set of stationary points of the G-module E/E_U, assuming $H^0(G, E/E_U) \neq 0$ we can take $x + E_U \in E/E_U$ not the identity so that

$$g \cdot (x + E_U) = x + E_U \text{ for all } g \in G$$
$$\Rightarrow (g-1) \cdot x \in E_U$$

Now we consider the G-submodule spanned by x and E_U. As $x \notin E_U$ this submodule is strictly bigger than E_U. Now we have that

$$(g-1)^n \cdot (x+e) = (g-1)^{n-1}((g-1)x + (g-1)e)$$

But $(g-1)x \in E_U$ so that $(g-1)x + (g-1)e \in E_U$. But as $E_U \neq E$ it has dimension at most $n-1$ and so $(g-1)^{n-1} = 0$ restricted to E_U. Therefore this is a unipotent submodule strictly bigger than E_U which contradicts maximality. □

The following splitting lemma will prove important in our characterization of completeness.

Lemma 2.11. If G is a nilpotent group and E a G-module with Fitting submodule E_U, then there exists a unique submodule F such that $E = E_U \oplus F$.

Proof. Let β be the induced representation $\beta : G \to GL(E/E_U)$ on $GL(E/E_U)$. By Lemma 2.10, $H^0(G, E/E_U) = 0$.

Claim. A submodule $F \subset E$ is complementary to E_U e.g. $E = E_U \oplus F$ if and only if $F = T(E/E_U)$ where $T : E/E_U \to E$ is an equivariant linear map with $P \circ T = 1_{E/E_U}$ with P the canonical projection map.

Now let $S : E/E_U \to E$ be a linear map satisfying $P \circ S = 1_{E/E_U}$. Then any such T as above can be uniquely written as $T = R + S$ with $R : E/E_U \to E_U$. It is T-equivariant if and only if

$$R + S = g \circ (R + S) \circ \beta(g)^{-1}$$

$$\Rightarrow R = g \circ R \circ \beta(g)^{-1} + g \circ S \circ \beta(g)^{-1} - S. \tag{2.1}$$

If we can show that there is a unique such T, then we have shown that E splits uniquely into $E = E_U \oplus F$. This will follow if we can show there is a unique R satisfying the above.

To do this we define a representation on $\mathrm{Hom}(E/E_U, E)$ by

$$\gamma : G \mapsto \mathrm{Hom}(E/E_U, E)$$
$$\gamma(g)(R) = g \circ R \circ \beta(g)^{-1}$$

and we define a crossed homomorphism for γ by $\mu(g) = g \circ S \circ \beta(g)^{-1} - S$. This defines a crossed homomorphism as

$$\begin{aligned}
\mu(gh) &= g \circ h \circ S \circ \beta(h)^{-1} \circ \beta(g)^{-1} - S \\
&= \gamma(g)(h \circ S \circ \beta(h)^{-1}) - S \\
&= \gamma(g)(\mu(h) + S) - S \\
&= \gamma(g)(\mu(h)) + g \circ S \circ \beta(g)^{-1} - S \\
&= \gamma(g)(\mu(h)) + \mu(g).
\end{aligned}$$

As P is equivariant and $P \circ S = 1_{E/E_U}$ we get

$$\begin{aligned} P \circ \mu(g) &= P \circ g \circ S \circ \beta(g)^{-1} - P \circ S \\ &= \beta(g) \circ P \circ S \circ \beta(g)^{-1} - 1 \\ &= \beta(g) \circ \beta(g)^{-1} - 1 \\ &= 0. \end{aligned}$$

Now we can we write (2.1) as

$$R = \gamma(g) \circ R + \mu(g)$$

which by Proposition 1.28 is saying that R is a fixed point of the affine representation induced by γ and μ. We wish now to show that this affine representation has a unique fixed point. This is equivalent by (TODO: define group group cohomology) Lemma 2.5 and Lemma 2.6 to $H^0(G, \mathrm{Hom}(E/E_U)) = 0$.

To this end we suppose that $R : E/E_U \mapsto E$ is fixed under all $\gamma(g)$ for $g \in G$. Then, letting $d = \dim(E_U)$ for any d elements $g_1, \ldots, g_d \in G$ we have

$$(1 - g_1) \circ \cdots \circ (1 - g_d)|_{E_U} = 0$$

as they act unipotently on E_U. It now follows by equivariance that

$$\begin{aligned} (1 - g_1) \circ \cdots \circ (1 - g_d) \circ R &= (1 - g_1) \circ \cdots \circ (R - g_d \circ R) \\ &= (1 - g_1) \circ \cdots \circ (R - R \circ \beta(g_d)) \\ &= (1 - g_1) \circ \cdots \circ R \circ (1 - \beta(g_d)) \\ &\Rightarrow R \circ (1 - \beta(g_1)) \circ \cdots \circ (1 - \beta(g_d)) = 0 \end{aligned}$$

so that R vanishes on vectors of the form

$$(1 - \beta(g_1)) \circ \cdots \circ (1 - \beta(g_d))x$$

for $x \in E/E_U$. Thus if all vectors are in this span, then the only such R fixed under the action of γ is the zero map. As $1 - \beta(g)$ is invertible on E/E_U (it acts without unipotent part), this is equivalent to showing that E/E_U is spanned by vectors of the form $(1 - \beta(g))x$ with $x \in E/E_U$. By definition, this is equivalent to $H_0(G, E/E_U) = 0$ which is true by 2.10 completing the proof. $\square$

This proves our result once we have cleared up that the stated claim is indeed true which we show in the following proposition.

Proposition 2.12. A submodule $F \subset E$ is complementary to E_U e.g. $E = E_U \oplus F$ if and only if $F = T(E/E_U)$ where $T : E/E_U \to E$ is an equivariant linear map with $P \circ T = 1_{E/E_U}$ with P the canonical projection map.

Proof. (TODO: format this better) Suppose $F = T(E/E_U)$, with T being G-equivariant so that F is a G-module. If $T(x + E_U) = e \in E$, then $P(e) = e + E_U = x + E_U \Rightarrow x \in E_U$ and so by linearity $e = 0$ and so $F \cap E_U = \{0\}$. Clearly, $E_U \oplus F \subseteq E$. For any $x \in E$, $x - T(x + E_U) \in E_U$. We can thus write $x = (x - T(x + E_U)) + T(x + E_U) \in E_U + F$.

Conversely, suppose that F is a G-submodule and $F \cap E_U = \{0\}$ and $F \oplus E_U = E$. Then its easy to see that $P|_F : F \mapsto E/E_U$ is bijective. It is G-equivariant as P is and F is a G-submodule. We can thus define T as $T = P^{-1}|_F$. $\square$

An important consequence of this theorem is that for *indecomposable affine representations* of a nilpotent group the affine representation is always unipotent. We recall the definition of an indecomposable affine representation now and then prove this result.

Definition 2.13 (Decomposable affine representation)**.** Let E be a G-module. An affine representation $\alpha : G \to \mathrm{Aff}(E)$ is said to be *decomposable* if there exists a decomposition $E = E_1 \oplus F$ with $E_1 \neq E$ such that the decomposition is invariant under the linear part of α and some coset $x + E_1$ is invariant under the entire affine holonomy α where $x \in E$.

Definition 2.14 (Indecomposable affine representation)**.** An affine representation $\alpha : G \to \mathrm{Aff}(E)$ is called *indecomposable* if it is not decomposable.

Lemma 2.15. Let $E_U \subset E$ be the Fitting submodule of an affine representation $\alpha : G \to \mathrm{Aff}(E)$. If G is nilpotent then there is a coset of E_U that is invariant under α.

Proof. Let $\mu : G \to E$ be the translational component of the affine representation $\alpha = (\lambda, \mu)$. We check that

$$\mu' : G \xrightarrow{\mu} E \xrightarrow{\pi} E/E_U$$

is a crossed homomorphism for $\lambda' = \pi \circ \lambda$. Letting $g, h \in G$ we compute

$$\begin{aligned}
\mu'(gh) &= \pi \circ \mu(gh) \\
&= \pi \circ (\mu(g) + \lambda(g)\mu(h)) \\
&= \pi \circ \mu(g) + \pi \circ \lambda(g)\mu(h) \\
&= \mu'(g) + \lambda'(g)\mu'(h).
\end{aligned}$$

Now, it follows that $\alpha' = (\lambda', \mu')$ is an affine representation of G on E/E_U. Furthermore, the following diagram commutes

$$\begin{tikzcd}
E \arrow[r, "\pi"] \arrow[d, "\alpha(g)"'] & E/E_U \arrow[d, "\alpha'(g)"] \\
E \arrow[r, "\pi"'] & E/E_U
\end{tikzcd}$$

for all $g \in G$. Hence π is a G-equivariant map with respect to the representations α and α'. Now Lemma 2.10 says that $H^0(G, E/E_U) = 0$ and hence by Lemma 2.5 we have that $H^1(G, E/E_U) = 0$ since G is nilpotent. Namely, the radiance obstruction $[c_{\alpha'}] \in H^1(G, E/E_U)$ vanishes and hence the affine representation α' fixes some $x + E_U$ where $x \in E$ and so α fixes that coset. $\square$

We can now prove the following theorem relating indecomposable affine representations to unipotence.

Lemma 2.16. Let $\alpha : G \to \mathrm{Aff}(E)$ be an indecomposable affine representation of a nilpotent group G. Then α is unipotent.

Proof. This follows as a corollary of our previous two lemmas. Lemma 2.11 tells us that there is a unique $F \subset E$ such that $E = E_U \oplus F$ where F is a G-submodule of E and E_U is the Fitting submodule. Now, Lemma 2.15 tells us that there is some coset of E_U invariant under the linear part of α. But this by definition means that α is decomposable unless $E_U = E$ which completes the proof as E_U is the unipotent part of E. □

2.2 Unipotent Representations and Completeness

2.2.1 Expansions in the linear holonomy

The aim of this section is to show that for a compact affine manifold, the holonomy group being unipotent is equivalent to completeness of the affine structure. Of geometrical importance to this will be the concept of an *expansion* in the linear holonomy.

Definition 2.17 (Expansion)**.** Let $T : V \to V$ be a linear map. Then T is said to be an expansion of V if for every eigenvalue λ of T we have $|\lambda| > 1$.

We now wish to prove Lemma 2.30 involving expansions in the linear holonomy group. However, in order to do so we will need to recall some of the theory of algebraic groups and the representation theory of nilpotent groups.

2.2.2 Algebraic groups and representation theory of Lie algebras

TODO: reference etc

Definition 2.18 (Algebraic Group)**.** An *algebraic group* is a matrix group that can be defined using polynomials.

If G is an algebraic group, then there is a unique irreducible component passing through the identity denoted by G°. The topology on G is taken to be the Zariski topology.

Remark 2.19. This unique irreducible component is called the *identity component* of G.

Proposition 2.20. Let G be an algebraic group. Then G° is a normal subgroup of finite index in G whose cosets are connected and irreducible components of G.

Proof. See section 7.3 of [Hum75]. □

Proposition 2.21. Let G be an algebraic group and H a normal subgroup. Then

$$\overline{[G, H]} = [\overline{G}, \overline{H}]$$

where $\overline{H}$ denotes the Zariski closure.

Proof. This follows from Proposition 2.4 of [Bor91]. □

This proposition leads to the following corollary.

Corollary 2.22. Let G be an algebraic group and N a normal subgroup that is nilpotent. Then $\overline{N}$ is also nilpotent.

Proof. By Proposition 2.21, a central series for N will pass to a central series of $\overline{N}$. □

The next result we will state is an important result from the representation theory of Lie algebras. It will guarantee us a basis for our representation such that the matrices of the representation are upper triangular.

Definition 2.23 (Weight Space)**.** Let $\mathfrak{g}$ be a Lie algebra and $\pi : \mathfrak{g} \mapsto \mathfrak{gl}_V$ be a representation on a vector space V. For $\lambda \in \mathfrak{g}^*$ we define the weight space of $\mathfrak{g}$ with respect to λ to be

$$V_\lambda^{\mathfrak{g}} = \{v \in V | \pi(g)v = \lambda(g)v, \quad \forall g \in \mathfrak{g}\}.$$

Remark 2.24. If $V_\lambda^{\mathfrak{g}} \neq 0$ we will say that λ is a weight for π.

Lemma 2.25 (Lie's theorem)**.** Let $\mathfrak{g}$ be a solvable Lie algebra and let π be a representation of $\mathfrak{g}$ on a finite dimensional, non-zero vector space V. Then there exists a weight $\lambda \in \mathfrak{g}^*$ for π.

Proof. This result can be found for instance in [Kac10]. □

Corollary 2.26. Let V be a vector space. If $\pi : \mathfrak{g} \mapsto \mathfrak{gl}(V)$ is a finite dimensional representation of a solvable Lie algebra, then there is a basis of V such that all linear transformations in $\pi(\mathfrak{g})$ are represented by upper triangular matrices.

We will need a slightly stronger theorem than Lie's theorem which gives a decomposition of our vector space into generalized weight spaces.

Definition 2.27 (Generalized Weight Space)**.** Let π be a representation of a Lie algebra $\mathfrak{g}$ over a finite dimensional vector space V. Then for $\lambda \in \mathfrak{g}^*$, we can associate to λ

$$V_\lambda = \{v \in V | (\pi(g) - \lambda(g))^n v = 0 \text{ for all } g \in \mathfrak{g}\}$$

where n depends on $g \in \mathfrak{g}$ and $v \in V$. If $V_\lambda \neq 0$ then we call it the *generalized weight space* associated to λ.

Remark 2.28. This tells us that $\pi(g) - \lambda(g)$ is a nilpotent operator so that $\pi(g) = \lambda(g) + N(g)$ with $N(g)$ nilpotent.

Lemma 2.29. Let $\mathfrak{g}$ be a nilpotent Lie algebra over $\mathbb{C}$. Let π be a representation of $\mathfrak{g}$ on a finite dimensional complex vector space V. Then there are finitely many generalized weight spaces V_{λ_k}, which are invariant under the action of the representation and $V = \oplus_k V_{\lambda_k}$.

Proof. See Proposition 3.2 in [War13]. □

We now have the required machinary to start our proof.

Lemma 2.30. Let G be a nilpotent group and $E = \mathbb{R}^n$ a G-module. Suppose that the linear holonomy group does not contain an expansion of E. Then for each $n \in \mathbb{N}_{>0}$, there is a C^r map $\varphi : E \mapsto \mathbb{R}$ for any $r \in \mathbb{Z}_{>0}$ which satisfies the following.

1. $\varphi > 0$ almost everywhere.
2. φ is G-invariant.
3. There exists some $a > 0$ such that
$$\varphi(e^t x) = e^{ta}\varphi(x)$$
for all $t \in \mathbb{R}$ and $x \in E$.

Proof. If we can prove this for a normal subgroup G_0 of finite index, then we will be able to construct our map φ as follows. Let $g_1G_0, \dots g_kG_0$ be the left cosets of G_0 and φ_0 a map satisfying properties 1 and 3 defined on G_0. Now define

$$\varphi(x) = \sum_{i=1}^{k} \varphi_0(g_i x).$$

This will be G-invariant as G_0 is a normal subgroup of G and multiplication by an element of G will simply permute the cosets.

As we want to eventually apply apply some results from complex Lie Theory, we complexify the dual space E^* as $F = \mathbb{C} \otimes E^*$. The contragradient representation of G on E^* extends to a representation on F in the obvious way. We will denote this as $\rho : G \mapsto GL(F)$. Now, $\rho(G)$ is a nilpotent subgroup of $GL(F)$ that passes through the identity.

Let H be the identity component of the algebraic closure of $\rho(G)$. As $\rho(G)$ is nilpotent, by Corollary 2.22 so is its Zariski closure and hence its identity component as subgroups of nilpotent groups are nilpotent. The subgroup H will be a normal subgroup of finite index by Proposition 2.20 and so $G_0 = \rho^{-1}(H)$ will also be a normal subgroup of finite index.

We can now obtain a Lie algebra representation of the Lie algebra of H on F by

$$di : \mathfrak{h} \mapsto \mathfrak{gl}(F)$$

where i is the inclusion map. Importantly, by construction on the underlying vector spaces $\rho(G_0) \subseteq di(\mathfrak{h})$. As H is a nilpotent Lie group, its Lie algebra is nilpotent hence solvable so we can apply Lemma 2.29 to obtain a decomposition $F = \oplus_{k=1}^m F_k$ which is $di(\mathfrak{h})$-invariant and hence ρ-invariant by construction when restricted to G_0.

Again, Lemma 2.29 yields that there is a basis B_k of each F_k that represents the operators $di(h)|_{F_k}$ as upper triangular complex matrices

$$di_k(h) = \lambda_k(h)I + N_k(h)$$

where di_k is the restriction to F_k and $N_k(h)$ is strictly upper triangular nilpotent matrix and $\lambda_k(h)$ are the eigenvalues of $\rho(h)$. The first basis vector in each B_k is killed by the strictly upper triangular matrix so that if $f_k \in F_k$ is the first basis vector in B_k

$$di_k(h)f_k = \lambda_k(h)f_k$$

for all $h \in H$.

Now we define the following group morphisms

$$\varphi_k : G_0 \mapsto \mathbb{R}$$
$$\varphi_k(g) = \log|\lambda_k(g)|$$

for $k = 1, \dots, m$.

This defines a map into $\mathbb{R}^m$ by

$$\varphi = (\varphi_1, \dots, \varphi_m) : G_0 \mapsto \mathbb{R}^m.$$

If G_0 does not contain any expansions of E then the image $\varphi(G_0)$ is disjoint from the set

$$P = \{y \in \mathbb{R}^m : y_k > 0, k = 1, \dots, m\}.$$

As it is a subgroup, it follows that its linear span span$\{\varphi(G_0)\}$ is also disjoint from P as if a point in the image was in the strictly negative orthant, it's inverse would lie in P. Therefore there exists a vector in the orthogonal complement of $\varphi(G_0)$ that lies in the closure of P. We denote this by $v = (v_1, \dots, v_m)$ and it satisfies

$$v_k \geq 0 \text{ for all } k,$$

$$\sum_{k=1}^{m} v_k = a \geq 0,$$

$$\sum_{k=1}^{m} v_k \varphi_k(g) = 0 \text{ for all } g \in G_0.$$

If we scale our v by a positive scalar $c > 0$ it will satisfy the same properties so that we can arrange that

$v_k = 0$ or $v_k > 2r$ for each k with r coming from the lemma.

Now for each k we have a vector $f_k \in \mathrm{Hom}(\mathbb{C} \otimes E, \mathbb{C})$ satisfy $di_k(h)f_k = \lambda_k(h)f_k$. We now include E into $\mathbb{C} \otimes E$ and construct the map

$$\psi : E \mapsto \mathbb{R}$$

$$\psi(x) = \prod_{k=1}^{m} |f_k(x)|^{v_k}.$$

Now ψ is C^r differentiable as $|f_k(x)|^{v_k} = (u_k(x)^2 + w_k(x)^2)^m$ with $m > r$. Properties 1) and 3) are satisfied by our construction of v_k. We now show that it is G_0-invariant.

$$\psi(g^{-1}x) = \prod_{k=1}^{m} |f_k(g^{-1}x)|^{v_k}$$

noting this is the contragradient representation and also that f_k is an eigenvector

$$\begin{aligned}\psi(g^{-1}x) &= \prod_{k=1}^{m} |g \circ f_k(x)|^{v_k} \\ &= \prod_{k=1}^{m} |\lambda(g) f_k(x)|^{v_k} \\ &= \prod_{k=1}^{m} |\lambda_k(g)|^{v_k} \prod_{k=1}^{m} |f_k(x)|^{v_k}\end{aligned}$$

Now, recalling that $\varphi_k(g) = \log |\lambda_k(g)| \Rightarrow e^{\varphi_k(g)} = |\lambda_k(g)|$ so we obtained

$$\begin{aligned}\prod_{k=1}^{m} |\lambda_k(g)|^{v_k} \prod_{k=1}^{m} |f_k(x)|^{v_k} &= e^{\sum_{k=1}^{m} v_k \varphi_k(g)} \psi(x) \\ &= \psi(x)\end{aligned}$$

completing the proof. □

2.2.3 Consequences of expansions in linear holonomy

The previous theorem leads to several important results regarding completeness of affine manifolds.

Theorem 2.31. Let M be a compact affine manifold. Let $E_0 \subset E$ be a proper linear subspace invariant under the affine holonomy and denote by Λ the linear holonomy. If the image of the linear holonomy group $\Lambda_1 \subseteq GL(E/E_0)$ is nilpotent, then some element of Λ_1 expands E/E_0.

Proof. Let $q : E \to E/E_0$ be the canonical projection. Denote by R the radiant vector field on E/E_0. If $\{y_1, \ldots, y_m\}$ are coordinates on E/E_0, then this appears as $R = \sum_i y_i \frac{\partial}{\partial y_i}$. Then in each affine chart there is a vector field X_i that is π-related to R. As M is compact, we can cover it in finitely many charts and using a partition of unity subordinate to that covering form the vector field

$$X = \sum_i \mu_i X_i$$

on M.

Now suppose that Λ_1 does not contain any expansions of E/E_0. Then by Lemma 2.30, there is a Λ_1-invariant C^1 map $\Psi : E/E_0 \to \mathbb{R}$. As the flow along the radiant vector field at a point z is given by $\phi_t(z) = e^t z$, our map satisfies

$$\begin{aligned}d\Psi_z R(Z) &= \frac{d}{dt} \psi(e^t z)|_{t=0} \\ &= \frac{d}{dt} e^{ta} \psi(z)|_{t=0} \\ &= a\Psi(z)\end{aligned}$$

for some $a > 0$ and all $z \in E/E_0$.

Now the composition

$$\tilde{f} : \tilde{M} \to E \to E/E_0 \to \mathbb{R}$$

is invariant under deck transformations. Indeed, let $\gamma \in \pi_1(M)$ then for $x \in \tilde{M}$

$$\Psi \circ \pi \circ \mathrm{dev}(\gamma \star x) = \Psi \circ \pi \circ \mathrm{hol}(\gamma) \circ \mathrm{dev}(x)$$

and this is invariant as E_0 is invariant under the holonomy group.

This implies that $\tilde{f}$ covers a map $f : M \mapsto \mathbb{R}$. As by Proposition 1.8, we can give charts in terms of the development pair, in affine coordiantates we have that $f = \Psi \circ \pi$. Now

$$\begin{aligned} df_y X(y) &= d(\Psi \circ \pi)_y X(y) \\ &= \frac{d}{dt} \Psi \circ \pi(\gamma(t))|_{t=0} \\ &= \frac{d}{dt} \Psi(e^t \tilde{y})|_{t=0} \\ &= a\Psi(\tilde{y}) \end{aligned}$$

where $\tilde{y} = \pi(y)$ and $\gamma(t)$ is an integral curve of X starting at y. This gives that

$$df_y X(y) = af(y).$$

It follows that if $\alpha : I \to M$ is an integral curve of X starting at $p \in M$ then

$$\begin{aligned} f(\alpha(t)) &= \Psi \circ \pi(\alpha(t)) \\ &= \Psi(\overline{\alpha(t)}) \\ &= \Psi(e^t \tilde{p}) \\ &= e^{ta}\Psi(\tilde{p}) \\ &= e^{ta} f(\alpha(c)). \end{aligned}$$

Now as $\Psi > 0$ almost everywhere, there exists $x_0 \in M$ with $f(x_0) > 0$ and the integral curve α_0 through x_0 is defined for all t as M is compact. Then $\lim_{t\to\infty} f(\alpha_0(t)) = \lim_{t\to\infty} e^{ta} f(\alpha_0(c)) \to \infty$. But as f is a continuous function and M is compact it must be bounded and so we reach a contradiction. $\square$

This proof leads to several important results. The first result follows almost immediately.

Corollary 2.32. Let M be a compact radiant manifold with nilpotent linear holonomy Λ. Then Λ contains an expansion of E.

Proof. We can take $E_0 = 0$ in Theorem 2.31 and the result follows immediately. $\square$

The next corollary follows from considering the Fitting submodule of E.

Corollary 2.33. Let M be a compact affine manifold with nilpotent affine holonomy group. If $F \neq 0$ then some element of the linear holonomy group expands F.

Proof. Recall from Lemma 2.11, that nilpotence of the affine holonomy gives a unique decomposition $E = E_U \oplus F$ which is invariant under the action of the holonomy group. If we take $E_0 = E_U$ in Theorem 2.31, then $F \cong E/E_U$ as a group module with respect to the linear holonomy group. □

2.3 Parallel volume

In [Mar63] it is conjectured that parallel volume is equivalent to completeness for affine manifolds. Using the previous results, we can now give a partial resolution of this conjecture for the case that our manifold has nilpotent holonomy. Orientable integral affine manifolds always possess a parallel volume form since their linear holonomy is in $\mathrm{SL}(\mathbb{Z}^2)$ and so the following results will prove important in our classification of such structures.

Theorem 2.34. ([FGH81]) Let M be a compact affine manifold with nilpotent affine holonomy group. Suppose that there is a parallel volume form on M. Then the linear holonomy is unipotent.

Proof. Again we will exploit that there is a unique decomposition $E = E_U \oplus F$ by Lemma 2.11. Now let g be any element of the linear holonomy group. As there is a parallel volume form by Lemma 1.35, $\det(g) = 1$. But by the unique decomposition

$$1 = \det(g) = \det(g|_{E_U})\det(g|_F)$$

But $g|_{E_U}$ is a unipotent operator and so has determinant 1. This forces $\det(g|_F) = 1$ so that F cannot contain an expansion. But now Corollary 2.33 forces $F = 0$ and so $E = E_U$. □

This will be used in conjunction with the following theorem in order to give equivalent conditions on the completeness of an affine manifold.

Theorem 2.35. Let M be an affine manifold. If M is compact and has unipotent holonomy then M is complete.

Proof. We pick a basepoint x_0 in M. Let $p : \tilde{M} \to M$ be the universal cover of M with $p(\tilde{x_0}) = x_0$. We now choose a developing map $\mathrm{dev} : \tilde{M} \to \mathbb{R}^n$ such that $\mathrm{dev}(\tilde{x_0}) = 0$. Let hol be the holonomy representation and $\mathrm{hol}(\pi_1(M)) = \Gamma$ be the affine holonomy group.

Now, suppose that the linear affine holonomy group is unipotent. Then there exists a linear flag

$$0 = F_0 \subset F_1 \subset \cdots \subset F_n = \mathbb{R}^n$$

preserved by the linear affine holonomy group as shown in Section 17.5 of [Hum12]. Furthermore, the induced action on the quotients F_k/F_{k-1} is trivial and so the restriction to F_k preserves a linear functional $l_k : F_k \to \mathbb{R}$ with kernel F_{k-1}.

This determines a family of parallel fields $\mathfrak{F}_k \subset TM$. In affine coordinates, each of these is spanned by a subset of the coordinate vector fields and so it is clear that they are integrable. Furthermore, each linear map l_i determines a parallel one-form ω_i defined on $\mathfrak{F}_i$ and vanishing on $\mathfrak{F}_{i-1}$.

We can construct vector fields X_i on M, tangent to $\mathfrak{F}_i$ such that $X_i(l_i) = 1$ using partitions of unity. These lift to vector fields $\tilde{X}_i$ on the universal cover.

We first show surjectivity of the developing map. Let $\gamma_i(t)$ be an integral curve of $\tilde{X}_i$ starting at the point x_i. As dev is a local diffeomorphism we have that

$$\begin{aligned} \tfrac{d}{dt}\mathrm{dev} \circ \gamma_i(t) = dev_* \circ \tilde{X}(\gamma_i(t)) \\ \Rightarrow \mathrm{dev} \circ \gamma_i(t) = dev_* \circ \tilde{X}_i(\gamma_i(t))t + \mathrm{dev} \circ \gamma_i(0) \end{aligned}$$

applying our linear functional l_i to this we obtained

$$\begin{aligned} l_i(\mathrm{dev} \circ \gamma_i(t)) &= l_i \circ dev_*(\tilde{X}_i(\gamma_i(t)))t + l_i(\mathrm{dev} \circ \gamma_i(0)) \\ &= l_i(\tilde{X}_i)t + l_i(\mathrm{dev} \circ \gamma_i(0)) = t + l_i(\mathrm{dev} \circ \gamma_i(0)) \qquad (*) \end{aligned}$$

Now we can start our inductive step. Taking $i = n$ we start at $\tilde{x}_0 \in \tilde{M}$ such that $\mathrm{dev}(\tilde{x}_0) = 0$. We flow along $\tilde{X}_n$ for time $t = l_n(v)$ until the point $\tilde{x}_1 \in \tilde{M}$ with $\mathrm{dev}(\tilde{x}_1) = v_1$. From (*) we obtain

$$\begin{aligned} l_n(v_1) = l_n(v) + 0 \\ \Rightarrow v - v_1 \in \ker l_n = F_{n-1} \end{aligned}$$

Now, we continue e.g. flow along $\tilde{X}_{n-1}$ for $t = l_{n-1}(v - v_1)$ ending at $\tilde{x}_2$ with $\mathrm{dev}(\tilde{x}_2) = v_2$. Then

$$\begin{aligned} l_{n-1}(v_2) = l_{n-1}(v - v_1) + l_n(v_1) \\ \Rightarrow l_{n-1}(v - v_2) = 0 \Rightarrow v - v_2 \in \ker l_{n-1} = F_{n-2} \end{aligned}$$

We continue this until we reach $F_0 = \{0\}$ showing surjectivity.

Now for injectivity we consider a path

$$\gamma_0 : [a, b] \mapsto \tilde{M}$$

which develops to a closed loop $\delta : [a, b] \mapsto \mathbb{R}^n$. We can deform γ_0 to a new path γ_1 such that it develops entirely inside F_{n-1}. We now define a new path $\gamma_t : [a, b] \mapsto \tilde{M}$ where $\gamma_t(s)$ is the image of $\gamma_0(s)$ at time $-tl_n(\delta(s))$ under the flow of $\tilde{X}_n$. Applying dev gives that

$$\begin{aligned} \mathrm{dev} \circ \gamma_t(s) = \mathrm{dev}(\gamma_0(s)) - tl_n(\delta(s))dev_*(\tilde{X}(\gamma_0(s))) \\ \Rightarrow l_n(\mathrm{dev} \circ \gamma_t(s)) = l_n(\mathrm{dev}(\gamma_0(s))) - tl_n(\delta(s))l_n \circ dev_*(\tilde{X}(\gamma_0(s))) \\ \Rightarrow l_n(\mathrm{dev} \circ \gamma_t(s)) = l_n(\mathrm{dev}(\gamma_0(s))) - tl_n(\delta(s)) \end{aligned}$$

At $t = 1$ this yields that $\mathrm{dev} \circ \gamma_t(s) \in \ker l_n = F_{n-1}$. Repeating this process we eventually get a curve that develops entirely in $F_0 = \{0\}$ showing injectivity. □

We summarize the previous results into the following theorem.

Theorem 2.36. Let M be a compact affine manifold with nilpotent affine holonomy representation. Then the following are equivalent

1. M is a complete affine manifold.
2. The linear holonomy is unipotent.
3. M has a parallel volume form.
4. The affine holonomy is irreducible.
5. The affine holonomy is indecomposable.

Proof. We have that 2) $\Rightarrow$ 1) by Theorem 2.35. We have that 3) $\Leftrightarrow$ 2) by Theorem 2.34. Now, 1) $\Rightarrow$ 4) by Theorem 1.21 and by definition 4) $\Rightarrow$ 5). Now 5) $\Rightarrow$ 2) by Lemma 2.16 and we are done. $\square$

Remark 2.37. We note that if the fundamental group of M is nilpotent, then its holonomy representation will also be nilpotent as this is preserved by group morphisms.

CHAPTER THREE

Lagrangian Fibrations and Integral Affine Manifolds

3.1 Symplectic Geometry

In this section we will recall some of the basics of symplectic geometry and Lagrangian submanifolds.

Definition 3.1 (Symplectic Manifold)**.** A symplectic manifold (M, ω) consists of a smooth manifold M and a closed 2-form ω such that ω is non-degenerate meaning the bi-linear form ω_p is non-degenerate on T_pM for all $p \in M$.

Symplectic geometry has its roots in classical mechanics as the phase space of a dynamical system naturally has the structure of a symplectic manifold. TODO: elaborate more on symplectic geometry - Hamiltons equations, applications to topology, Poisson geometry.

Non-degeneracy of the symplectic form gives a canonical way to identify the tangent and cotangent bundle of a symplectic manifold. This is done via the map

$$\tilde{\omega}|_p : T_pM \to T_p^*M$$
$$v \mapsto \omega_p(v, \cdot)$$

for $v \in T_pM$. This is an isomorphism by non-degeneracy of the symplectic form. This identification leads to the notion of a Hamiltonian vector field.

Definition 3.2 (Hamiltonian Vector Field)**.** To any $H \in C^\infty(M)$ we can associate a unique Hamiltonian vector field X_H by

$$\iota_{X_H}\omega := \omega(X_H, \cdot) = dH$$

For mathematicians there is generally no distinguished choice of Hamiltonian function H. However, physicists often pick a Hamiltonian function based on physical criteria and are generally interested in studying the flow of this Hamiltonian vector field which, for instance, recovers Hamilton's equations.

3.1.1 The Cotangent Bundle

One of the most important examples of symplectic manifolds is the cotangent bundle, which naturally inherits the structure of a symplectic manifold. In fact, Darboux's theorem states that every symplectic manifold looks locally like a cotangent bundle. This class of symplectic manifolds will play a big role throughout this thesis and we will see now exactly how the cotangent bundle obtains a symplectic structure.

Example 3.3 (Cotangent Bundle)**.** Let Q be any smooth manifold and $M = T^*Q$ its cotangent bundle. We will now illustrate that there is a canonical one-form $\theta \in \Omega^1(T^*Q)$ and show that the exterior derivative $-d\theta$ defines a symplectic structure on T^*Q.

Let $\pi : T^*Q \to Q$ be the canonical projection map. Then for $X_p \in T_p(T^*Q)$ we define $\langle \theta_p, X_p \rangle := \langle p, d_p\pi(X_p) \rangle$ where $\langle \cdot, \cdot \rangle$ denotes contraction of a vector and covector. This canonical one-form satisfies the following important property.

Lemma 3.4. The canonical one-form θ is the unique one-form such that for any other one-form $\alpha \in \Omega^1(Q)$

$$\alpha = \alpha^*\theta$$

where we view α as a section $\alpha : Q \to T^*Q$.

Proof. Let $w \in T_qQ$. Then we have that

$$\begin{aligned} \langle (\alpha^*\theta)_q, w \rangle &= \langle \theta_{\alpha_q}, d_q\alpha(w) \rangle \\ &= \langle \alpha_q, w \rangle \end{aligned}$$

Uniqueness is obtained by noting that every tangent vector $v \in T_p(T^*Q)$ can be written in the form $d_q\alpha(w)$ except for those in the kernel of $d_p\pi$. But the vectors not in this kernel form an open, dense set and so by continuity we get uniqueness. □

We can also find an expression for this canonical one-form in coordinates.

Lemma 3.5. In local cotangent coordinates $q_1, \ldots, q_n, p_1, \ldots, p_n$ (TODO: define these) on T^*Q, the canonical one-form is given by

$$\theta|_{T^*U} = \sum_j p_j dq_j.$$

Proof. We let $\alpha = \sum_j \alpha_j dq_j$ be a 1-form on U. We have that $\alpha^*p_j = \alpha_j$ and $\alpha^*q_j = q_j$ which gives that

$$\begin{aligned} \alpha^*\theta|_{T^*U} &= \alpha^* \sum_j p_j dq_j \\ &= \sum_j \alpha_j dq_j \\ &= \alpha. \end{aligned}$$ □

Putting this together we obtain a symplectic form on T^*Q.

Theorem 3.6. The 2-form $-d\theta$ defines a symplectic form on T^*Q.

Proof. In local coordinates we have that $\omega = \sum_j dq_j \wedge dp_j$ which is clearly closed and non-degenerate. □

We will also wish to speak about maps between the symplectic manifolds. The correct notion of "isomorphism" in this category is the following.

Definition 3.7. Let (M, ω) and (N, ω') be two symplectic manifolds. Then a symplectomorphism between M and N is a diffeomorphism

$$\varphi : M \to N$$

such that $\omega = \varphi^* \omega'$.

3.1.2 Lagrangian Submanifolds

The goal of this thesis is to classify Lagrangian fibrations over compact surfaces. In this section we will introduce the concept of Lagrangian submanifolds and prove some basic results regarding these.

Definition 3.8 (Symplectic Orthogonal)**.** Let (M, ω) be a symplectic manifold and let $N \subset M$ be a submanifold. The symplectic orthogonal at a point $p \in M$ is

$$(T_p N)^\omega := \{v \in T_p M \mid \omega(v, w) = 0, \forall w \in N\}$$

- If $(T_p N)^\omega \subset T_p N$ then $T_p N$ is said to be *coisotropic*.
- If $T_p N \subset (T_p N)^\omega$ then $T_p N$ is said to be *isotropic*.
- If $(T_p N)^\omega = T_p N$ then $T_p N$ is said to be *Lagrangian*.
- If $(T_p N)^\omega \cap T_p N =$ then $T_p N$ is said to be *symplectic*.

A submanifold N of a symplectic manifold M is said to be coisotropic (isotropic, Lagrangian, symplectic) if its tangent space is a coisotropic (isotropic, Lagrangian, symplectic) subspace of $T_p M$ at every point $p \in M$.

Remark 3.9. We note that a submanifold $\iota : N \to M$ is isotropic if $\iota^* \omega = 0$ and furthermore, it is Lagrangian if we also have $\dim N = \frac{1}{2} \dim M$.

We will make use of the following lemma.

Lemma 3.10. The graph $\Gamma_\alpha \subset T^*Q$ of a one-form α is Lagrangian if and only if α is closed.

Proof. It is clear that $\dim \Gamma_\alpha = \frac{1}{2} \dim T^*Q$. We compute

$$\begin{aligned} \alpha^* \omega &= -\alpha^* d\theta \\ &= -d\alpha^* \theta \\ &= -d\alpha \end{aligned}$$

and so $\alpha^* \omega = 0$ if and only if α is closed completing the proof by the previous remark. □

The cotangent bundle T^*Q also gives rise to another Lagrangian submanifold.

Example 3.11. Let $\pi : T^*Q \to Q$ and fix $x \in Q$. Then $\pi^{-1}(x) = T_x^*Q$ is a Lagrangian submanifold inside (T^*Q, ω). It is easy to see that $\dim T_x^*Q = \frac{1}{2} \dim T^*Q$. Letting $i : T_x^*Q \hookrightarrow$

T^*Q be the inclusion and considering local coordinates $(q_1, \ldots, q_n, p_1 \ldots p_n)$ around x and writing $\omega = \sum_j dq_j \wedge dp_j$ it follows that $\omega|_{T_x^*Q} = 0$ as $i^*(\sum_j dq_j \wedge dp_j) = \sum_j d(i^*q_j) \wedge d(i^*p_j)$ and $d(i^*q_i) = 0$ as i^*q_i is constant.

3.2 Lagrangian Fibrations

One of the goals of this thesis is to classify the Lagrangian fibrations over closed surfaces. In this section we will give an overview of these objects and give some examples and basic results surrounding them. We will then describe how the base space of a Lagrangian fibration inherits an integral affine structure.

3.2.1 Definition of Lagrangian Fibration

Definition 3.12. A Lagrangian fibration $\pi : (M, \omega) \to B$ is a surjective map whose regular fibres are Lagrangian submanifolds of (M, ω).

Example 3.13. We seen in Lemma 3.10 that the fibres of $\pi : (T^*Q, \omega_{can}) \to Q$ are Lagrangian submanifolds and hence this is a Lagrangian fibration.

Example 3.14. Let $Q = (\mathbb{R}/\mathbb{Z})^n = \mathbb{T}^n$ be the n-torus. Then we have that $T^*(\mathbb{T}^n) = \mathbb{T}^n \times \mathbb{R}^n$ and the map

$$\pi : T^*(\mathbb{T}^n) \to \mathbb{R}^n$$

is a Lagrangian fibration with fibre $\mathbb{T}^n$.

Definition 3.15 (Shift by a one-form.)**.** Let $\pi : (T^*Q, \omega_{can}) \to Q$ and fix $\alpha \in \Omega^1(Q)$. We define the *shift by* α to be the map

$$s_\alpha : T^*Q \to T^*Q$$

$$\beta_q \mapsto \beta_q + \alpha_q.$$

Proposition 3.16. This defines a symplectomorphism of T^*Q if and only if $d\alpha = 0$.

Proof. This map is a fibrewise translation and hence is obviously a diffeomorphism. We now check when it preserves the symplectic form. We calculate

$$\begin{aligned} s_\alpha^*\omega_{can} &= -s_\alpha^* d\theta \\ &= -ds_\alpha^*\theta. \end{aligned}$$

We calculate

$$\begin{aligned} s_\alpha^*\theta(v)|_{\beta_q} &= \theta_{\beta_q+\alpha_q}(ds_\alpha(v)) \\ &= (\beta_q + \alpha_q)(d(\pi \circ s_\alpha)(v)) \\ &= (\beta_q + \alpha_q)(v) \\ &= \theta_{\beta_q}(v) + \alpha_q(v) \end{aligned}$$

and so $s_\alpha^*\theta = \theta + \alpha$. Now

$$\begin{aligned} -ds_\alpha^*\theta &= -d\theta - d\alpha \\ &= \omega_{can} - d\alpha \end{aligned}$$

and hence this defines a symplectomorphism if and only if $d\alpha = 0$. □

Remark 3.17. We now note that given any Lagrangian section of T^*Q, i.e. a section determined by a closed one-form, we can shift this section to the zero section since shifts by closed one-forms are symplectomorphisms. This implies that if we consider T^*Q as an abstract Lagrangian bundle it has no marked zero section and so the fibres do not have a canonical vector space structure, only the structure of an affine space.

We now give the notion of equivalence of Lagrangian bundles.

Definition 3.18. Let $M \to B$ and $N \to B$ be two Lagrangian fibrations over B. Then a *Lagrangian equivalence* between $M \to B$ and $N \to B$ is given by

$$\begin{tikzcd} M \arrow[r, "\Phi"] \arrow[dr] & N \arrow[d] \\ & B \end{tikzcd}$$

where Φ is a fibrewise symplectomorphism i.e. a symplectomorphism that is also a bundle map.

Remark 3.19. There is no loss in generality in assuming that Φ covers the identity on B. Indeed if $f : B \to B$ is a diffeomorphism then there exists a fibrewise symplectomorphism M and N covering f if and only if the exists a fibrewise symplectomorphsim between M and the pullback bundle f^*N covering the identity on B.

3.3 Integral Affine Structure on the Base

We now discuss how the base space of a Lagrangian fibration inherits an integral affine structure. This structure will be an invariant of the fibration.

3.3.1 Torus Bundles

Throughout this section let Q be a manifold and let G be a Lie group.

Definition 3.20. Suppose that $\dim Q = \dim G$. We say that Q is a *principal homogeneous G-space* if there is a free, transitive action of G on Q.

Remark 3.21. For any $q \in Q$ we have that $Q = G \cdot q \cong G$ as the action is transitive. For any $q' \in Q$ there exists a $g \in Q$ such that $g \cdot q = q'$. Therefore, any two such identifications differ by an element of G.

Definition 3.22. If $G \cong (\mathbb{R}/\mathbb{Z})^n$ and Q is a principal homogeneous G-space, we call Q an *affine torus*. If G is a vector space we call Q an *affine vector space*.

Now we consider the case that $G := V$ is a vector space. Suppose now that the action is transitive but *not necessarily free*. We define

$$V_q := \{v \in V \mid v \cdot q = q\}$$

.

Proposition 3.23. V_q is a discrete subgroup of V independent of $q \in Q$.

Proof. As a topological space, $Q = V/V_q$. Consider the map

$$\phi_q : V \to Q$$

$$v \mapsto v \cdot q$$

Taking the differential $(d\phi_q)_0 : T_0 V \to T_0 Q$ and applying the rank-nullity theorem we find that $\dim V_q = 0$ and hence it is a discrete subgroup. We now show it is independent of q.

For any $q' \in Q$, by transitivity of the action, the exists $w \in V$ such that $w \cdot q = q'$ and so for $v \in V_q$, $w \in V_{q'}$

$$\begin{aligned} (w^{-1} v w) \cdot q &= q \\ \Rightarrow (w^{-1} v) \cdot q' &= q \\ \Rightarrow v \cdot q' &= w \cdot q \\ \Rightarrow v \cdot q' &= q \end{aligned}$$

and hence $V_q = V_{q'}$ ☐

Remark 3.24. This implies that V_q is a lattice, e.g. $V_q \cong \mathbb{Z}^n$. We therefore have that $V/V_q \cong \mathbb{R}^n/\mathbb{Z}^k \cong \mathbb{R}^{n-k} \times \mathbb{T}^k$.

Remark 3.25. It follows from our previous definitions that Q is a product of an affine torus and an affine vector space.

Definition 3.26 (Group Bundle)**.** A group bundle $\pi : \mathcal{G} \to B$ is a fibre bundle such that each fibre is equipped with a group structure and there are charts $\pi^{-1}(U) \cong U \times G$ that are fibrewise isomorphisms with a fixed group G.

We can define the notion of an action of a group bundle on any fibre bundle over the same base.

Definition 3.27. Let $\mathcal{G} \to B$ be a group bundle and $M \to B$ be a fibre bundle. An action of $\mathcal{G} \to B$ on $M \to B$ is a smooth map

$$\varphi : \mathcal{G} \times_B M \to M$$

that is a fibrewise group action.

Example 3.28. Let $\mathcal{T} \to B$ be a torus bundle. Then an affine torus bundle $M \to B$ is a fiber bundle with a free, transitive action of $\mathcal{T} \to B$. We therefore have by Remark 3.24 that if $M \to B$ is a fiber bundle with compact fibres and $E \to B$ is a vector bundle with a transitive action on $M \to B$ such that $\dim M = \dim E$, then M inherits the structure of an affine torus bundle.

Suppose that we have an action as in the above example. Then if we let

$$\Lambda = \bigsqcup_p \Lambda_p$$

be the bundle of stabilisers of the action of $E \to B$ on $M \to B$, e.g. $\Lambda_p \subset E_p$ is the isotropy group at $p \in B$, then we can form a torus bundle

$$\mathcal{T} := E/\Lambda \to B$$

and in this way $M \to B$ becomes an affine torus bundle since $\mathcal{T}$ now acts freely on M as we have quotiented out the fixed points.

Suppose now that we have a global section $\sigma : B \to M$. Then $M \to B$ becomes a torus bundle as we can use the global section to pick out an identity. (TODO: elaborate on this)

Theorem 3.29. Let (M, ω) be a symplectic manifold and let $\pi : M \to B$ be a Lagrangian fibration with compact connected fibres. Then, there is a canonical, fibrewise transitive action of the vector bundle $T^*B \to B$ on $M \to B$. Therefore, $M \to B$ can be canonically equipped with the structure of an affine torus bundle.

Proof. Let $VM \subset TM$ be the vertical bundle i.e. $V_pM = \ker d_p\pi$. Since $\pi : M \to B$ is a Lagrangian fibration it follows that $VM \subset TM$ is a Lagrangian subbundle as it is tangent to the fibres of the fibration. Let $\alpha \in \Omega^1(B)$ be any 1-form and let $X_\alpha \in \mathfrak{X}$ be the vector field defined by

$$\iota_{X_\alpha}\omega = -\pi^*\alpha.$$

Let $Y \in \mathfrak{X}(M)$ be a vertical vector field i.e. $d\pi(Y(p)) = 0$ for all $p \in M$. Then we have

$$\begin{aligned} \omega(X_\alpha, Y) &= \iota_Y \iota_{X_\alpha}\omega \\ &= -\iota_Y \pi^*\alpha \\ &= 0 \end{aligned}$$

as Y is vertical. Now as $VM^\omega = VM$, this implies that X_α is vertical. We note that this implies that X_α is constant along the fibres and so its value at $m \in M$ depends only on the value of $\alpha_{\pi(m)}$. We have thus assigned to each $\alpha_{\pi(m)}$ with $m \in M$, a vector in the vertical bundle VM by $X_\alpha(m)$ depending only on the value of $\pi(m)$. Moreover, this is assignment is linear since

$$\begin{aligned} \iota_{X_{a\alpha+b\beta}}\omega &= -\pi^*(a\alpha + b\beta) \\ &= -a\pi^*(\alpha) - b\pi^*(\beta) \end{aligned}$$

for $a, b \in \mathbb{R}$ and $\alpha, \beta \in \Omega^1(B)$. We thus have a isomorphism

$$V_mM \cong T^*_{\pi(m)}B$$

and putting together all these isomorphisms yields a bundle isomorphism

$$VM \cong \pi^*T^*B$$

covering the identity. Now, we let F^t_α denote the flow of X_α and we will denote by $F_\alpha := F^1_\alpha :$ $M \to M$ the flow at time 1. This gives a fibrewise action on $M \to B$ as the X_α are vertical. We get a bundle map

$$T^*B \times_B M \to M,$$

$$(\alpha_b, m) \mapsto F_\alpha(m)$$

and we claim that this is a vector bundle action. Transitivity is clear as F_α is a fibrewise diffeomorphism. It remains to show that the action is commutative i.e. that the flows of the X_α commute. We let $\alpha, \beta \in \Omega^1(B)$ be two one-forms with corresponding vector fields X_α, X_β. Then, using the formula from Cartan calculus,

$$\begin{aligned}
\iota_{[X_\alpha, X_\beta]}\omega &= (L_{X_\alpha}\iota_{X_\beta} - \iota_{X_\beta}L_{X_\alpha})\omega \\
&= -L_{X_\alpha}\pi^*\beta - \iota_{X_\beta} d\iota_{X_\alpha}\omega \\
&= -L_{X_\alpha}\pi^*\beta + \iota_{X_\beta}\pi^* d\alpha \\
&= -(d\iota_{X_\alpha}\pi^*\beta + \iota_{X_\alpha}\pi^* d\beta) + \iota_{X_\beta}\pi^* d\alpha \\
&= 0
\end{aligned}$$

since $d\pi(X_\alpha) = d\pi(X_\beta) = 0$. Now, by non-degeneracy of ω, it follows that $[X_\alpha, X_\beta] = 0$ for all $\alpha, \beta \in \Omega^1(B)$ and hence the flows commute. We therefore have a vector bundle action of $T^*B \to B$ on $M \to B$ and hence $M \to B$ has the structure of an affine torus bundle. ☐

It is through this action that the base space of a Lagrangian fibration inherits an integral affine structure. Indeed the bundle of stabilisers for this action $\Lambda \subset T^*B$ is a discrete sub-bundle of the cotangent space and it will turn out that this will locally be spanned by closed one forms. Since by Proposition 1.40 this is equivalent to the base space being an integral affine manifold, this will be exactly how B inherits an integral affine structure.

We in fact wish to show that Λ is a Lagrangian submanifold of T^*B. We will first need the following lemma.

Lemma 3.30. Let $\alpha \in \Omega^1(B)$ be a one-form and let X_α be its corresponding vector field on M and denote by F_α^t its flow. Then

$$(F_\alpha^t)^*\omega = \omega - t\pi^* d\alpha$$

and so F_α^1 is a symplectomorphism if and only if $d\alpha = 0$.

Proof. We start by integrating

$$\int_0^t \frac{\partial}{\partial t}(F_\alpha^t)^*\omega = (F_\alpha^t)^*\omega - \omega$$

On the otherhand, calculating this derivative yields

$$\begin{aligned}
\frac{\partial}{\partial t}(F_\alpha^t)^*\omega &= (F_\alpha^t)^* L_{X_\alpha}\omega \\
&= (F_\alpha^t)^* d\iota_{X_\alpha}\omega \\
&= -(F_\alpha^t)^*\pi^* d\alpha \\
&= -(\pi \circ F_\alpha^t)^* d\alpha \\
&= -\pi^* d\alpha
\end{aligned}$$

Now $\int_0^t -\pi^* d\alpha = -t\pi^* d\alpha$ and so

$$(F_\alpha^t)^*\omega - \omega = -t\pi^* d\alpha$$

as required. ☐

We can now show the following proposition regarding the bundle of stabilisers Λ.

Proposition 3.31. The lattice $\Lambda \subset T^*B$ is a Lagrangian submanifold.

Proof. Since by assumption our fibres are compact, by Remark 3.24 we must have that $\Lambda_p \cong \mathbb{Z}^n$. Now let $U \subseteq B$ be an open set on the base such that $\Lambda|_U \cong U \times \mathbb{Z}^n$. Now, the image of any section of $\Lambda|_U$ is given by the graph of a one-form $\alpha : B \to T^*B$ such that its corresponding time one flow F_α is the identity on $\pi^{-1}(U)$. But by Lemma 3.30 this implies that α is closed and by Lemma 3.10 the image of each section is a Lagrangian submanifold of T^*B. □

We are now able to see how the base space of a Lagrangian fibration inherits an integral affine structure.

Corollary 3.32. The base space of any Lagrangian fibration $M \to B$ inherits an integral affine structure.

Proof. By Proposition 3.31, $\Lambda \subset T^*B$ is a discrete lattice in T^*B locally spanned by closed one-forms. By Proposition 1.40 this is equivalent to an integral affine structure on B □

The lattice Λ will be an important object of study throughout the remainder of this thesis. We introduce some terminology for it now.

Definition 3.33 (Period Lattice)**.** Let B be an integral affine manifold and let $\Lambda \subset T^*B$ be the corresponding lattice in the cotangent bundle. Then we will call Λ the period lattice associated to the integral affine structure on B.

Theorem 3.34. The symplectic form on T^*B descends to T^*B/Λ and the projection map $\tau : T^*B/\Lambda \to B$ is a Lagrangian fibration.

Proof. The local sections of Λ are given by closed one-forms. By Proposition 3.16, the shift by a closed form is a symplectomorphism and hence preserves the symplectic form. (TODO: maybe need to explain better why Λ acts by fibrewise affine shifts of closed one-forms) □

Remark 3.35. We have thus constructed a torus bundle $\tau : T^*B/\Lambda \to B$ which acts fibrewise on our original Lagrangian fibration $M \to B$. Indeed, this torus bundle is itself a Lagrangian fibration and inherits a symplectic form from T^*B. The torus bundle $\tau : T^*B/\Lambda \to B$ always has a global section, namely the projection of the zero section. However, our original Lagrangian fibration $M \to B$ has only the structure of an *affine torus bundle*. It becomes a torus bundle itself only if it admits a global section $\sigma : B \to M$ in which case it identifies with the torus bundle $T^*B/\Lambda \to B$. There are in general, obstructions to the existence of such a global section.

If we assume that the obstructions to the existence of a global section vanish, then $M \to B$ itself becomes a torus bundle and moreover, we can identify it with $T^*B/\Lambda \to B$ in a way that respects the group structure.

Lemma 3.36. Let $\sigma : B \to M$ be a global section of $M \to B$ and $\alpha \in \Omega^1(B)$ be a one-form. Then this determines a unique, fibre bundle isomorphism $A : T^*B/\Lambda \to M$ that is T^*B/Λ-equivariant such that the following diagram commutes.

$$\begin{tikzcd} T^*B/\Lambda \arrow[r, "A"] \arrow[d, "\hat{F}_\alpha"'] & M \arrow[d, "F_\alpha"] \\ T^*B/\Lambda \arrow[r, "A"'] & M \end{tikzcd}$$

Here $\hat{F}_\alpha$ is determined by the action of T^*B/Λ on itself (it is namely a Lagrangian fibration over B).

Proof. We define $A : T^*B/\Lambda \to M$ by

$$\begin{aligned} A_b : T^*_bB/\Lambda_b &\to M_b \\ [\alpha_b] &\mapsto F_\alpha(\sigma(b)). \end{aligned}$$

This map is injective as we have quotiented out the fixed point set of this action and hence a bijection by dimension arguments. We saw that this action is a fibrewise diffeomorphism in the proof of Theorem 3.29 and so this defines a bundle isomorphism. We now show equivariance. This follows from the following calculation.

$$\begin{aligned} A \circ \hat{F}_\alpha([\beta]) &= A([\alpha] + [\beta]) \\ &= F_{\alpha+\beta}(\sigma(b)) \\ &= F_\alpha \circ F_\beta(\sigma(b)) \\ &= F_\alpha \circ A([\beta]) \end{aligned}$$

for $b \in B$ and $\alpha, \beta \in \Omega^1(B)$. □

We note that this gives only a fibrewise, equivariant bundle isomorphism between $\tau : T^*B/\Lambda$ and $M \to B$. However, we have not yet discussed when the symplectic structure is preserved. We discuss this now.

Proposition 3.37. The cohomology class $[\sigma^*\omega] \in H^2(B)$ does not depend on the choice of σ.

Proof. Since the action of T^*B on M is transitive, any other section $\hat{\sigma}$ differs by an action of T^*B e.g. $\hat{\sigma} = F_\alpha \circ \sigma$. It follows that

$$\begin{aligned} \hat{\sigma}^*\omega &= (F_\alpha \circ \sigma)^*\omega \\ &= \sigma^*(F_\alpha^*\omega) \\ &= \sigma^*(\omega - \pi^* d\alpha) \\ &= \sigma^*\omega - d\alpha. \end{aligned}$$

□

Lemma 3.38. Suppose that $\sigma : B \to M$ is a section with $[\sigma^*\omega] = 0 \in H^2(B)$. Then $A : T^*B/\Lambda \to M$ is a symplectomorphism.

Proof. We can assume without loss of generality that $\sigma^*\omega = 0$ as otherwise we can shift it to the zero section by a closed form which is a symplectomorphism. It is enough to show that the map $\hat{A} : T^*B \to M$ covering A is a local symplectomorphism since the symplectic form descends to the quotient. The map $\hat{A}$ is determined by the condition

$$\hat{A} \circ \alpha = F_\alpha \circ \sigma$$

for every $\alpha \in \Omega^1(B)$. We show that the symplectic form is preserved. We compute the pullback

$$\begin{aligned}
\alpha^*\hat{A}^*\omega &= (F_\alpha \circ \sigma)^*\omega \\
&= \sigma^* F_\alpha^* \omega \\
&= \sigma^*(\omega - d\pi^*\alpha) \\
&= -d\sigma^*\pi^*\alpha \\
&= -d\alpha \\
&= \alpha^*(-d\theta) \\
&= \alpha^*\omega.
\end{aligned}$$

Since this holds for any $\alpha \in \Omega^1(B)$ it follows that $\hat{A}^*\omega = \omega$. ☐

3.3.2 Action Angle Coordinates

In this section, we will assume that there exists a global Lagrangian section $\sigma : B \to M$ and that the cohomology class $[\sigma^*\omega] = 0 \in H^2(B)$ so that $M \to B$ and $T^*B/\Lambda \to B$ are fibrewise symplectomorphic. If we assume also that the base space B is simply connected then we have that the linear holonomy representation of B is trivial and so the group bundle $\Lambda \to B$ is trivial. Fixing a basepoint $b \in B$ we pick some isomorphism $\Lambda_b \cong \mathbb{Z}^n$ and then for any other $b' \in B$, we identify $\Lambda_b \cong \Lambda_{b'}$ by picking a path between them. Since by Corollary 3.32, B is itself an integral affine manifold, it hence possesses a flat, torsion free connection and so we use this connection to parallel transport a basis of Λ_b to a basis of $\Lambda_{b'}$.

Since B is assumed to be simply connected, this gives an isomorphism

$$\Lambda \cong B \times \mathbb{Z}^n$$

that depends on our choice of base point. Any two such isomorphisms will differ by an element of $GL(n, \mathbb{Z})$. Let $\beta_1, \dots, \beta_n$ be a global frame for $B \times \mathbb{Z}^n$. We note that the β_i will be closed since sections of $\Lambda \subset T^*B$ are given by closed one-forms. Since Λ is a full rank lattice, this also defines a global frame for T^*B and hence we have trivializations

$$T^*B \cong B \times \mathbb{R}^n$$

and

$$M \cong T^*B/\Lambda \cong B \times (\mathbb{R}/\mathbb{Z})^n.$$

Definition 3.39. We define *angle coordinates* on $M \cong B \times (\mathbb{R}/\mathbb{Z})^n$ by

$$s : M \cong B \times (\mathbb{R}/\mathbb{Z})^n \to (\mathbb{R}/\mathbb{Z})^n$$

where s is the projection map.

Recall that we can associate vector fields on M to the $\beta_i \in \Omega^1(B)$ by $\iota_{X_{\beta_i}}\omega = -\pi^*\beta_i$ and a quick calculation using Cartans formula shows that these vector fields are symplectic e.g. $L_{X_{\beta_i}}\omega = 0$ since both β_i and ω are closed. Denoting their corresponding flow by F_i^t, in terms of the angle coordinates we have that

$$s_j(F_i^t(p)) = \begin{cases} s_j(p), & i \neq j, \\ s_j(p) + t, & i = j \end{cases}$$

for $p \in M$. (TODO: prove this probably.)

Moreover, since B is simply connected the β_i are in fact exact e.g. $\beta_i = dI_i$. Since the β_i form a global frame, $dI_p = (dI_1, \ldots, dI_n)_p$ is full rank at each $p \in M$ and hence we can use them to define coordinates on the base.

Definition 3.40. We define *action coordinates* on M by $\pi^*I_1, \ldots, \pi^*I_n$.

(TODO: prove this) In these action-angle coordinates we can write the symplectic form as

$$\omega = \sum_i dI_i \wedge ds_i$$

TODO: There is some good stuff in the Duistermaat paper that can also go here and maybe some of the explanations are better.

CHAPTER FOUR

Classification of Integral Affine Closed Surfaces

As we have seen in the previous chapter, the integral affine stucture induced by a regular Lagrangian torus fibration is an invariant of the base space of the fibration. Moreover, given an integral affine structure on a manifold B, we can construct a Lagrangian fibration by taking the quotient of T^*B by the lattice inducing the integral affine structure. We shall see in Chapter 5 that after having fixed an integral affine structure on B, all Lagrangian fibrations over B will locally look like $T^*B/\Lambda \to B$ where $\Lambda \subset T^*B$ is the period lattice inducing the integral affine structure.

In order to classify the regular Lagranigan torus fibrations over a manifold B, the first step is to classify the integral affine structures on B. The goal of this chapter is to carry this out for compact surfaces.

4.1 Classification of integral affine circles

It is easy to classify the integral affine structures on the circle S^1 and so for completeness we will also give the classification of these.

The fundamental group of S^1 is the integers $\mathbb{Z}$ and hence by Theorem 2.36 any affine structure will be complete if it possesses a parallel volume form. The holonomy representation hol : $\pi_1(S^1) \to \mathrm{Aff}_{\mathbb{Z}}(\mathbb{R})$ is determined by the image of $1 \in \mathbb{Z}$ and so the possible affine holonomy groups are

$$A = \langle(1, a)\rangle \quad \text{or} \quad B = \langle(-1, b)\rangle$$

with $a, b \in \mathbb{R}$. We note that $B \cong \mathbb{Z}/2\mathbb{Z}$ and in fact, this affine structure would be radiant as $\frac{b}{2}$ is a fixed point. However, Corollary 2.32 says that the linear holonomy would contain an expansion but all the eigenvalues of the elements of B have modulus one. Hence B cannot occur as the affine holonomy group of an integral affine structure over S^1.

Now, since the affine holonomy group A preserves a parallel volume form, by Theorem 2.36, this affine structure is complete. The integral affine structures on S^1 are therefore of the form $S^1 = \mathbb{R}/A$. By Lemma 1.9, complete integral affine structures are equivalent when their affine holonomy groups are conjugate. The following lemma classifies the integral affine structures on S^1.

Lemma 4.1. The integral affine structures on S^1 are determined by a positive real number $a > 0$ and are of the form $\mathbb{R}/A$ where

$$A = \langle (1, a) \rangle.$$

Proof. Our previous discussion showed that all integral affine structures on S^1 are of the form $\mathbb{R}/A$ where the translational component is any real number. It remains to check when two structures are equivalent. We note that the only possible linear affine holonomy is $\{\pm 1\}$ and that if it is 1 conjugating A leaves it unchanged. We therefore compute the conjugation by an element with negative linear holonomy. Letting $b \in \mathbb{R}$ we compute

$$\begin{aligned}(-1, b)(1, a)(-1, b)^{-1} &= (-1, b)(1, a)(-1, b)\\ &= (1, -a)\end{aligned}$$

but this generates the same affine holonomy group as $(1, a)$ completing the proof since if $a \neq 0$ the action is clearly free. □

4.2 Some basic results

We now move on to classifying the integral affine structures over compact surfaces which requires a bit more work. We first discuss our approach to the classification problem and state some important results in the theory of affine manifolds. The first task is to determine the possible compact surfaces that can admit an affine structure. The following theorem classifies the compact, connected 2-manifolds.

Theorem 4.2. Every compact, connected 2-manifold is homeomorphic to the sphere, the connected sum of tori or the connected sum of $\mathbb{R}P^2$.

Proof. This can be found in [Hat02]. □

We make use of the following theorem from Milnor relating to the existence of a flat connection on closed orientable surfaces.

Theorem 4.3 (Milnor [Mil58])**.** Let M^2 be a closed orientable surface of genus g. Then if $g \geq 2$ it does not possess a connection with zero curvature.

Corollary 4.4. The possible orientable closed surfaces admitting an integral affine structure are the sphere S^2 or the torus $\mathbb{T}^2$.

Proof. This follows immediately from the classification of closed 2-surfaces and the previous theorem. □

However, the following simple argument shows that the sphere S^2 does not admit such a structure.

Proposition 4.5. The sphere S^2 does not admit an affine structure.

Proof. Suppose there was such an affine structure. Then

$$\text{dev} : S^2 \to \mathbb{R}^2$$

is a local diffeomorphism and hence an open map. However dev is also continuous, and so $\text{dev}(S^2)$ is a compact subset of $\mathbb{R}^2$ and hence closed. Now, as $\text{dev}(S^2) \neq \emptyset$, by connectivity $\text{dev}(S^2) = \mathbb{R}^2$ but $\mathbb{R}^2$ is not compact. □

We have the following immediate corollary.

Corollary 4.6. The only closed, orientable surface admitting an (integral) affine structure is $\mathbb{T}^2$.

We will deal with the non-orientable case by passing to the orientable double cover.

Lemma 4.7. Every closed, non-orientable surface has a double cover that is orientable. Furthermore, the genus of the double cover is given by

$$g = \bar{g} - 1$$

where $\bar{g}$ is the number of connected sums of $\mathbb{R}P^2$.

Proof. This is proved in [Hat02]. □

This lemma leads to the following corollary.

Corollary 4.8. The only closed surfaces admitting an (integral) affine structure are the torus $\mathbb{T}^2$ and the Klein bottle $\mathbb{K}^2$.

Proof. As any covering space of an integral affine structure inherits an integral affine structure by (TODO: do this in (G,X)-structure section), it follows that the orientable double cover of a non-orientable integral affine surface must be $\mathbb{T}^2$. Furthermore, it must have non-orientable genus $\bar{g} = 1$ or 2. If $\bar{g} = 1$ our surface is homeomorphic to $\mathbb{R}P^2$ which is covered by S^2 and hence admits no such structure. If $\bar{g} = 2$ we get the Klein bottle which is covered by $\mathbb{T}^2$ and hence it can admit such a structure. □

4.3 Classification of Integral Affine structures on the torus

4.3.1 Completeness and freeness of the action

We now look at how we can classify the integral affine structures on the torus. The following observation will greatly simplify the classification process.

Lemma 4.9. Any integral affine structure on the torus is complete.

Proof. We have that the fundamental group $\pi_1(\mathbb{T}^2) = \mathbb{Z} \times \mathbb{Z}$ is abelian and hence nilpotent. As the torus is orientable, our linear holonomy group $\Gamma \subseteq SL(2, \mathbb{Z})$ by Corollary 1.36 and now by Theorem 2.36 this is complete. □

(TODO: This might be a good place for discussion on completeness stuff. Auslander-Markus show that every complete affine manifold is quotient by linear holonomy).

From the above discussion and (TODO: recap relevant results), classifying complete integral affine structures on a surface M amounts to classifying injective homomorphisms from the fundamental group into the group $\mathrm{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$ which induce a free and proper action on $\mathbb{R}^2$.

Lemma 4.10. If $g = (A, b) \in \mathrm{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$ acts freely on $\mathbb{R}^2$, then

$$\text{if } \det(A) = 1 \text{ then } \mathrm{Tr}(A) = 2$$

and,

$$\text{if } \det(A) = -1 \text{ then } \mathrm{Tr}(A) = -1.$$

Proof. Since the action by g is free, this implies

$$Ax + b = x$$

has no solution. But now,

$$(A - I)x = -b$$

has no solutions for all $x \in \mathbb{R}^2$ and so $\det(A - I) = 0$. Calculating this determinant yields

$$1 - \mathrm{Tr}(A) + \det(A) = 0$$

and substituting in the appropriate values of $\det(A)$ gives the required results.

□

We note also that 1 is an eigenvalue of such an A as by the above proof $\det(A - 1 \cdot I) = 0$. Using this result we can find a nice basis to express the linear part of our element $g \in \mathrm{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$.

Lemma 4.11. Let $g = (A, b) \in \mathrm{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$ act freely. Then there is an integral basis of $\mathbb{R}^2$ such that

$$A = \begin{pmatrix} 1 & n \\ 0 & \pm 1 \end{pmatrix}$$

Proof. By our previous lemma, A must have 1 as an eigenvalue. We can take an eigenvector e_1 such that e_1 is integral, i.e. its components are integers. Now as $\mathrm{Tr}(A) = 2$ or 0, it follows that we can pick our second basis vector such that

$$A = \begin{pmatrix} 1 & n \\ 0 & \pm 1 \end{pmatrix}$$

since we need $A \in GL(2, \mathbb{Z})$. Furthermore, by changing orientation if necessary we may assume $n \geq 0$. □

Lemma 4.12. Let $H \subset \mathrm{Aff}_{\mathbb{Z}}(\mathbb{R}^2)^+$ be a subset of the elements of determinant one and suppose H acts freely on $\mathbb{R}^2$. Then there is an integral basis such that for all $(A, b) \in H$ we can write

$$A = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$$

for $n \in \mathbb{Z}$.

Proof. Let $(A, a), (B, b) \in H$. Now by Lemma 4.11, we can write

$$A = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

with $n \neq 0$. Now by Lemma 4.10, we have that

$$\begin{aligned} \mathrm{Tr}(A^k B) &= a + nck + d \\ &= \mathrm{Tr}(B) + nck \\ &= 2. \end{aligned}$$

As $n \neq 0$ and this holds for all $k \in \mathbb{Z}$, it follows that $c = 0$.

But now, $\det B = ad = 1$ and $\mathrm{Tr}(B) = a + d = 2$ and so $a = d = 1$ giving the required result as $b \in \mathbb{Z}$. □

We have the following corollary of this result.

Corollary 4.13. For any two elements $(A, a), (B, b) \in H$, there exists relatively prime $(k, l) \neq (0, 0)$ such that $A^k B^l = 1$.

Proof. We simply take $k = \frac{m}{(m,n)}$ and $l = \frac{-n}{(m,n)}$ and then

$$\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}^k \begin{pmatrix} 1 & m \\ 0 & 1 \end{pmatrix}^l = 1$$

. □

4.3.2 Finding the possible generators

We have now built up the required machinery to begin the classification of the integral affine structures on the torus $\mathbb{T}^2$.

Theorem 4.14. Any complete integral affine structure on $\mathbb{T}^2$ has its affine holonomy group generated by one of the following subgroups of $\mathrm{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$

1. $\mathcal{A}_{a,b,c,d}$ - These integral affine structures are the quotient of $\mathbb{R}^2$ by a lattice of pure translations. The generators in $\mathrm{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$ are given by
$$\left(1, \begin{pmatrix} a \\ c \end{pmatrix}\right) \text{ and } \left(1, \begin{pmatrix} b \\ d \end{pmatrix}\right)$$
with $a, b, c, d \in \mathbb{R}$ and $ad - bc \neq 0$.

2. $\mathcal{B}_{n,a,b}$ - The generators in $\mathrm{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$ are given by
$$\left(1, \begin{pmatrix} a \\ 0 \end{pmatrix}\right) \text{ and } \left(\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 \\ b \end{pmatrix}\right)$$
with $a, b > 0$ and $n \in \mathbb{N}$.

Proof. Let $\pi_1(\mathbb{T}^2) = \langle \tilde{C}, \tilde{D} \rangle$ and denote by C, D the images of these generators under the affine holonomy representation. We wish to show that these generators reduce to the form of either $\mathcal{A}_{a,b,c,d}$ or $\mathcal{B}_{n,a,b}$. We can do this by

1. Changing the generators $\tilde{C}, \tilde{D}$ of the $\pi_1(\mathbb{T}^2)$.
2. Conjugating the images of these generators by some $g \in \mathrm{Aff}^+_{\mathbb{Z}}(\mathbb{R})$ - this corresponds to pre-composing the developing map by g which preserves the affine structure.

If both $C = \left(1, \begin{pmatrix} a \\ c \end{pmatrix}\right)$ and $D = \left(1, \begin{pmatrix} b \\ d \end{pmatrix}\right)$ are pure translations, e.g. the linear component is the identity, injectivity forces that our translational vectors are linearly independent, otherwise $\mathrm{hol}(k\tilde{C} + m\tilde{D}) = 0$ for some $k, m \neq 0 \in \mathbb{R}$. In this case our lattice in $\mathbb{R}^2$ would be one-dimensional. This gives the condition that $ad - bc \neq 0$ and completes the first class of lattices.

Now suppose that C, D are not pure translations. Then by Corollary 4.13, there exists relatively prime $k, l \in \mathbb{Z}$ such that $C^k D^l$ is a translation. We can therefore change generators in $\mathbb{Z}^2$ to $\tilde{A} = \tilde{C}^k \tilde{D}^l$ and $\tilde{B} = \begin{pmatrix} a \\ b \end{pmatrix}$. In the basis $\tilde{C}, \tilde{D}$ we have that the matrix
$$M = \begin{pmatrix} k & a \\ l & b \end{pmatrix}$$
has $\det M = kb - la$ and since k, l are relatively prime by Bezout's theorem we can always pick $a, b \in \mathbb{Z}$ such that this determinant is 1 and hence an automorphism of our lattice.

We denote the images of our new generators under the affine holonomy representation by
$$A' = \left(I, a = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}\right) \text{ and } B' = \left(B, b = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}\right).$$

Now by Lemma 4.12, we can find a basis of $\mathbb{R}^2$ such that $B = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$ with $n > 0$. Now $\mathrm{hol}(\pi_1(\mathbb{T}^2))$ is abelian so we require that $A'B' = B'A'$ which gives the condition
$$Ba = a$$

this implies that

$$\begin{aligned}\begin{pmatrix}1 & n\\ 0 & 1\end{pmatrix}\begin{pmatrix}a_1\\ a_2\end{pmatrix} &= \begin{pmatrix}a_1 + na_2\\ a_2\end{pmatrix}\\ &= \begin{pmatrix}a_1\\ a_2\end{pmatrix}\end{aligned}$$

which gives $a_2 = 0$ as $n > 0$ and so $a = \begin{pmatrix}a_1\\ 0\end{pmatrix}$. We note that $a_1 \neq 0$ as otherwise the action is not free. For freeness, we also require that $A^l B^k$ acts freely for all $k, l \in \mathbb{Z}$. We check

$$\begin{aligned}\begin{pmatrix}1 & n\\ 0 & 1\end{pmatrix}^k \begin{pmatrix}x\\ y\end{pmatrix} + \begin{pmatrix}kb_1 + la_1\\ kb_2\end{pmatrix} &= \begin{pmatrix}x + kny + kb_1 + la_1\\ y + kb_2\end{pmatrix}\\ &= \begin{pmatrix}x\\ y\end{pmatrix}\end{aligned}$$

If $b_2 = 0$ then there is a fixed point for $y = \dfrac{-kb_1 - la_1}{kn} \in \mathbb{R}$ so we must have $b_2 \neq 0$. Conversely, if $b_2 \neq 0$ and the action is not free, this requires $k = 0$ and so $x + la_1 = a_1$ for $l \neq 0$ but $a_1 \neq 0$ and hence there are no solutions. Now, to show that this takes the form of $\mathcal{B}_{n,a,b}$, we conjugate by

$$T = \left(I, \begin{pmatrix}0\\ \frac{-b_1}{n}\end{pmatrix}\right)$$

this fixes A' and so we compute

$$T^{-1}B'T = \left(B, \begin{pmatrix}0\\ b_2\end{pmatrix}\right)$$

and so we may assume $b_1 = 0$. We can also make the change $A' \mapsto A'^{-1}$ and $B' \mapsto B'^{-1}$ and assume $a_1, b_2 > 0$ which completes the proof. □

4.3.3 When are two structures equivalent?

This gives the possible integral affine structures on the 2-torus. However we still have to tackle the question of when 2 lattices give the same integral affine structure. This occurs when the subgroups of the integral affine group determined by the holonomy representation are conjugate to eachother.

We have the following classification result.

Lemma 4.15. Lattices from different series do not induce equivalent integral affine structures.

Proof. Lattices in series $\mathcal{A}_{a,b,c,d}$ are generated by pure translations and hence cannot be conjugated into lattices of the form $\mathcal{B}_{n,a,b}$.

□

Theorem 4.16. Lattices $\mathcal{A}_{a,b,c,d}$ and $\mathcal{A}_{a',b',c',d'}$ induce equivalent integral affine structures if and only if

$$X = GX'H$$

where $X = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $X' = \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix}$ with $G, H \in GL(2, \mathbb{Z})$.

Proof. Performing lattice automorphisms in $\pi_1(\mathbb{T}^2)$ and conjugating in the image of the generators in $\mathrm{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$ preserves the integral affine structure.

Let $\pi_1(\mathbb{T}^2) = \langle v_1, v_2 \rangle$ with $\mathrm{hol}(v_1) = \left(I, \begin{pmatrix} a \\ c \end{pmatrix}\right)$ and $\mathrm{hol}(v_2) = \left(I, \begin{pmatrix} b \\ d \end{pmatrix}\right)$ and set $X = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Let $H \in GL(2, \mathbb{Z})$ and $g = (G, w) \in \mathrm{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$. We note that conjugating a pure translation (I, v) by g gives the element (I, Gv). We compute

$$\mathrm{hol}(Hv_1) = \left(I, XH \begin{pmatrix} 1 \\ 0 \end{pmatrix}\right)$$

and

$$\mathrm{hol}(Hv_2) = \left(I, XH \begin{pmatrix} 0 \\ 1 \end{pmatrix}\right).$$

This corresponds to an automorphism of our lattice in $\pi_1(\mathbb{T}^2)$. We can also conjugate generators by $g = (G, w) \in \mathrm{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$ which gives

$$g \circ \mathrm{hol}(Hv_1) \circ g^{-1} = \left(I, GXH \begin{pmatrix} 1 \\ 0 \end{pmatrix}\right)$$

and

$$g \circ \mathrm{hol}(Hv_2) \circ g^{-1} = \left(I, GXH \begin{pmatrix} 0 \\ 1 \end{pmatrix}\right)$$

and so it follows that the integral affine structures are equivalent if and only if $X' = GXH$. TODO: finish this proof. ☐

Theorem 4.17. Lattices in $\mathcal{B}_{n,a,b}$ never induce the same integral affine structure.

Proof. Suppose that $\mathcal{B}_{n,a,b} = \langle (1, \bar{a}), (B, \bar{b}) \rangle$ and $\mathcal{B}_{n',a',b'} = \langle (1, \bar{a}'), (B', \bar{b}') \rangle$ induce equivalent integral affine structures. We must have that

$$g\tilde{A}g^{-1} = \tilde{A}'^k \tilde{B}'^r \tag{4.1}$$

$$g\tilde{B}g^{-1} = \tilde{A}'^l \tilde{B}'^s \tag{4.2}$$

where $\tilde{A}, \tilde{B}$ and $\tilde{A}', \tilde{B}'$ are the images of the generators of $\pi_1(\mathbb{T}^2)$ of the structures $\mathcal{B}_{n,a,b}$ and $\mathcal{B}_{n',a',b'}$ and $g = (G = [g_{ij}], \bar{h}) \in \mathrm{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$. The matrix

$$M = \begin{pmatrix} k & l \\ r & s \end{pmatrix} \in GL(2, \mathbb{Z})$$

corresponds to an automorphism of our lattice. This implies that

$$GBG^{-1} = \begin{pmatrix} 1 & * \\ 0 & 1 \end{pmatrix}$$

where $B = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$. Computing this condition yields

$$\begin{pmatrix} * & * \\ g_{21}g_{22} - ng_{21}^2 - g_{21}g_{22} & * \end{pmatrix} = \begin{pmatrix} 1 & * \\ 0 & 1 \end{pmatrix}$$

and so $g_{21}g_{22} - ng_{21}^2 - g_{21}g_{22} = 0$ which gives that $g_{21} = 0$.

Now, inspecting the linear part of $g\tilde{A}g^{-1}$ as the linear part of $\tilde{A}$ is the identity this must also be a translation i.e. $\tilde{A}'^k \tilde{B}'^r$ is a translation. Again, inspecting the linear components we must have that $r = 0$. Our matrix M corresponding to the automorphism of our lattice now looks like

$$M = \begin{pmatrix} k & l \\ 0 & s \end{pmatrix}$$

and so $k, s = \pm 1$ as this matrix needs to have determinant ± 1. Now from Equation 4.1 we obtain

$$\begin{aligned} (G, \bar{h}) \left(I, \begin{pmatrix} a \\ 0 \end{pmatrix} \right) (G^{-1}, -G^{-1}\bar{h}) &= \left(I, G \begin{pmatrix} a \\ 0 \end{pmatrix} \right) \\ &= \left(I, \begin{pmatrix} g_{11}a \\ 0 \end{pmatrix} \right) \\ &= \left(I, \begin{pmatrix} ka' \\ 0 \end{pmatrix} \right) \end{aligned}$$

where the last equality comes from the RHS of Equation 4.1. This now gives that

$$g_{11}a = ka' \Rightarrow g_{11}a = \pm a'$$

as $k = \pm 1$. But $a, a' > 0$ by the classification and so $g_{11} = k$ and $a = a'$.

Next we inspect Equation 4.2. The matrix part gives that

$$GBG^{-1} = B'^s.$$

Computing this gives

$$\begin{aligned} \begin{pmatrix} g_{11} & g_{12} \\ 0 & g_{22} \end{pmatrix} \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} \begin{pmatrix} g_{11} & -g_{12}g_{11}g_{22} \\ 0 & g_{22} \end{pmatrix} &= \begin{pmatrix} 1 & g_{11}g_{22}n \\ 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 1 & n's \\ 0 & 1 \end{pmatrix}. \end{aligned}$$

Where the last equality comes again from the RHS of Equation 4.2. Comparing matrix entries yields $g_{11}g_{22}n = n's$. Using that $g_{11} = \pm 1$, $g_{22} = \pm 1$ and $s = \pm 1$ we have that $n = \pm n'$. But both $n, n' \in \mathbb{N}$ by the classification and so we must have $n = n'$. This now gives that $s = g_{11}g_{22}$. This was the linear part of Equation 4.2 so we now analyze the translational component.

Comparing the translational parts yields

$$-GBG^{-1}\bar{h} + G\bar{b} + \bar{h} = s\bar{b}' + l\bar{a}'. \tag{4.3}$$

We can factorise the LHS as

$$-GBG^{-1}h + G\bar{b} + \bar{h} = G(I - B)G^{-1}\bar{h} + G\bar{b}.$$

Now we simplify

$$G(I-B)G^{-1}\bar{h} = \begin{pmatrix} x \\ 0 \end{pmatrix}$$

where $x \in \mathbb{R}$ is arbitrary depending on $\bar{h}$. Now Equation 4.3 becomes

$$\begin{pmatrix} x \\ 0 \end{pmatrix} + G\bar{b} = s\bar{b}' + l\bar{a}'.$$

Expanding this we are left with

$$\begin{pmatrix} x + g_{12}b \\ g_{22}b \end{pmatrix} = \begin{pmatrix} la' \\ sb' \end{pmatrix}.$$

This gives $g_{22}b = sb'$ and again as $b, b' > 0$ and $g_{22}, s = \pm 1$ we have $b = b'$ and now we are finished as we showed $a = a'$, $b = b'$ and $n = n'$.

□

This completes the classification of the integral affine structures on the torus $\mathbb{T}^2$. We note that lattices in the series $\mathcal{A}_{a,b,c,d}$ induce a Euclidean structure on the torus - i.e. their holonomy representation is a subgroup of $O(n) \rtimes \mathbb{R}^n$. This implies that there is a Riemannian metric preserved by this structure. However, the integral affine structures of $\mathcal{B}_{n,a,b}$ never induce a Euclidean structure on the torus.

Next, we will look at the classification of integral affine structures over the Klein bottle, hence completing the classification of all integral affine structures over closed surfaces.

4.4 Classification of Integral Affine structures on the Klein Bottle

4.4.1 Completeness of integral affine structures on K^2

We now wish to classify the integral affine structures on the Klein bottle. We would hope to proceed as before by calculating all of the injective homomorphisms form $\Gamma = \pi_1(K^2) \to \mathrm{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$ that induce a free and proper action on $\mathbb{R}^2$. However, in order to do this we would need to know that every integral affine structure on K^2 is complete. The following lemma takes care of this fact.

Lemma 4.18. Let N be an integral affine manifold and suppose that there is a regular covering $M \to N$ by a compact orientable integral affine manifold whose fundamental group is nilpotent. Then any integral affine structure on N is complete.

Proof. We can pull back the integral affine structure on N to M. A regular covering induces an injection of fundamental groups $\iota : \pi_1(M) \mapsto \pi_1(N)$. We can induce the holonomy representation on M by $\mathrm{hol} \circ \iota$ where hol is the affine holonomy representation induced by the integral affine structure on N. As M and N share a common universal cover X, the integral affine structure on N induces a developing map $\mathrm{dev} : X \mapsto \mathbb{R}^n$ and this induced holonomy representation will remain equivariant with respect to this developing map as the action of $\pi_1(\mathbb{T}^2)$ on the universal cover is a restriction of the action of $\pi_1(K^2)$.

From Proposition 1.8 this defines an integral affine structure on M which has the same developing map as that of N. Therefore, if the integral affine structure on N is incomplete then the integral affine structure on M will also be. However, M is a compact orientable integral affine manifold with nilpotent fundamental group and hence by Theorem 2.36 is complete. It follows that any integral affine structure on N must also be complete. ☐

Throughout this section we let $\pi_1(K^2) = \Gamma$ have presentation

$$\Gamma = \langle a, b \mid aba = b \rangle$$

and we note that $\langle a, b^2 \rangle \subset \Gamma$ is abelian. Indeed, the oriented double cover of K^2 is the torus $\mathbb{T}^2$ and so there is an injection of fundamental groups $\pi_1(\mathbb{T}^2) \hookrightarrow \pi_1(K^2)$. This abelian subgroup can be realised as the aforementioned injection. As in the previous proposition, the induced integral affine structure on $\mathbb{T}^2$ via pullback will have holonomy representation $\mathrm{hol} \circ \iota$ where ι is the injection of fundamental groups and hol is the holonomy representation of the affine structure on $\mathbb{T}^2$ - indeed this will be equivariant with respect to the same developing map on their shared universal cover. This shows the following proposition.

Proposition 4.19. Let K^2 be equipped with an integral affine structure with holonomy representation

$$\mathrm{hol} : \Gamma \to \mathrm{Aff}_{\mathbb{Z}}(\mathbb{R}^2).$$

Then if $\mathrm{hol}(a) = g \in \mathrm{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$ and $\mathrm{hol}(b) = h \in \mathrm{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$ then the induced holonomy representation on the torus $\mathbb{T}^2$ will be given by

$$\tilde{\mathrm{hol}} : \pi_1(\mathbb{T})^2 = \langle x, y \rangle \to \mathrm{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$$

$$x \mapsto g, \quad y \mapsto h^2$$

Remark 4.20. This implies that once we classify the injective homomorphisms from $\Gamma \to \mathrm{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$, we can easily check which integral affine structure they induce on the torus.

4.4.2 Determining the linear holonomy

We now start are classification of the injective group morphisms from $\Gamma \to \mathrm{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$. We will denote also by $a = (A, \bar{x})$ and $b = (B, \bar{y})$ the images of our generators of Γ under the holonomy representation where $A, B \in GL(2, \mathbb{Z})$ and $\bar{x}, \bar{y} \in \mathbb{R}^2$.

We will make use of the following in our classification

1. We must have that $\det A = 1$ and $\det B^2 = 1$ as these are generators form $\pi_1(\mathbb{T}^2)$.
2. Since b reverses orientation we in fact have $\det B = -1$
3. The action given by the composition $\pi_1(\mathbb{T}^2) \hookrightarrow \Gamma \to \mathrm{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$ must also act freely and properly as this defines the induced integral affine structure on $\mathbb{T}^2$ by Proposition 4.19.

We will now make use of Lemma 4.10 to get restrictions on the matrix B. Namely, this lemma implies that $\mathrm{Tr}\ B = 0$. By Lemma 1 of [FS83], our matrix B must be conjugate to one of

$$B_1 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad \text{or} \quad B_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Now we must find the matrices $A \in GL(2, \mathbb{Z})$ such that $\det A = 1$, $\mathrm{Tr}\ A = 2$ which also satisfy the group relation $ABA = B$ where $B = B_1, B_2$. We perform this calculation now.

Proposition 4.21. Let $B = B_1$. Then we have the following choices for our matrix A.

1. $A = I$.
2. $A_1 = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$.
3. $A_2 = \begin{pmatrix} 1 & 0 \\ p & 1 \end{pmatrix}$.

Proof. Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in GL(2, \mathbb{Z})$. The trace condition implies that $d = 2 - a$. We note that $B^{-1} = B$. Calculating the group condition $ABA = B$ we obtain

$$\begin{aligned} ABA &= \begin{pmatrix} a^2 - bc & ab - bd \\ ca - cd & cb - d^2 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \end{aligned}$$

Comparing matrix entries we find

$$a = d$$

since $d = 2 - a$ this implies $a = d = 1$. Our matrix A now has the form

$$A = \begin{pmatrix} 1 & b \\ c & 1 \end{pmatrix}.$$

From $\det A = 1$ we find that either $b = 0$ or $c = 0$ which completes the proof. □

Now we perform the same calculation for $B = B_2$.

Proposition 4.22. Let $B = B_2$. Then we have the following choices for our matrix A.

1. $A = I$.
2. $A_3 = \begin{pmatrix} 1+n & n \\ -n & 1-n \end{pmatrix}$.
3. $A_4 = \begin{pmatrix} 1-n & n \\ -n & 1+n \end{pmatrix}$.

Proof. Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in GL(2, \mathbb{Z})$. Again we must have that $d = 2 - a$ by the trace condition. Calculating the group law $ABA = B$ yields

$$\begin{aligned} ABA &= \begin{pmatrix} ab + ac & b^2 + ad \\ ad + c^2 & bd + cd \end{pmatrix} \\ &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \end{aligned}$$

Comparing entries yields that $a(b + c) = 0$ and $d(b + c) = 0$. Hoiwever, the trace condition implies that a and d can't both be zero and so $b = -c$. Our matrix A now has the form

$$A = \begin{pmatrix} a & b \\ -b & d \end{pmatrix}$$

Calculating the determinant condition and using $d = 2 - a$ yields

$$\begin{aligned} b^2 &= 1 - ad \\ &= (a - 1)^2 \end{aligned}$$

and so

$$b = a - 1 \quad \text{or} \quad b = 1 - a$$

The first case yields

$$A = \begin{pmatrix} n+1 & n \\ -n & 1-n \end{pmatrix}$$

where $n = a - 1$ and the second case yields

$$A = \begin{pmatrix} 1-n & n \\ -n & 1+n \end{pmatrix}$$

where $n = 1 - a$. The case where $n = 0$ corresponds to the identity and this completes our proof. $\square$

This gives the possible choices of A and B that satisfy the group law and Lemma 4.10. We must now deduce the translational component in each case and check that the induced action is free and proper.

4.4.3 Translational component and freeness of the action

To find the translational components we impose the group law $aba = b$. When checking freeness, we can often use Proposition 4.19 to take shortcuts as we know that a, b^2 must generate one of the integral affine structures on the torus and hence act freely.

We will now compute the translational components and check whether the action is free.

Lemma 4.23. Let $B = B_1$. Then the possible choices of A that induce an integral affine structure on K^2 are $A = I, A_1$.

Proof.

Case 1 ($A = I$, $B = B_1$)**.** Let $a = (I, \bar{x})$ and $b = (B_1, \bar{b})$ with $\bar{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ and $\bar{b} = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$ be our generators in $\mathrm{Aff}_{\mathbb{Z}}(\mathbb{R}^2)$. They must satisfy the relation $aba = b$. Calculating this we find

$$(B, B\bar{x} + \bar{x} + \bar{b}) = (B, \bar{b})$$

The translational component gives $B\bar{x} = -\bar{x}$ and so $\bar{x}$ lives in the -1-eigenspace. This implies that $x_1 = 0$. We are free to conjugate both our generators as this preserves the integral affine structure and so we will do so by $g = \left(I, \begin{pmatrix} 0 \\ \frac{y_2}{2} \end{pmatrix}\right)$ Conjugating a leaves it unchanged so we calculate

$$g^{-1} \circ b \circ g = \left(B, \begin{pmatrix} y_1 \\ 0 \end{pmatrix}\right)$$

and so we may assume that $y_2 = 0$ by performing this conjugation. Our generators now look like

$$a = \left(I, \begin{pmatrix} 0 \\ x_2 \end{pmatrix}\right), \quad b = \left(B, \begin{pmatrix} y_1 \\ 0 \end{pmatrix}\right)$$

Since B alone clearly does not act freely we can't have that $y_1 = 0$ and if $x_2 = 0$ then our lattice is one-dimensional and hence the holonomy representation is not injective. Therefore, $x_2, y_1 \neq 0$. Furthermore, by changing generators $a \mapsto a^{-1}$ and $b \mapsto b^{-1}$ we can assume $x_2, y_1 > 0$. It remains to check that this action is free. We must check that $a^q b^n$ acts freely for all $q, n \in \mathbb{Z}$ not both zero. However, we now note that as $B^{2k} = I$ for $k \in \mathbb{Z}$, by the torus classification of Theorem 4.16, we have already shown that $a^q b^{2k}$ acts freely. We therefore must check the case of $a^q b^{2k+1}$. We compute that

$$a^q = \left(I, \begin{pmatrix} 0 \\ qx_2 \end{pmatrix}\right), \quad b^{2k+1} = \left(B, \begin{pmatrix} (2k+1)y_1 \\ 0 \end{pmatrix}\right)$$

For the action to be free we need $a^q b^{2k+1} \circ z = z$ to have no solutions for all $z \in \mathbb{R}^2$. We compute this action

$$\begin{aligned} a^q b^{2k+1} \circ z &= B \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} + \begin{pmatrix} (2k+1)y_1 \\ x_2 \end{pmatrix} \\ &= \begin{pmatrix} (2k+1)y_1 + z_1 \\ x_2 - z_2 \end{pmatrix} \end{aligned}$$

and so, for there to be a fixed point we need

$$(2k+1)y_1 + z_1 = z_1$$

which gives $(2k+1)y_1 = 0$. But now $y_1 > 0$ and $2k+1 = 0 \Rightarrow k = -\frac{1}{2} \notin \mathbb{Z}$ and so this action can have no fixed points. The induced integral affine structure on the torus is given by

$$a = \left(1, \begin{pmatrix} 0 \\ x_2 \end{pmatrix}\right), \quad b^2 = \left(1, \begin{pmatrix} 2y_1 \\ 0 \end{pmatrix}\right)$$

Case 2 ($A = A_1$, $B = B_1$)**.** Let $a = \left(A_1, \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right)$ and $b = \left(B, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}\right)$ As $A_1^2 \neq I$ and $B^2 = I$, by the torus classification we again obtain that $x_1 = 0$. We calculate the group law $aba = b$. This gives us

$$\begin{aligned} aba &= \left(B, \begin{pmatrix} y_1 + n(y_2 - x_2) \\ y_2 \end{pmatrix}\right) \\ &= \left(B, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}\right) \end{aligned}$$

which imposes the restriction $n(y_2 - x_2) = 0$ and since $n \neq 0$ this implies $y_2 = x_2$. We can also assume that $x_2 > 0$ by conjugating the generators by $(_I, \bar{0})$ and that $y_1 > 0$ by changing generators $b \mapsto b^{-1}$ if necessary. Now we check freeness of the action.

We calculate that

$$a^q = \left(\begin{pmatrix} 1 & qn \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} \frac{nx_2 q(q-1)}{2} \\ qx_2 \end{pmatrix}\right)$$

and

$$b^k = \begin{cases} \left(B, \begin{pmatrix} ky_1 \\ x_2 \end{pmatrix}\right), \text{ k odd} \\ \left(I, \begin{pmatrix} ky_1 \\ 0 \end{pmatrix}\right), \text{ k even} \end{cases}$$

Since the case of k being even is covered by the torus classification we must only check freeness for k being odd.

For k odd we compute

$$a^q b^k = \left(\begin{pmatrix} 1 & -qn \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} \dfrac{nx_2 q(q-1)}{2} + qnx_2 + ky_1 \\ (q+1)x_2 \end{pmatrix}\right)$$

calculating this action on $z = \begin{pmatrix} z_1 \\ z_2 \end{pmatrix}$ yields

$$(a^q b^k) \circ z = \begin{pmatrix} \dfrac{nx_2 q(q-1)}{2} + qnx_2 + ky_1 + z_1 - qnz_2 \\ (q+1)x_2 - z_2 \end{pmatrix}$$

Solving the fixed point equation, the bottom entry of the vector yields

$$z_2 = \frac{q+1}{2} x_2 \tag{4.4}$$

Solving the fixed point equation in the top component yields

$$z_2 = \frac{x_2(q-1)}{2} x_2 + \frac{ky_1}{qn}$$

Substituting Equation 4.4 for z_2 yields $ky_1 = 0$ which implies $k = 0$ as $y_1 > 0$. This now implies that a^q would need to have a fixed point but one checks

$$a^q \circ z = \begin{pmatrix} * \\ z_2 + qx_2 \end{pmatrix}$$

and so we require that $qx_2 = 0$ but $x_2 > 0$ and hence the only element with a fixed point occurs when $k = q = 0$ which is the identity and hence the action is free.

The induced integral affine structure on $\mathbb{T}^2$ is given by

$$a = \left(A_1, \begin{pmatrix} 0 \\ x_2 \end{pmatrix} \right), \quad b^2 = \left(I, \begin{pmatrix} 2y_1 \\ 0 \end{pmatrix} \right)$$

with $x_2, y_1 > 0$

Case 3 ($A = A_2$, $B = B_1$)**.** Let $a = \left(A_1, \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \right)$ and $b = \left(B, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \right)$. In this case the action will fail to be free. Again, by considering the induced structure on the torus it follows that $x_1 = 0$. We calculate aba again which yields

$$\begin{aligned} aba &= \left(B, \begin{pmatrix} y_1 \\ py_1 + y_2 \end{pmatrix} \right) \\ &= \left(B, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \right) \end{aligned}$$

and so we must have $py_1 = 0$ and since $p > 0$ we have $y_1 = 0$. But now we calculate b^2 as

$$b^2 = \left(I, \begin{pmatrix} 2y_1 \\ 0 \end{pmatrix} \right)$$

but since $y_1 = 0$ this cannot induce an integral affine structure on the torus hence the action is not free.

□

This completes the case where $B = B_1$. Next, we tackle the case of $B = B_2$.

Lemma 4.24. Let $B = B_2$. Then the possible choices of A that induce an integral affine structure on K^2 are $A = I, A_4$.

Proof.

Case 1 ($A = I$, $B = B_2$)**.** We calculate the group law $aba = b$ which yields

$$\begin{aligned} aba &= \left(B, \begin{pmatrix} x_1 + x_2 + y_1 \\ x_2 + y_2 + x_1 \end{pmatrix}\right) \\ &= \left(B, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}\right) \end{aligned}$$

This implies that $x_1 = -x_2$. We conjugate a, b by

$$c = \left(-I, \begin{pmatrix} -\frac{y_2}{2} \\ -\frac{y_2}{2} \end{pmatrix}\right)$$

which gives use the new generators

$$a = \left(I, \begin{pmatrix} -x_1 \\ x_1 \end{pmatrix}\right), \quad b = \left(B, \begin{pmatrix} -y_2 - y_1 \\ 0 \end{pmatrix}\right)$$

and by switching generators $a \mapsto a^{-1}$ if necessary we may assume that $x_1 < 0$ so that we will switch the signs on the translational component of a. We will do the same for the translational component of b and assume $y_2 = 0$ and $y_1 > 0$ after relabelling $-y_2 - y_1 \to y_1$. This finishes our generating set which we write as

$$a = \left(I, \begin{pmatrix} x_1 \\ -x_1 \end{pmatrix}\right), \quad b = \left(B, \begin{pmatrix} y_1 \\ 0 \end{pmatrix}\right)$$

with $x_1, y_1 > 0$. We now check freeness of the action. Again since $B^{2k} = I$, the even case is covered by the torus classification and so we will check freeness of words $a^q b^k$ with k being odd. We calculate that

$$\begin{aligned} a^q &= \left(I, \begin{pmatrix} qx_1 \\ -qx_1 \end{pmatrix}\right) \\ b^k &= \left(B, \begin{pmatrix} \frac{k+1}{2} y_1 \\ \frac{k-1}{2} y_1 \end{pmatrix}\right), \text{ for k odd} \\ a^q b^k &= \left(B, \begin{pmatrix} \frac{k+1}{2} y_1 + qx_1 \\ \frac{k-1}{2} y_1 - qx_1 \end{pmatrix}\right), \text{ for k odd.} \end{aligned}$$

Now let $z = \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} \in \mathbb{R}^2$ and consider the fixed point equation $(a^q b^k) \circ z = z$. This gives

$$\begin{pmatrix} \frac{k+1}{2} y_1 + qx_1 + z_2 \\ \frac{k-1}{2} y_1 - qx_1 + z_1 \end{pmatrix} = \begin{pmatrix} z_1 \\ z_2 \end{pmatrix}$$

Solving this gives $\frac{k+1}{2} y_1 + \frac{k-1}{2} y_1 = 0 \Rightarrow k y_1 = 0$. But since $y_1 > 0$ we must have $k = 0$. So if the action is not free we must have that a^q acts non-freely for some $q \neq 0$ but this action is by translations and hence is free and so $A = I, B = B_2$ does induce an integral affine structure. The induced structure on the torus is given by

$$a = \left(I, \begin{pmatrix} x_1 \\ -x_1 \end{pmatrix}\right), \quad b^2 = \left(I, \begin{pmatrix} y_1 \\ y_1 \end{pmatrix}\right)$$

$x_1, y_1 > 0$.

Case 2 ($A = A_3$, $B = B_2$)**.** We show that this cannot induce an integral affine structure on the torus. We conjugate A and B by

$$G = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

which puts them in the form

$$A = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} -1 & 1 \\ 0 & 1 \end{pmatrix}$$

and so we have $B^2 = I$. Hence, the integral affine structure on $\mathbb{T}^2$ should belong to the second family. However calculating

$$b^2 = \left(I, \begin{pmatrix} y_2 \\ 2y_2 \end{pmatrix} \right)$$

we require that $2y_2 = 0 \Rightarrow y_2 = 0$ but then b^2 does not act freely and hence does not induce an integral affine structure on the torus. Therefore, this case cannot induce an integral affine structure on K^2.

Case 3 ($A = A_4$, $B = B_2$)**.** Again, we conjugate A, B by the matrix

$$G = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}$$

and assume that

$$a = \left(A = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \right), \quad b = \left(B = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \right)$$

Using that $B^2 = I$ and a, b^2 must induce an integral affine structure on the torus it follows that $x_1 = 0$ since the induced structure belongs to the second family. Now, calculating the group law $aba = b$ yields

$$\left(\begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} y_1 + ny_2 + x_2(1-n) \\ y_2 \end{pmatrix} \right) = \left(B, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \right)$$

and so $ny_2 + (1-n)x_2 = 0 \Rightarrow y_2 = \frac{n-1}{n}x_2$. Our generators are therefore

$$a = \left(\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 \\ x_2 \end{pmatrix} \right), \quad b = \left(\begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} y_1 \\ \frac{n-1}{n}x_2 \end{pmatrix} \right)$$

and the induced structure on the torus is

$$a = \left(\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 \\ x_2 \end{pmatrix} \right), \quad b^2 = \left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 2y_1 + \frac{n-1}{n}x_2 \\ 0 \end{pmatrix} \right)$$

with $y_1 > 0$ and so a necessary condition is $2y_1 \neq -\frac{n-1}{n}x_2$. In fact, we shall now show that this is sufficient. We calculate that

$$a^k = \left(\begin{pmatrix} 1 & kn \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} \frac{nk(k-1)}{2}x_2 \\ kx_2 \end{pmatrix} \right), \quad b^q = \left(B, \begin{pmatrix} qy_1 + \frac{q-1}{2}y_2 \\ y_2 \end{pmatrix} \right), \text{ for q odd}$$

and

$$a^k b^q = \left(\begin{pmatrix} 1 & 1-kn \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} qy_1 + \frac{q-1}{2}y_2 + kny_2 + \frac{nk(k-1)}{2}x_2 \\ y_2 + kx_2 \end{pmatrix} \right)$$

Letting $v_1 := qy_1 + \frac{q-1}{2}y_2 + kny_2 + \frac{nk(k-1)}{2}x_2$ and $v_2 := y_2 + kx_2$ and $z = \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} \in \mathbb{R}^2$, the fixed point equation yields

$$\begin{pmatrix} (1-kn)z_2 \\ -2z_2 \end{pmatrix} = -\begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$$

This yields the equation $(1-kn)v_2 + 2v_1 = 0$. Solving this equation and using that $y_2 = \frac{n-1}{n}x_2$ we obtain

$$q(2y_1 + y_2) = 0$$

It is quick to check that a^k acts freely so if the action is not free then $q \neq 0$, and this gives the requirement

$$2y_1 \neq -\frac{n-1}{n}x_2$$

and so this condition is sufficient and necessary.

□

This completes the classification of the integral affine structures on the torus. We summarize the results of the previous two lemmas into the following theorem.

Theorem 4.25. The possible integral integral affine structures on the Klein bottle are

1. $a = \left(I, \begin{pmatrix} 0 \\ x_2 \end{pmatrix}\right), \quad b = \left(\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} y_1 \\ 0 \end{pmatrix}\right)$, with $x_2, y_1 > 0$.
2. $a = \left(\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 \\ x_2 \end{pmatrix}\right), \quad b = \left(\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} y_1 \\ x_2 \end{pmatrix}\right)$, with $x_2, y_1 > 0$.
3. $a = \left(I, \begin{pmatrix} x_1 \\ -x_1 \end{pmatrix}\right), \quad b = \left(\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} y_1 \\ 0 \end{pmatrix}\right)$, with $x_1, y_1 > 0$.
4. $a = \left(\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 \\ x_2 \end{pmatrix}\right), \quad b = \left(\begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} y_1 \\ \frac{n-1}{n}x_2 \end{pmatrix}\right)$, with $y_1 > 0$ and $2y_1 \neq -\frac{n-1}{n}x_2$.

We note that families 1) and 3) induce a Euclidean structure on the Klein bottle, e.g. they preserves some Riemannian metric as we have that the group they generate is a subset of $O(n) \rtimes \mathbb{R}^2$.

We now wish to classify the corresponding Lagrangian fibrations for each of the integral affine structures on the two torus and the Klein bottle. Below are some pictures of the different families of integral affine structures on both the torus and the Klein bottle. (TODO: Lattices for the Klein Bottle)

4.4.4 Fundamental domains for $\mathbb{T}^2$ lattices.

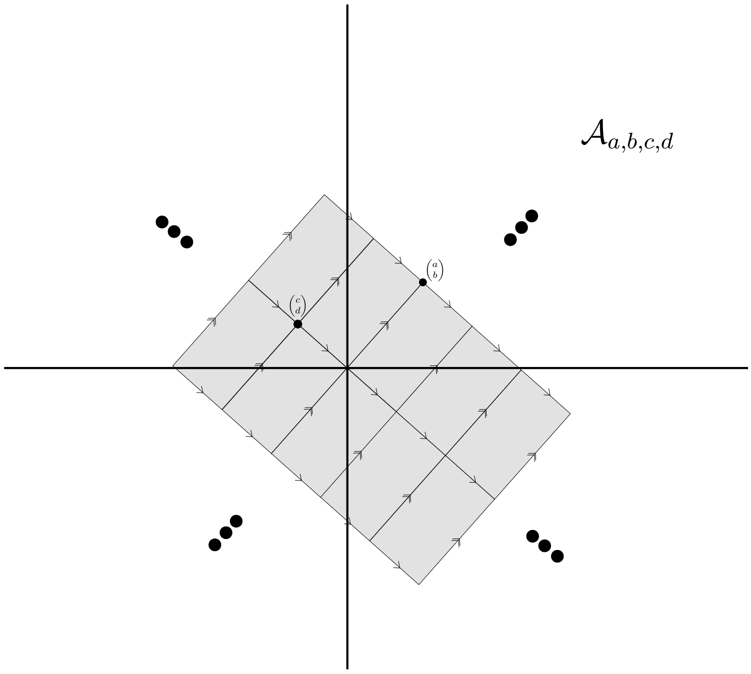

Figure 4.1: Fundamental domain for lattice $\mathcal{A}_{a,b,c,d}$

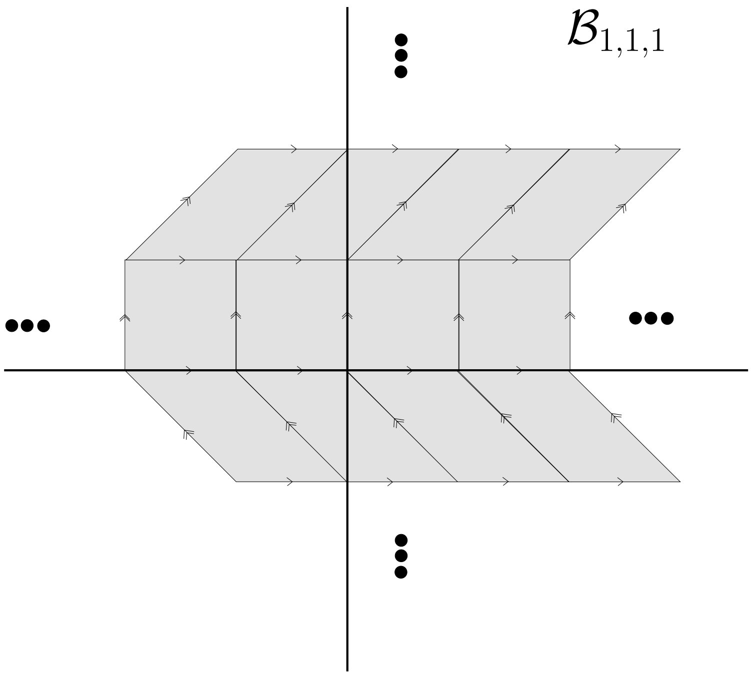

Figure 4.2: Fundamental domain for lattice $\mathcal{B}_{1,1,1}$

CHAPTER FIVE

Classification of Lagrangian Fibrations over Closed Surfaces

5.1 The Classification Problem

Now that we have classified the integral affine structures on closed surfaces, we can look to classify the Lagrangian fibrations $E \to B$ over these surfaces. The integral affine structure on the base B is one invariant of a Lagrangian fibration however it is not the only one. We have seen that there is a correspondence between integral affine structures on B and full rank lattices in T^*B locally spanned by closed one-forms. This lattice $\Lambda \subset T^*B$ gives rise to a Lagrangian fibration $\pi : T^*B/\Lambda \to B$ which is a canonical Lagrangian fibration we can associate to B once we have fixed the integral affine structure. There are in general, other Lagrangian fibrations over B which are in some sense *twistings* of this canonical fibration.

We wish to classify these fibrations up to fibrewise symplectomorphism, covering the identity. We shall see that there are obstructions both to a Lagrangian bundle over B being topologically equivalent to this canonical Lagrangian fibration $\pi : T^*B/\Lambda \to B$ as well as obstructions to a Lagrangian bundle over B being symplectically equivalent to the canonical Lagrangian bundle. Since the problem of classifying these bundles will be an issue of stitching together locally defined Lagrangian fibrations into a global one, it will be helpful to do this in the language of sheaves. (TODO: this is written a bit badly).

We shall make use of the following sheaves on B.

Definition 5.1. Let B be a closed integral affine surface with period lattice $\Lambda \subset T^*B$.

1. Λ is the sheaf of sections of the period lattice $\Lambda \subset T^*B$.
2. $\Lambda\mathcal{T}^*B$ is the sheaf of Lagrangian sections of T^*B and $\mathcal{T}^*B$ is the sheaf of all sections of T^*B.
3. $\mathcal{Z}(T^*B/\Lambda)$ is the sheaf of Lagrangian sections of T^*B/Λ and T^*B/Λ is the sheaf of all sections of T^*B/Λ.
4. Λ^2B is the sheaf of closed 2-forms on B.

Remark 5.2. Let $\Omega^1(B)$ denote the sheaf of differential forms on B. Then the differential

$$d : \Omega^1(B) \to \Lambda^2(B)$$

descends to a well-defined differential

$$d_\Lambda : \mathcal{T}^*B/\Lambda \to \Lambda^2(B)$$

on the sections of T^*B/Λ as sections of Λ are given by closed one forms and hence vanish under this differential. This leads to the following notion of latticed forms.

Definition 5.3 (Latticed one form)**.** Fix a period lattice $\Lambda \subset T^*B$. A *latticed one form* is a section of T^*B/Λ. A lattice one form will be called closed if it determines a Lagrangian submnaifold of the symplectic manifold $(T^*B/\Lambda, \tilde{\omega}_{can})$

Since the differential descends to T^*B/Λ, it makes sense to consider elements in $H^2(B, \mathbb{R})$ which arise from this differential.

Definition 5.4. Let s be a latticed one form on B. Then the cohomology class $[ds] \in H^2(B, \mathbb{R})$ will be called a *cohomology invariant of the* s

We now work towards showing when adding closed two forms to $(T^*B/\Lambda, \tilde{\omega}_{can})$ produces equivalent Lagrangian bundles. This will turn out to depend on these cohomology invariant of s.

Proposition 5.5. A fibrewise affine of T^*B/Λ is determined by a latticed one form and any latticed one form determines a . Moreover, this is a Lagrangian equivalence if the latticed one form is closed.

Proof. Let $F : T^*B/\Lambda \to T^*B/\Lambda$ be a fibrewise affine . On each fibre, this takes the form $F_b(x) = x + c$ for some $c \in T_b^*B/\Lambda_b$ and hence is determined by a smooth section of T^*B/Λ.

Conversely, given a section $\alpha : B \to T^*B/\Lambda$ we define an affine by

$$F_\alpha : T^*B/\Lambda \to T^*B/\Lambda$$

$$x_b \mapsto x_b + \alpha_b$$

. Proposition 3.16 yields that this is a Lagrangian equivalence if and only if α is closed. □

Remark 5.6. The contents of Theorem 3.29 show that if $E \to B$ is a Lagrangian fibration with period lattice Λ, then a latticed one-form also determines a fibrewise of E. Indeed, since over contaractible open sets, $E \to B$ identifies with $T^*B/\Lambda \to B$, the converse can also be shown to be true.

Proposition 5.7. Let $\rho : (E, \omega) \to B$ be a Lagrangian fibration with period lattice Λ. Adding the pullback of closed two-forms $\alpha, \beta \in \Omega^2(B)$ to E is a Lagrangian equivalence if and only if

$$[\alpha] - [\beta] = [d\eta] \in H^2(B)$$

where $[d\eta]$ is a cohomological invarian of a s.

Proof. We again make use of Proposition 3.16. Let η be a one-form and consider the s_η : $T^*B \to T^*B$. Then

$$s_\eta^*(\omega_{can} + \pi^*\alpha) = \omega_{can} + \pi^*(\alpha - d\eta)$$

and so $\pi^*(\alpha + d\eta) = \pi^*\beta$ if $\pi^*(\alpha - \beta - d\eta) = 0$ and so this gives $\alpha - \beta = d\eta$. Descending to the quotient T^*B/Λ, as sections of Λ are given by closed forms it follows that we can alter $d\eta$ by a closed form. (TODO: This proof is a bit loose. We are implicitly using the fact that $E \to B$ locally looks like $T^*B/\Lambda \to B$. Can make this more explicit).

Conversely, suppose we have a Lagrangian equivalence between $(T^*B, \omega + \pi^*\alpha)$ and $(T^*B, \omega + \pi^*\beta)$, then Proposition 1 of [Mis96] shows that this is determined by the of some one-form α on the base and the result follows by the descension to the quotient. (TODO: I will need to add this proof, it should have uses elsewhere too) $\square$

5.2 Luttinger Surgery

Luttinger surgery was first introduced in [Lut95] and provides a way to modify a symplectic four-manifold (M, ω) along an embedded Lagrangian tori $L \subset M$ such that the symplectic structure is preserved and the geometry is altered only in a neighborhood of the torus. Topologically it is equivalent to Dehn surgery with the framing chosen to be the Weinstein framing. A framing is, roughly speaking (TODO: roughly? I think it is up to some invariance condition), a paramaterization of the tubular neighborhood of the torus $L \subset M$. For a Lagrangian submanifold there is a canonical choice of such a framing given by the Weinstein-Lagrangian tubular neighborhood theorem.

Theorem 5.8 ([Wei71])**.** Let (M, ω) be a symplectic manifold and $L \subset M$ be a Lagrangian submanifold. Then there exist tubular neighborhoods $U \subset M$ and $V \subset T^*L$ of L together with a symplectomorphism:

$$f : (U, \omega|_U) \to (V, \omega_{\text{can}}|_V)$$

with $f|_L = Id_L$.

The surgery construction works for general Lagrangian tori in a symplectic four-manifold but we will focus on the case where our Lagrangian torus is given by a fibre of a Lagrangian fibration.

Let B be the base space of such a Lagrangian fibration and let $T^*B/\Lambda \to B$ be the symplectic reference bundle. We view B as the quotient $\mathbb{R}^2/\langle A, B\rangle$ where A, B are the generators of the affine holonomy group. Let $p \in \mathbb{R}^2$ be in the interior of a fundamental domain for the quotient and pick a small disc D of radius 2ϵ centered at this point and a neighborhood U of D. Over this neighborhood, our fibration looks like $\pi : U \times T^2 \to U$. In the case of the torus, the symplectic form on $U \times T^2$ is inherited from $U \times T^2 \subset (T^*T^2, \omega_{can})$ (TODO: what about Klein bottle??).

Now, if we take $\partial\pi^{-1}(D) = \partial(T^2 \times D)$ this is a three torus $T^2 \times S^1$. If we remove the interior of this region from the total space of the fibration, we are left with a manifold with boundary T^3. The surgery now amounts to specifying an attaching map which glues together the boundary of the removed $T^2 \times D$ back into the boundary created in the total space of the fibration. This attaching map is given by a diffeomorphism $F : T^3 \to T^3$. Of course, so far this attaching map has not taken into account the symplectic structure of the total space of the fibration. Indeed, the three torus has odd dimension and hence it does not make sense to ask that the attaching map F itself be a symplectomorphism. However, we can ask that F is the restriction of a symplectomorphism of the bundle $\pi^{-1}(N)$ where N is a small annulus around the boundary of the disc ∂D^2.

Of course, one way to define such a map is to pick a Lagrangian section of $\pi^{-1}(N)$ which is given by a closed one-form α, and perform a shift by this one-form. Then, by Proposition 3.16 this will be a symplectomorphism and since the preserves the fibres, its restriction to ∂D^2 will induce a map $F_\alpha : T^3 \to T^3$. A section of our bundle over the boundary of the disc is a map $f : S^1 \to T^2$ and is determined up to homotopy by an element of $\pi_1(T^2) = \mathbb{Z}^2$ as such a map is a loop in T^2. The pair of integers (m, n) represent the winding number of such a map. Let $(x, y) = (r\cos\varphi, r\sin\varphi)$ be coordinates on N centered at p. Then we can define a representative section for any (m, n) to be

$$s = \frac{1}{2\pi}(m\varphi dx + n\varphi dy).$$

One checks that after winding once around the circle i.e. $\varphi \mapsto \varphi + 2\pi$ we have that our section changes by $mdx + ndy$ and hence corresponds to the winding number given by (m, n). This indeed defines a section of the torus bundle since it is locally given by the symplectic reference bundle and hence over U looks like $T^*U/\langle dx, dy\rangle_{\mathbb{Z}}$.

In principle, one could perform the surgery by performing a shift using this section. The issue is, this section is not Lagrangian. Indeed one calculates

$$ds = \frac{1}{2\pi}(md\varphi \wedge dx + nd\varphi \wedge dy).$$

In order to make this closed, one can add the correction term

$$\beta = \frac{1}{2\pi}(mx + ny)d\varphi$$

to s. It is easily checked that this does not alter the homotopy class of the section s as incrementing φ by 2π does not alter β. Hence, the section $F_{m,n} = s + \beta$ determines a symplectomorphism on a neighborhood of the bundle over ∂D^2 given by performing a shift by the closed one-form $F_{m,n}$ as in Definition 3.15. This completes the surgery construction.

5.2.1 Calculating Cohomological Invariant of s

We now wish to calculate the cohomological invariants of s for the integral affine structure of Theorem 4.14. Since a class $[d\eta] \in H^2(\mathbb{T}^2, \mathbb{R})$ is determined by the value $\int_{\mathbb{T}^2} d\eta$ we are left to calculate the value of this integral.

Theorem 5.9. We have that

1. For the lattice $\mathcal{A}_{a,b,c,d}$, the cohomological invariant of s are determined by integer covectors $\tilde{m} = kdx + rdy$ and $\tilde{n} = ldx + sdy$ subject to no restrictions. Two invariants of s are equaivalent if the quantity $ka + lb + rc + sd$ is the same in both cases.
2. For the lattice $\mathcal{B}_{n,a,b}$, the cohomological invariant of s are determined by integer covectors $\tilde{m} = kdx + rdy$ and $\tilde{n} = sdy$. Two invariants of s are equaivalent if the quantity $ka + sb$ is the same in both cases.

Proof.

Case 1 ($\mathcal{A}_{a,b,c,d}$)**.** Recall that in this case the fundamental domain for the torus is generated by

$$g = \left(1, \begin{pmatrix} a \\ c \end{pmatrix}\right) \text{ and } h = \left(1, \begin{pmatrix} b \\ d \end{pmatrix}\right)$$

where the translational components are linearly independent. To construct a latticed one-form, we need the value of the one-form to agree with the identifications induced by the translational actions up to taking the quotient by the lattice $\Lambda \subset T^*B$. Namely, given the value of a latticed one form $\alpha \in \Gamma(T^*B/\Lambda)$, if we fix a chart $U \subset \mathbb{T}^2$ around p and let $\alpha_p = \alpha_1 dx + \alpha_2 dy \in T_p^*\mathbb{T}^2$, translating by one of our generators g or h we should get back $\alpha_p + kdx + rdy$ for some integers $k, r \in \mathbb{Z}$ as in affine coordinates (x, y), the lattice Λ is spanned by dx and dy over $\mathbb{Z}$.

The invariant we wish to calculate is $\int_{\mathbb{T}^2} d\alpha$ where α is a latticed one-form. By Stokes theorem, letting U denote a fundamental domain for $\mathbb{T}^2$,

$$\begin{aligned}\int_U d\alpha &= \int_{\partial U} \alpha \\ &= \int_A \alpha - \int_{A'} \alpha + \int_B \alpha - \int_{B'} \alpha\end{aligned}$$

where A, A', B, B' are the boundaries of the fundamental domain depicted in figure (TODO: image of fundamental domain.)

For example, translating α along A by g should return α up to some integer covector. Denoting by $\tilde{m} = (k, r)$ and $\tilde{n} = (l, s)$, the integer covectors obtained from transporting along A and B respectively, we can then translate $g^*\alpha + \hat{n}$ by h along B'. This should return the same one-form as first translating along B and then translating along A'. or our first lattice, we see that this always holds true and so there is no restrictions on the covectors $\bar{m}$ and $\bar{n}$.

Now we calculate the required integral. We note that on A and A', the one form α differs by $\bar{m}$ and so

$$\begin{aligned}\int_A \alpha - \int_{A'} \alpha &= \int_A \alpha - \int_A \alpha + \bar{m} \\ &= -\int_A \bar{m}\end{aligned}$$

Similarly, we have $\int_B \alpha - \int_{B'} = -\int_B \bar{n}$. Putting this together yields

$$\begin{aligned}\int_U d\alpha &= \int_A \alpha - \int_{A'} \alpha + \int_B \alpha - \int_{B'} \\ &= -(\int_A \bar{m} + \int_B \bar{n}) \\ &= -(\int_{(0,0)}^{(a,c)} kdx + rdy + \int_{(0,0)}^{(b,d)} ldx + sdy) \\ &= -(ka + rc + lb + sd)\end{aligned}$$

with $k, l, r, s \in \mathbb{Z}$ which gives the required invariants.

Case 2 ($\mathcal{B}_{n,a,b}$)**.** Recall that for this family our fundamental domain is generated by

$$g = \left(1, \begin{pmatrix} a \\ 0 \end{pmatrix}\right) \text{ and } h = \left(\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 \\ b \end{pmatrix}\right)$$

In this case, our integer covectors $\tilde{m}$ and $\tilde{n}$ are not arbitrary. We must have that

$$h^*(g^*\alpha + \tilde{n}) + \tilde{m} = g^*(h^* + \bar{m}) + \tilde{n}$$

this implies that $h^*n = n$ or more algebraically

$$\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}^T \begin{pmatrix} l \\ s \end{pmatrix} = \begin{pmatrix} l \\ s \end{pmatrix}$$

This implies that $\tilde{n} = \begin{pmatrix} 0 \\ s \end{pmatrix}$ and this is the only restriction on our integer co-vectors $\tilde{m}$ and $\tilde{n}$. We now calculate the integral over our fundamental domain. We let $\alpha = \alpha_1(x, y)dx + \alpha_2(x, y)dy$.

Again, applying Stokes theorem, we have that

$$\begin{aligned}\int_{\partial U}\alpha &= \int_A \alpha - \int_{A'}\alpha + \int_B \alpha - \int_{B'}\alpha\\ &= \int_A \alpha - \int_A h^*\alpha + \tilde{m} + \int_B \alpha - \int_B g^*\alpha + \tilde{n}\\ &= \int_A \alpha - h^*\alpha - \tilde{m} - \int_B \tilde{n}\end{aligned}$$

Now, recalling that $h^* = \begin{pmatrix} 1 & 0 \\ n & 1 \end{pmatrix}$ we have that

$$\begin{aligned}\alpha - h^*\alpha &= (\alpha_1(x,y) - \alpha_1(x,y))dx + n\alpha_1(x,y)dy\\ &= n\alpha_1(x,y)dy\end{aligned}$$

Now, we have that

$$\begin{aligned}\int_A n\alpha_1(x,y)dy &= \int_{(0,0)}^{(a,0)} n\alpha_1(x,y)dy\\ &= 0\end{aligned}$$

as there is no dx contribution. Hence, our invariant becomes

$$\begin{aligned}\int_{\partial U}\alpha &= -(\int_A \tilde{m} + \int_B \tilde{n})\\ &= -(\int_{(0,0)}^{(a,0)} \tilde{m} + \int_{(0,0)}^{(0,b)} \tilde{n})\\ &= -(ka + sb)\end{aligned}$$

where $k, s \in \mathbb{Z}$ and this completes the calculation of the cohomological invariants of .

□

5.3 Chern Class and Lagrangian Class

The strategy in classifying the Lagrangian bundles over B will be to start with the topological reference bundle T^*B/Λ and using this, produce new bundles by gluing these reference bundles together. We will then show that we can essentially obtain any Lagrangian bunlde over B in this way. In [Dui80], Duistermaat defined the *Chern class* of a Lagrangian fibration. It measures the obstruction for a Lagrangian fibration to be topologically equivalent to T^*B/Λ. We now illustrate how this class is defined.

We have seen previously that given a regular Lagrangian fibration with compact, connected fibres, there is a canonical period lattice $\Lambda \subset T^*B$ and a fibrewise transitive action of $T^*B/\Lambda \to B$ on $E \to B$. This endowed $E \to B$ with the structure of an *affine* torus bundle, however we could not yet identify it with the torus bundle T^*B/Λ. The obstruction to topologically identifying these two fibrations is the obstruction to $E \to B$ admitting a global section. We can attempt to answer when this is the case.

If we cover the base space B with open sets $\{U_i\}$ such that each U_i is contractible, then, restricting the fibration to some U_i, all the obstructions to this fibration admitting global action angle coordinates vanish or in other words, locally this can be identified with T^*B/Λ. Hence, if we pick our U_i small enough we can choose sections $s_i : U_i \to E$. Since the action of T^*B/Λ

on $E \to B$ is transitive, on the intersections $U_i \cap U_j$, we know that $s_i = \mu_{ij} + s_j$ whereby μ_{ij} is a section of T^*B/Λ and $\mu_{ij} + s_j$ denotes the abelian vector space action of this section on s_j. We now make the following claim.

Proposition 5.10. The sections μ_{ij} satisfy the cocycle condition and hence define a cohomology class $[\mu] \in H^1(\mathcal{T}^*B/\Lambda)$. Moreover, the cohomology class does not depend on the choice of sections s_i.

Proof. We show that $\mu_{jk} + \mu_{ij} = \mu_{ik}$. We have that

$$\begin{aligned}\mu_{jk} + \mu_{ij} &= s_j - s_k + s_i - s_j \\ &= s_i - s_k\end{aligned}$$

and so the condition is immediate. Suppose we choose different sections s_i', s_j'. Since the T^*B/Λ action is transitive we have

$$s_i' = s_i + \sigma_i$$

We now compute

$$\begin{aligned}\mu_{ij}' - \mu_{ij} &= s_i' - s_j' - s_i + s_j \\ &= s_i + \sigma_i - s_j - \sigma_j - s_i + s_j \\ &= \sigma_i - \sigma_j\end{aligned}$$

and hence the difference is a Cech co-boundary and so is zero in cohomology as required. □

It is clear that if μ itself equals to zero, then we have a global section of $M \to B$. Indeed if this is the case, then our sections are literally equal on the overlaps and hence there is no problem gluing them together to obtain a global section. However, this condition turns out to be in some sense too strong. Indeed, if we loosen the condition to $[\mu] = 0 \in H^1(\mathcal{T}^*B/\Lambda)$, it turns out that we will be able to *redefine* our sections so that they equal on the overlap and hence glue together. The following Lemma proves this.

Lemma 5.11. Suppose that $[\mu] = 0 \in H^1(\mathcal{T}^*B/\Lambda)$. Then there exists a global section of $M \to B$.

Proof. If $[\mu] = 0$, then it follows that $[\mu]$ is a coboundary and so $\mu_{ij} = \sigma_i - \sigma_j$ where $\sigma_i : U_i \to T^*B/\Lambda$. We redefine our sections as

$$s_i' = s_i - \sigma_i$$

Now, on intersections we have that

$$\begin{aligned}s_i' - s_j' &= s_i - \sigma_i - s_j + \sigma_j \\ &= s_i - s_j - (\sigma_i - \sigma_j) \\ &= s_i - s_j - \mu_{ij} \\ &= 0\end{aligned}$$

and hence we have no problem gluing our new local sections into a global section of $M \to B$. □

Hence, the obstruction the the existence of a global section of $M \to B$ lies in the sheaf cohomology group $H^1(\mathcal{T}^*B/\Lambda)$. In practice, this group may not be so easy to calculate. We would hence like to relate this sheaf cohomology group to a nicer cohomology group. Namely, we will be able to show that there is an isomorphism $H^1(\mathcal{T}^*B/\Lambda) \cong H^2(\Lambda)$. The sheaf of sections of the period lattice Λ is a locally constant sheaf as there are local trivialisations of the period lattice such that it looks like $U \times \mathbb{Z}^n$. Locally constant sheaves are equivalent to objects called *local systems*.

Calculating the cohomology of these local systems turns out to be much nicer than calculating sheaf cohomology more generally. For example, it is possible to compute this cohomology using cellular methods. Moreover, in the case that the base manifold in question is an Eilenberg-MacLane $K(G, 1)$-space, we can if fact use group cohomology to calculate this cohomology group which we shall later see. For now, we shall look at how we can obtain this isomorphism.

5.3.1 The Sheaf Theory Approach

From our discussion in the previous section, the obstruction to $E \to B$ topologically identifying with T^*B/Λ lived in $H^1(\mathcal{T}^*B/\Lambda)$. However, we noted that this may not be easy to calculate in general and so we wished to relate it to a nicer cohomology group. We will see now how we can make use of some standard sheaf theory to do this. A reference for some of the standard sheaf cohomology results used here is [Bre97].

Lemma 5.12. The following diagram of sheaves is exact in all rows and columns.

$$\begin{tikzcd}
 & & 0 \arrow[d] & 0 \arrow[d] & \\
0 \arrow[r] & \Lambda \arrow[r, "i"] \arrow[d, equal] & \Lambda\mathcal{T}^*B \arrow[r, "\mathrm{pr}"] \arrow[d] & \Lambda\mathcal{T}^*B/\Lambda \arrow[r] \arrow[d, "j"] & 0 \\
0 \arrow[r] & \Lambda \arrow[r] & \mathcal{T}^*B \arrow[r, "\mathrm{pr}"] \arrow[d, "d"] & \mathcal{T}^*B/\Lambda \arrow[r] \arrow[d, "d"] & 0 \\
 & & \Lambda^2 B \arrow[r, equal] \arrow[d] & \Lambda^2 B \arrow[d] & \\
 & & 0 & 0 &
\end{tikzcd}$$

Proof. It is clear the the two horizontal sequences given by the inclusion and projection maps are exact. We show that

$$0 \to \Lambda\mathcal{T}^*B \to \mathcal{T}^*B \to \Lambda^2B \to 0$$

is exact. The map $\mathcal{T}^*B \to \mathcal{T}^*B$ is the inclusion map and so is injective. It is also clear that the differential map d is surjective onto the space of closed 2-forms by definition. The kernel of the differential d is the closed 1-forms on T^*B but these are given by Lagrangian sections. The other vertical sequence follows by the same argument and Remark 5.2. $\square$

Lemma 5.13. The following two sheaf cohomological facts hold;

1. $\mathcal{T}^*B$ is a fine sheaf so that $H^i(\mathcal{T}^*B) = 0$ for $i > 0$.
2. $H^i(\Lambda\mathcal{T}^*B) \cong H^{i+1}(B, \mathbb{R})$ for $i > 0$

Proof. (TODO: source for fine sheaf or else proof). Denoting by $C^\infty(B)$ the sheaf of smooth functions on B this is a fine sheaf. We have a short exact sequence

$$0 \to \mathbb{R} \to C^\infty(B) \to \Lambda\mathcal{T}^*B \to 0$$

where $\mathbb{R} \to C^\infty(B)$ is the inclusion and $C^\infty(B) \to \Lambda\mathcal{T}^*B$ is the given by the differential. Since $df = 0$ if and only if f is a constant function, this sequence is exact. Taking the associate long exact sequence in sheaf cohomology we obtain

$$\cdots \to H^i(B, \mathbb{R}) \to H^i(C^\infty(B)) \to H^i(\Lambda\mathcal{T}^*B) \to H^{i+1}(B, \mathbb{R}) \to \ldots$$

Now since $C^\infty(B)$ is a fine sheaf, for $i > 0$ we obtain $0 \to H^i(\Lambda\mathcal{T}^*B) \to H^{i+1}(B, \mathbb{R}) \to 0$ giving the required isomorphism. $\square$

5.3.2 The Chern Class

We can now show the isomorphism $H^1(T^*B/\Lambda) \cong H^2(\Lambda)$ which will lead to the definition of the Chern class.

Theorem 5.14. There is an isomorphism $\delta : H^1(T^*B/\Lambda) \to H^2(\Lambda)$.

Proof. We had the short exact sequence

$$0 \to \Lambda \to T^*B \to T^*B/\Lambda \to 0$$

from Lemma 5.12. Taking the associated long exact sequence in cohomology yields

$$\cdots \to H^i(\Lambda) \to H^i(T^*B) \to H^i(T^*B/\Lambda) \to H^{i+1}(\Lambda) \to \ldots$$

Again, by Lemma 5.12, we have that $H^i(T^*B) = 0$ for $i > 0$ which gives an isomorphism $H^i(T^*B/\Lambda) \cong H^{i+1}(\Lambda)$ for $i > 0$. $\square$

Definition 5.15 (Chern Class)**.** The image of $[\mu] \in H^1(T^*B/\Lambda)$ in $H^2(\Lambda)$ under δ is called the *Chern class* and we shall denote it by $[\mu_C]$.

The Chern class was defined by Duistermaat in [Dui80]. The vanishing of this class implies that the Lagrangian fibration $E \to B$ is topologically equivalent to $T^*B/\Lambda \to B$. We note that this already depends on the integral affine structure on B as this determines our period lattice Λ. It is a feature of the regularity of these Lagrangian torus fibrations that such a structure exists. Indeed, if we allow our fibration to have singular fibres, there is no longer a canonical symplectic reference bundle to compare our fibration $E \to B$ with. Indeed, in this case one must fix a so called *reference system* for which to compare our Lagrangian fibration to and then in a similar way one may define the Chern class of such a singular fibration. More details on the singular case can be found in [Zun01].

The above discussion shows that given a regular Lagrangian torus fibration, we can associate to it the invariant $[\mu_C] \in H^2(\Lambda)$. However, we have not yet answered whether the converse of this statement holds true - that is given a class $[\eta] \in H^2(\Lambda)$ can this be realised as the Chern class of some Lagrangian torus fibration?

To answer this question, we consider the short exact sequence

$$0 \to \Lambda \to \mathcal{Z}(T^*B) \to \mathcal{Z}(T^*B/\Lambda) \to 0$$

and take the long exact sequence in sheaf cohomology. This leads to the exact sequence

$$\dots \to H^1(\mathcal{Z}(T^*B/\Lambda)) \xrightarrow{\Delta} H^2(\Lambda) \xrightarrow{\mathcal{D}_{(B,\mathcal{A})}} H^2(\mathcal{Z}(T^*B)) \cong H^3(B,\mathbb{R}) \to \dots$$

The map $\mathcal{D}_{(B,\mathcal{A})}$ was considered in [DD87].

They showed that a class $[\eta] \in H^2(\Lambda)$ is realisable as the Chern class of a Lagrangian torus fibration if and only if $\mathcal{D}_{(B,\mathcal{A})}(\eta) = 0$. They proved this using methods from Poisson geometry. One can show that Lagrangian torus fibrations are the same thing as complete symplectic realisations of the Poisson zero structure on the base space of a manifold. In their paper, they considered more generally symplectic realisations of regular Poisson structures of which the zero Poisson structure is a special case. Indeed there is a cohomology class determined by the characteristic foliation $\mathcal{F}_\pi$ and they showed that under the map $\mathcal{D}_{(B,\mathcal{A})}$, the Chern class must map to this cohomology class. In the case of the Poisson zero structure, the characteristic foliation of $(B, \pi \equiv 0)$ consists of just the points of B and this cohomology class will vanish. For more information on Poisson geometry see the following sources [CFM21].

5.3.3 The Lagrangian Class

This answers the question of which $[\eta] \in H^2(\Lambda)$ can arise as the Chern class of some Lagrangian fibration. However, this invariant is only topologically sharp. Once we have fixed an integral affine manifold B and a Chern class $[\mu_C] \in H^2(\Lambda)$, we can have Lagrangian fibrations over B with the same Chern class $[\mu_C]$ but which are not Lagrangian equivalent to each other. If we recall how the Chern class was constructed this fact should not be surprising. Indeed, we did not take into account the symplectic structure of our Lagrangian fibration in the construction of our Chern class since we allowed the sections of T^*B/Λ to be arbitrary so it would be too much to hope for that we would achieve a symplectic classification. We can fix this problem by choosing our sections to be sections of $\mathcal{Z}(T^*B/\Lambda)$ as we shall now see.

Definition 5.16. Let $M \to B$ be a Lagrangian fibration with period lattice Λ. Let $\{U_i\}_{i\in\mathbb{N}}$ be a trivializing cover and let $s_i : U_i \to \mathcal{Z}(T^*B/\Lambda)$. Then $\mu_{ij} = s_i - s_j$ defines a one-cocycle independent of the choice of section and hence a cohomology class $[\mu] \in H^1(\mathcal{Z}(T^*B/\Lambda))$. The class $[\mu]$ is called the *Lagrangian class* of $M \to B$. [Zun01]

If the class $[\mu] = 0$ the we can obtain a global *Lagrangian* section of $M \to B$ and hence we obtain that $M \to B$ is Lagrangian equivalent to $T^*B/\Lambda \to B$. In fact, the following proposition shows that every element of this cohomology group will be the Lagrangian class of some Lagrangian torus fibration.

Proposition 5.17. Every class $[\mu] \in H^1(\mathcal{Z}(T^*B/\Lambda))$ can be realised as the Lagrangian class of some Lagrangian torus fibration.

Proof. Suppose we are given some $[\mu] \in \mathcal{Z}(T^*B/\Lambda)$. Then picking a good cover $\{U_i\}_{i\in\mathbb{N}}$ of B we may assume $[\mu]$ can be represented by a one-cocycle $\alpha_{ij} : U_i \cap U_j \to \mathcal{Z}(T^*B/\Lambda)$. Now, each α_{ij} determines a fibrewise affine

$$s_{\alpha_{ij}} : T^*(U_i \cap U_j)/\Lambda \to T^*(U_i \cap U_j)/\Lambda$$

$$(b, m) \mapsto (b, m + \alpha_{ij}(b))$$

where the additive action of α_{ij} comes from the action of T^*B on $M \to B$. On triples $U_i \cap U_j \cap U_j$ we require that transitionaiing a point in the fibre by $s_{\alpha_{ij}} \circ s_{\alpha_{jk}} = s_{\alpha_{ik}}$ but this holds precisely because α_{ij} satisfies the cocyle condition. By Proposition 3.16, the $s_{\alpha_{ij}}$ is a symplectomorphism of the Lagrangian fibration $T^*(U_i \cap U_j)/\Lambda$ as α is closed. Hence, the transition maps are symplectomorphisms and so our local forms glue together to form a global symplectic form on $M = \bigsqcup_i T^*U_i/\Lambda$ and this becomes a Lagrangian fibration as locally in each chart U_i it is by construction a Lagrangian fibration. $\square$

Remark 5.18. In fact, taking the long exact sequence in cohomology of the first row of Lemma 5.12, we obtain exactness at $H^1(\mathcal{Z}(T^*B/\Lambda)) \xrightarrow{\Delta} H^2(\Lambda)$ and hence every class $\Delta([\mu]) \in H^2(\Lambda)$ lies in the kernel of $\mathcal{D}_{(B,\mathcal{A})}$ the Dazord-Delzant homomorphism and hence can be realised as a Lagrangian torus fibration. However, this alone tells us only that $\Delta([\mu])$ arises as the Chern class of a Lagrangian torus fibration.

We seen earlier that an element $[\mu] \in H^2(\Lambda)$ is the Chern class of a Lagrangian torus fibration if and only if it vanished under the Delzant-Dazord homomorphism. In fact, If we consider the case of surfaces we have the following result.

Proposition 5.19. Let M be an integral affine surface. Then every class $[\mu] \in H^2(\Lambda)$ can be realised as the Chern class of a Lagrangian torus fibration.

Proof. From the short exact sequence

$$0 \to \Lambda \to \mathcal{Z}(T^*B) \to \mathcal{Z}(T^*B/\Lambda) \to 0$$

we obtain the long exact sequence in cohomology

$$\begin{aligned} H^1(\Lambda) \to H^1(\mathcal{Z}(T^*B)) \cong H^2(M, \mathbb{R}) \xrightarrow{d} H^1(\mathcal{Z}(T^*B/\Lambda)) \xrightarrow{\Delta} H^2(\Lambda) \\ \to H^1(\mathcal{Z}(T^*B)) \cong H^3(M, \mathbb{R}) = 0 \end{aligned}$$

where the isomorphisms $H^i(\mathcal{Z}(T^*B)) \cong H^{i+1}(M, \mathbb{R})$ are from Lemma 5.12. By exactness this yields that $H^2(\Lambda) \cong \dfrac{H^1(\mathcal{Z}(T^*B/\Lambda))}{\ker \Delta}$. By Proposition 5.17, every element of $\dfrac{H^1(\mathcal{Z}(T^*B/\Lambda))}{\ker \Delta}$ arises as the Lagrangian class of a Lagrangian torus fibration and hence as this group is isomorphic to $H^2(\Lambda)$ the result follows. $\square$

Proposition 5.20. The group $\ker \Delta = \dfrac{H^2(B, R)}{\text{Im } H^1(\Lambda)}$.

Proof. By exactness $\ker \Delta = \text{Im } d$. We have that $d(H^2(M, \mathbb{R})) \cong \dfrac{H^2(M, \mathbb{R})}{\ker d}$. There is a map $j : H^1(\Lambda) \to H^2(B, \mathbb{R})$ as in the l.e.s. of Proposition 5.17 and by exactness $\ker d = j(H^1(\Lambda))$ and this gives the required result. $\square$

Remark 5.21. Once we have fixed a Chern class we can ask which Lagrangian classes map to this Chern class? The group $\ker \Delta = \dfrac{H^2(B, R)}{\text{Im } H^1(\Lambda)}$ paramaterizes the Lagrangian class $[\mu]$ which map to a fixed Chern class $[\mu_C]$.

Proposition 5.22. The group Im $H^1(\Lambda) \subset H^2(B, \mathbb{R})$ is generated by the cohomological invariants of of Definition 5.4.

Proof. Applying the long exact sequence in cohomology to the second row of Lemma 5.12, we obtain that

$$H^0(T^*B/\Lambda) \xrightarrow{\delta} H^1(\Lambda) \to H^1(T^*B) = 0$$

and so δ is surjective. We therefore have that Im $H^1(\Lambda) \subset H^2(B, R)$ is the same as Im $H^0(T^*B/\Lambda) \subset H^2(\Lambda)$ under the image of δ composed with the map $j : H^1(\Lambda) \to H^2(B, \mathbb{R})$ from Proposition 5.20.

It remains to show that $j \circ \delta$ coincides with the differential $d : H^0(T^*B/\Lambda) \to H^2(B, \mathbb{R})$ since by definition $H^0(\mathcal{Z}(T^*B/\Lambda))$ is the space of latticed one-forms. This follows from another version of $H^1(\mathcal{Z}(T^*B/\Lambda)) \cong H^2(B, \mathbb{R})$. Again, applying the long exact sequence in cohomology to the short exact sequence

$$0 \to \mathcal{Z}(T^*B) \to T^*B \to \Lambda^2 B \to 0$$

yields the long exact sequence

$$0 \to H^0(\mathcal{Z}(T^*B)) \to H^0(T^*B) \xrightarrow{d} H^0(\Lambda^2 B) \to H^1(\mathcal{Z}(T^*B)) \to 0$$

since T^*B is a fine sheaf. This gives that

$$H^1(\mathcal{Z}(T^*B)) \cong \frac{H^0(\Lambda^2 B)}{dH^0(T^*B)}$$

and this is the de Rham cohomology group $H^2(B, \mathbb{R})$ by definition.

Now, we have the following short exact sequences

$$0 \longrightarrow \mathcal{Z}(T^*B/\Lambda) \longrightarrow T^*B/\Lambda \xrightarrow{d} \Lambda^2 B \longrightarrow 0$$

from the long exact sequence in cohomology, there is a map $d : H^0(T^*B/\Lambda) \to H^0(\Lambda^2 B)$. The required result now follows from commutativity of the following square.

$$\begin{tikzcd} H^0(T^*B/\Lambda) \arrow[r, "d"] \arrow[d, "\delta"] & H^0(\Lambda^2 B) \arrow[d, "\pi"] \\ H^1(\Lambda) \arrow[r, "j"] & H^2(B, R) \end{tikzcd}$$

where $\pi \circ d$ is the required differential. (TODO: This proof is a bit of a mess.) ☐

The following theorem summarizes the main results of this section.

Theorem 5.23. Let B be a smooth manifold. The the Lagrangian fibrations over B are determined by

1. The integral affine structure on B. This determines a symplectic reference bundle T^*B/Λ.
2. Fixing the integral affine structure on B, the Lagrangian fibrations over B are topologically determined by a Chern class $[\mu_C] \in H^2(\Lambda)$. The Chern class determines the topological type of the Lagrangian torus fibration and its vanishing implies the fibration is topologically equibvalent to T^*B/Λ. An element $[\mu_C] \in H^2(\Lambda)$ is the Chern class of a Lagrangian torus fibration if and only if $\mathcal{D}_{(B,\mathcal{A})}([\mu_C]) = 0$ where $\mathcal{D}_{(B,\mathcal{A})}$ is the Delzant-Dazord homomorphism.
3. The obstruction to our bundle being symplectically equivalent to T^*B/Λ is given by the Lagrangian class $[\mu] \in H^1(\mathcal{Z}(T^*B/\Lambda))$. Every Lagrangian class arises as the Lagrangian class of some Lagrangian torus fibration

5.4 Computing the Chern Class

After the discussion in the previous section, our strategy will be to compute all of the possible Chern classes for our integral affine closed surfaces. Once we have fixed a Chern class, we can then look at the possible Lagrangian classes that map to it. A lot of the complexity in this process will be in the calculation of the sheaf cohomology groups. Fortunately, since the sheaf of sections of the period lattice Λ is a locally constant sheaf, its sheaf cohomology will be equal to the cohomology of the so called *local system* it determines.

We shall now recall some basic facts and results around the theory of local systems

5.4.1 Local Systems

Recall that a constant sheaf $\mathcal{F}$ over a topological space X, is a sheaf such that for each open set $U \subset X$ we have that $\mathcal{F}(U) = R$ for some fixed group/ring/module R.

Definition 5.24 (Local system)**.** A local system on a topological space X, is a locally constant sheaf $\mathcal{F}$ - e.g. for each $x \in X$, there is a neighborhood $U \subset X$ such that for all $V \subseteq U$ we have that $\mathcal{F}|_V$ is a constant sheaf for some fixed group/ring/module R.

A key feature of these local systems is that if our topological space is nice enough, for example a manifold M, then the local systems over M are in bijection with representations of the fundamental group. In fact, an even stronger condition than this holds. It turns out that this bijection can be upgraded to an equivalence of categories and hence the study of local systems over these topological spaces is equivalent to studying the representations of its fundamental group.

Indeed, given a local system $\mathcal{F}$ over M with fixed module a finite dimensional vector space V, one can associate an "algebraic" monodromy to it. Namely, given $[\gamma] \in \pi_1(M, x)$ we can associate an automorphism of the stalk $\mathcal{F}_x \cong V$. In this way we obtain a monodromy map

$$\rho : \pi_1(M, x) \to \mathrm{Aut}(\mathcal{F}_x)$$

and this is how we associate a representation of the fundamental group to the local system.

Conversely, given a representation $\rho : \pi_1(M, x) \to \mathrm{Aut}(V)$ we can associate to it a locally constant sheaf as follows. Let $\pi : \tilde{M} \to M$ be the universal cover of M and denote by $V_{\tilde{M}}$ the constant sheaf on $\tilde{M}$. Then we define the sections $\mathcal{F}(U)$ to be sections of $V_{\tilde{M}}$ over $\pi^{-1}(U)$ that are equivariant with respect to the action of the fundamental group $\pi_1(M, x)$ which acts on $\tilde{M}$ by deck transformations. This determines a locally constant sheaf as if we pick a small enough neighborhood $U \subset M$ that is evenly covered, then the sheaf is constant on U as $\pi|_{U_\gamma}$ is a homeomorphism and $\pi \circ \gamma = \pi$ where γ is a deck transformation. For more details see [ES09].

5.4.2 Cohomology of Local Sytems

We could calculate the cohomology of a local system by calculating its sheaf cohomology directly. However, there turns out to be a nicer way to calculate the cohomology in this case which is amenable to cellular methods.

In fact, the equivalence between local systems and representations of the fundamental group leads to two different ways of calculating the cohomology of the local system. We now briefly illustrate how to calculate through the two different perspectives.

The singular cohomology approach

The first approach we take is the singular cohomology approach. Given a representation $\rho : \pi_1(M) \to \mathrm{Aut}(V)$ our construction of a local system in fact give a vector bundle E with fibre V by

$$E := (\tilde{M} \times V)/\pi_1(M)$$

where $\gamma \cdot (\tilde{m}, v) = (\gamma \cdot \tilde{x}, \rho(\gamma)v)$ see [DK01]. Let $\sigma : \Delta^n \to X$ be a singular k-simplex in X and denote by $e_0 = (1, 0, \dots, 0)$ the base point of σ and $E_{\sigma(e_0)}$ the fibre above $\sigma(e_0) \in X$. We let $S^k(M; E)$ the set of formal sums of functions f that assign to each singular simplex $\sigma : \Delta^k \to X$, an element $f(\sigma) \in E_{\sigma(e_0)}$. Then $S^k(M; E)$ is an abelian group and we can associate a coboundary operator to it by

$$\delta : S^k(X; E) \to S^{k+1}(X; E)$$

given by

$$\delta(f)(\sigma) = (-1)^k \left(\gamma_\sigma^{-1}(f(\partial_0 \sigma)) + \sum_{i=1}^{k+1} (-1)^i c(\partial_i \sigma) \right)$$

where the ∂_i are the usual boundary maps for singular simplices and

$$\gamma_\sigma : E_{\sigma(0,1,0,\dots,0)} \to E_{\sigma(1,0,0,\dots,0)}$$

is an isomorphism induced by the path

$$\tilde{\gamma}_\sigma : [0, 1] \to X$$

$$t \mapsto \sigma(t, 1 - t, 0 \dots, 0)$$

via path lifting as $\rho : E \to X$ is a covering space.

A computation shows that $\delta^2 = 0$ and we will denote the corresponding cohomology by $H^*_{\mathrm{sing}}(M, V)$ where it is understood that V is viewed as a $\pi_1(M)$-module through the action of the representation.

Definition 5.25. Let $\rho : \pi_1(M) \to \mathrm{Aut}(V)$ determine a local system. The *singular cohomology of the local system* is the cohomology of the complex $(S^K(X; E), \delta)$

The equivariant or Borel cohomology approach

Equivariant or Borel cohomology was introduced by Borel in [Bor60] and encodes information about group actions on a topological space X. It has connections to Lie groupoid cohomology and has found applications throughout algebraic geometry and differential geometry (TODO: probably need to refer to some references here). The description that we now give of the cohomology of a local system is a special case of this equivariant cohomology.

We have that there is a right action of the fundamental group by deck transformations on the universal cover $\tilde{M}$ of our manifold M. Denoting by $C^*(\tilde{M})$ the free abelian group of simplices $\sigma : \Delta^n \to \tilde{M}$, this action turns $C^*(\tilde{M})$ into a right $\pi_1(M)$-module. Namely, given a simplex and $\gamma \in \pi_1(M)$ we can define this right action as $\sigma \cdot \gamma = \gamma \star \sigma$ where $\gamma \star \sigma$ is the action of γ by deck transformations and extend to the group ring by linearity. Now, we have that the fixed vector space of our local system V is a right $\pi_1(M)$-module via the representation $\rho : \pi_1(M) \to \mathrm{Aut}(V)$. We would like to apply $\mathrm{Hom}_{\mathbb{Z}[\pi_1(M)]}(-, V)$ to our chain complex of right $\mathbb{Z}[\pi_1(M)]$-modules $C^*(\tilde{M})$ but we cannot do so as $\mathrm{Hom}_{\mathbb{Z}[\pi_1(M)]}(-,-)$ must take in either two right modules or two left modules. However, we can solve this problem by turning $C^*(\tilde{M})$ into a left $\mathbb{Z}[\pi_1(M)]$-module by defining the action of $g \in \mathbb{Z}[\pi_1(M)]$ to be $g \cdot z = z \cdot g^{-1}$. Applying $\mathrm{Hom}_{\mathbb{Z}[\pi_1(M)]}(-, V)$ yields a co-complex.

Definition 5.26. The cohomology of this complex is called the *cohomology of M with twisted coefficients* and is the same as the equivariant cohomology of the universal cover $\tilde{M}$. Fixing a representation $\rho : \pi_1(M) \to \mathrm{Aut}(V)$ we shall denote this cohomology as $H^i(M, V_\rho)$.

The two approaches described can be seen to be equivalent. We shall not give the proof of this but it can be found in Theorem 3.5 of [Whi78].

5.4.3 Relation to group cohomology

There is yet another approach to be taken in this case that our manifold is a "nice" space by which we mean it is an Eilenber Mac-Lane space of type $K(G, 1)$ as of Definition 1.20.

A local system is determined by a representation of the fundamental group on an abelian group V and hence, one might speculate that the group cohomology of $H^i(\pi_1(M), V)$ where we view V as a $\pi_1(M)$-module is the same as the cohomology of the local system determined by it. However, this is not the case in general although there are relationships between the two. The following lemma is a special case of when the two coincide.

Lemma 5.27. Let M be an Eilenberg-MacLane space of type $K(G, 1)$ and let $\pi_1(M) \to \mathrm{Aut}(V)$ be a representation. Then the group cohomology $H^i(\pi_1(M), V)$ and the cohomology of the local system coincide.

Proof. I will give the proof in the case of M being a CW-complex although this is not strictly needed [Whi78]. For $n \geq 2$ we have that S^n is simply connected and so given a map $f : S^n \to M$ there is a unique lift to the universal cover.

$$\begin{tikzcd} & \widetilde{M} \arrow[d, "\pi"] \\ S^n \arrow[ur, dashed, "\widetilde{f}"] \arrow[r, "f"] & M \end{tikzcd}$$

and hence we get a map

$$\varphi : \pi_n(M) \to \pi_n(\tilde{M})$$
$$[f] \mapsto [\tilde{f}]$$

The covering map also gives a map in the other direction and these maps are easily seen to be inverses of one another. This leads to an isomorphism $\pi_n(M) \cong \pi_n(\tilde{M})$ for $n \geq 2$.

Hence, if M is an Eilenberg-MacLane space of type $K(G,1)$ then its universal cover has $\pi_n(\tilde{M}) = 0$ for all $n \in \mathbb{N}$. Now if we consider the chain complex

$$\cdots \to C^2(\tilde{M}) \to C^1(\tilde{M}) \to C^0(\tilde{M}) \xrightarrow{\epsilon} \mathbb{Z} \to 0$$

where ϵ is the augmentation map. By Whiteheads lemma (Corollary 7.34 [DK01]), $\tilde{M}$ is homotopy equivalent to a point and so $H^i(C^*(M)) = 0$ for $i > 0$ and $H^0(C^*(M)) = 0$. Hence the above sequence is exact and provides a free resolution of $\mathbb{Z}$ by $\pi_1(M)$-modules hence computing the group cohomology after applying the $\mathrm{Hom}_{\mathbb{Z}[\pi_1(M)]}(-, V)$ functor. □

Remark 5.28. If M is a $K(G,1)$ space and path connected, then M is often called *aspherical*.

We now make the following observation which we shall use when computing the Chern class of a Lagrangian torus fibration over a complete integral affine manifold.

Lemma 5.29. Let M be a complete affine manifold. Then M is aspherical and hence the cohomology of the local system dtermined by its linear holonomy representation, is equal to the group cohomology of $\pi_1(M)$ with respect to the linear holonomy representation.

Proof. Since M is complete, the developing map $\mathrm{dev} : \tilde{M} \to \mathbb{R}^n$ is a diffeomorphism. Since $\mathbb{R}^n$ is contractible it follows that $\tilde{M}$ is also and hence by Lemma 5.27, the group cohomology and the cohomology of the local system coincide. □

In the case of the Chern class, this lived in $H^2(\Lambda)$ where Λ is the locally constant sheaf of sections of the period lattice. Our previous discussion shows that in the case of a complete integral affine manifold with linear holonomy representation $Lin \circ hol : pi_1(M) \to \mathrm{Aut}(\mathbb{Z}^n)$, the sheaf cohomology of Λ coincides with the group cohomology of $\pi_1(M)$ with respect to this linear holonomy representation once one verifies that these local systems do indeed coincide. In group cohomology, the group H^2 classifies objects called *group extensions*. We quickly recap these now.

Definition 5.30. A *group extension* is a short exact sequence of groups

$$0 \to A \to E \xrightarrow{\pi} G \to 1$$

where A is abelian.

There is a notion of equivalence for group extensions.

Definition 5.31. An equivalence of extensions is a group isomorphism $\varphi : E \to E'$ such that the following diagram commutes:

$$\begin{tikzcd} 0 \arrow[r] & A \arrow[r] \arrow[d, equal] & E \arrow[r, "\pi"] \arrow[d, "\varphi"] & G \arrow[r] \arrow[d, equal] & 1 \\ 0 \arrow[r] & A \arrow[r] & E' \arrow[r, "\pi'"] & G \arrow[r] & 1 \end{tikzcd}$$

Given a group extension of G by A then there is a unique G-module structure induced on A. Equivalent group extensions induce the same G-module structure on A. Fixing a G-module A, there may be multiple extensions that induce the given G-module structure. We have the following classification result.

Theorem 5.32 (TODO: Cite GT notes)**.** Fix a G-module A. Let $\epsilon(G, A)$ denote the set of extensions which induce the given G-module structure on A. Then we have that there is a bijection

$$H^2(G, A) \Leftrightarrow \epsilon(G, A)$$

This establishes the following perspective on the Chern classes of a Lagrangian fibration over a closed integral affine surface.

Proposition 5.33. Let M be a closed integral affine surface. Then there is a one-to-one correspondence between Lagrangian fibrations over M with a given Chern class and isomorphism classes of group extensions

$$0 \to \mathbb{Z}^n \to E \to \pi_1(M) \to 1$$

Proof. This follows from the fact that every Chern class of an integral affine closed surface is realisable as the Chern class of a Lagrangian fibration and the bijection established in the previous theorem. □

Remark 5.34. Under this correspondence, the symplectic reference fibration correspondes to the group extension class of

$$0 \to \mathbb{Z}^n \to \mathbb{Z}^n \rtimes \pi_1(M) \to \pi_1(M) \to 1$$

What remains to show in this section, is that calculating the cohomology of the local system in the sense of the two different definitions given, is equivalent to calculating the sheaf cohomology of the locally constant sheaf. We will not give the proof but we will outline where one can find the required equivalences.

Theorem 5.35. Let M a manifold and let $\mathcal{F}$ be a local system over M with fixed abelian group V. Then the following group cohomologies are isomorphic.

1. The sheaf cohomology of the locally constant sheaf $\mathcal{F}$ over M.
2. The singular cohomology of the local system as in Definition 5.25.
3. The cohomology of M with twisted coefficients as in Definition 5.26.
4. The Cech cohomology of the locally constant sheaf $\mathcal{F}$

Proof. For 1) $\Leftrightarrow$ 2) see Theorem 1.1 in Chapter 3 of [Bre97]. For 2) $\Leftrightarrow$ 3) see Theorem 3.4 of [Whi78]. For 4) $\Leftrightarrow$ 1) see Theorem 4.13 of [Bre97]. $\square$

We also note the following important fact about the homotopy invariance of the cohomology of locally constant sheaves.

Theorem 5.36 (Theorem 21.5. [Dah20])**.** Let $f : M \to N$ be a homotopy equivalence. Then for every locally constant sheaf $\mathcal{F}$ on Y, we have that

$$H^*(Y, \mathcal{F}) \cong H^*(X, f^{-1}\mathcal{F})$$

5.4.4 Classifying Lagrangian fibrations over S^1

Before we give the classification of regular Lagrangian torus fibrations over compact surfaces, we can first write sown the classification over S^1.

Lemma 5.37. The Lagrangian fibrations over S^1 are determined by a positive number $y > 0$.

Proof. Since S^1 has dimension one $H^2(\Lambda) = 0$ for any period lattice $\Lambda \subset T^*S^1$. One way to see this is by noting that $\pi_1(S^1) = \mathbb{Z}$ and the group cohomology of $\mathbb{Z}$ vanishes for $i > 1$. Hence, every Lagrangian fibration admits a global section, and since the de Rham cohomology $H^2(S^1) = 0$ there are no other considerations. Hence, the Lagrangian fibrations over S^1 are all topologically trivial, admit a global section and are of the form $T^*S^1/\Lambda_y \to S^1$ where Λ_y is the lattice determined by $y > 0$. They are classified by the integral affine structure on S^1 and by Lemma 4.1 are therefore determined by a number $y > 0$. $\square$

Remark 5.38. An integral affine structure on S^1 is equivalent to a full rank lattice in $T^*S^1 = S^1 \times \mathbb{R}$ which is just determined by multiples of some number $y \in \mathbb{R}$. This determines the symplectic volume of the space T^*B/Λ_y. Different choices of y change the symplectic volume of T^*B/Λ_y and hence these bundles cannot be symplectomorphic for different choices of y.

5.4.5 Computing $H^2(\Lambda)$ for the torus

We now use the group cohomology approach in order to calculate $H^2(\Lambda)$ for our compact integral affine surfaces. For $\mathbb{T}^2$, we have that $\pi_1(\mathbb{T}^2) = \mathbb{Z} \oplus \mathbb{Z}$. The group ring for $G = \mathbb{Z} \oplus \mathbb{Z}$ is given by $\mathbb{Z}[G] = \mathbb{Z}[x, x^{-1}, y, y^{-1}]$ where x and y are the generators for $\mathbb{Z} \oplus \mathbb{Z}$.

We will make use of the Koszul resolution which provides us with a nice free resolution of $\mathbb{Z}$ in this case.

Definition 5.39. Let R^n be a free R-module with basis $e_1, \dots, e_n$. The k-th exterior power $\Lambda^k(R^n)$ is the free R-module generated by

$$e_{i_1} \wedge \cdots \wedge e_{i_k}, \qquad 1 \le i_1 < \cdots < i_k \le n,$$

where the wedge product is alternating:

$$e_i \wedge e_j = -\, e_j \wedge e_i$$

Lemma 5.40 (Koszul resolution [Wei94])**.** If $\mathbf{x}= (x_1, \ldots, x_n)$ is a regular sequence in R, then the following sequence is exact:

$$0 \longrightarrow \Lambda^n(R^n) \longrightarrow \cdots \longrightarrow \Lambda^2(R^n) \longrightarrow R^n \xrightarrow{x} R \longrightarrow R/I \longrightarrow 0.$$

We have that $(x-1, y-1)$ is a regular sequence in $\mathbb{Z}[G]$ as $x-1$ is a non-zero divisor in $\mathbb{Z}[G]$ and $y-1$ is a non-zero divisor in $\mathbb{Z}[G]/(x-1) = \mathbb{Z}[y, y^{-1}]$. Since $Z[x, x^{-1}, y, y^{-1}]/\langle x-1, y-1\rangle \cong \mathbb{Z}$, the Koszul complex yields the following exact sequence

$$0 \mapsto \Lambda^2(\mathbb{Z}[G]^2) \xrightarrow{d_2} \mathbb{Z}[G]^2 \xrightarrow{d_1} \mathbb{Z}[G] \to \mathbb{Z} \to 0$$

where $\Lambda^2(\mathbb{Z}[G]^2) = \mathbb{Z}[G]$ and

$$\begin{aligned} d_1(a, b) &= (x-1)a + (y-1)b, \\ d_2(c) &= ((y-1)c, -(x-1)c) \end{aligned}$$

We apply $Hom_{\mathbb{Z}[G]}(-, \mathbb{Z}^2)$ to the deleted resolution which yields

$$0 \mapsto \mathbb{Z}^2 \xrightarrow{\tilde{d_1}} \mathbb{Z}^2 \oplus \mathbb{Z}^2 \xrightarrow{\tilde{d_2}} \mathbb{Z}^2 \to 0$$

This immediately yields the following.

Lemma 5.41. Let $G = \mathbb{Z}^2 = \langle x, y\rangle$ and let $\mathbb{Z}^2$ be a G-module. then the second group cohomology group is

$$H^2(G, \mathbb{Z}^2) \cong \frac{\mathbb{Z}^2}{(x-1)\mathbb{Z}^2 + (y-1)\mathbb{Z}^2}$$

Proof. This follows from the above resolution.

$$im(\tilde{d_2}) = -(x-1)\mathbb{Z}^2 + (y-1)\mathbb{Z}^2$$

and the kernel of $\mathbb{Z}^2 \to 0$ is all of $\mathbb{Z}^2$. ☐

For the first family $\mathcal{A}_{a,b,c,d}$, the linear holonomy induces the trivial representation of G on $\mathbb{Z}^2$. We therefore have the following.

Lemma 5.42. For $\mathcal{A}_{a,b,c,d}$, the cohomology group $H^2(G, \mathbb{Z}^2) = \mathbb{Z}^2$.

Proof. Both $(x-1)\mathbb{Z} = 0$ and $(y-1)\mathbb{Z} = 0$ as x and y act trivially. Therefore,

$$\begin{aligned} H^2(G, \mathbb{Z}^2) &= \frac{\mathbb{Z}^2}{(x-1)\mathbb{Z}^2 + (y-1)\mathbb{Z}^2} \\ &= \mathbb{Z}^2 \end{aligned}$$

☐

Lemma 5.43. For $\mathcal{B}_{n,a,b}$, the cohomology group $H^2(G, \mathbb{Z}^2) = \dfrac{\mathbb{Z}}{n\mathbb{Z}} \oplus \mathbb{Z}$.

Proof. This amounts to computing the action of $(x-1)$ and $(y-1)$ on $\mathbb{Z}^2$ induced by the linear holonomy representation. From our classification in Theorem 4.14 we have that x acts trivially and y acts by $\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$. We calculate

$$y \cdot (l, m) - (l, m) = (nl, 0)$$

and so $(y-1)\mathbb{Z}^2 = n\mathbb{Z} \oplus 0$. This gives

$$\begin{aligned} H^2(G, \mathbb{Z}^2) &= \frac{\mathbb{Z}^2}{n\mathbb{Z} \oplus 0} \\ &= \frac{\mathbb{Z}}{n\mathbb{Z}} \oplus \mathbb{Z} \end{aligned}$$

□

Since $H^3(\mathbb{T}^2, \mathbb{R}) = 0$, it follows from Proposition 5.19 that every element in $H^2(\Lambda)$ is the Chern class of some Lagrangian fibration.

5.5 Computing $H^2(\Lambda)$ for the Klein bottle

5.5.1 Spectral Sequences

The fundamental group of the Klein bottle is $\pi_1(K^2) = \mathbb{Z} \rtimes \mathbb{Z}$. In this case, there is a nice spectral sequence that reduces computation of this semi-direct product, to the computation of group cohomologies of $\mathbb{Z}$ with respect to a representation on a certain first cohomology group. We will give a quick outline of spectral sequences with a focus on how to use them to calculate this group cohomology. We will closely follow [McC01].

Consider generally, a cohomolgy theory and set $H^* = \bigoplus_i H^i$. The goal of a spectral sequence is to approximate the graded object H^* or in some nice cases, to directly compute this. We make the assumption that H^* has a filtration

$$H^* = F^0 H^* \supseteq \cdots \supseteq F^n H^* \supseteq F^{n+1} H^* \supseteq \cdots \supseteq \{0\}$$

and for now we will assume that H^* is a graded vector space for simplicity. Given a filtration F^* of H^* it can be collapsed into another graded vector space called *the associated graded vector spcae* defined by

$$E_0^p(H^*) = F^p H^* / F^{p+1} H^*$$

In the case that H^* is a vector space we can recover H^* up to isomorphism by

$$H^* \cong \bigoplus_{p=0}^{\infty} E_0^p(H^*) = \bigoplus_{p=0}^{\infty} F^p H^* / F^{p+1} H^*$$

However if H^* is not a graded vector space there can be extension problems. For example, although we may know the graded pieces making up H^* we may not know how they piece together to give H^* a module structure. That this holds for vector spaces is due to the fact that every short exact sequence of vector spaces splits wehereas this is not true for modules.

For example, if we know only the graded pieces $E_0^p(H^*) = F^pH^*/F^{p+1}H^*$ and say we know $F^{p+1}H^*$ and want to determine $F^{p+1}H^*$, then there is a s.e.s

$$0 \to F^{p+1}H^* \to F^pH^* \to E_0^p(H^*) = F^pH^*/F^{p+1}H^* \to 0$$

and so, in general, F^pH^* can be any extension of $E_0^p(H^*)$ by $F^{p+1}H^*$. These are all split in the vector space case but more generally this is not true and so to determine $F^0H^* = H^*$ we must solve all of these extension problems. I will not discuss this extension problem in much more detail as in our computations we will have only one graded piece to consider and hence there is no extension problem.

Now, the idea is that H^* may not be easily computed but we can try to approximate it by the associated graded vector space to some filtration F^* of H^*. We now observe that $E_0^*(H^*)$ is bigraded in the following sense. If we define $F^pH^r = F^pH^* \cap H^r$ then we have that

$$E_0^{p,q} = \frac{F^pH^{p+q}}{F^{p+1}H^{p+q}}$$

The index q is called *the complementary degree* and the index p is called the *filtration degree*.

We note that in the case of vector spaces we can recover H^r as the direct sum of the pieces of $E_0^{p,q}$ of total degree r e.g. where $p+q=r$. We can now give the definition of a spectral sequence.

Definition 5.44. A first quadrant spectral sequence is a sequence of differential bigraded vector spaces. For every $r \in \mathbb{N}$ and $p, q > 0$ we have a vector space $E_r^{p,q}$ equipped with a linear map

$$d_r : E_r^{p,q} \to E^{p+r,q-r+1}$$

with $d_r^2 = 0$ a differential of bidegree $(r, 1-r)$. Furthermore, for all $rgeq1$ we have that $E_{r+1}^{*,*} = H(E_r^{*,*}, d_r)$ where $H(E_r^{*,*}, d_r)$ denotes taking the cohomology at the $E_r^{*,*}$ spot.

Remark 5.45. We call $E_r^{*,*}$ the E_r-page. Turning a page e.g. going from the E_r-page to the E_{r+1}-page corresponds to taking cohomology. The pages E_r may be pictured as a lattice with each point a vector space and arrows between vector spaces corresponding to the differential d_r.

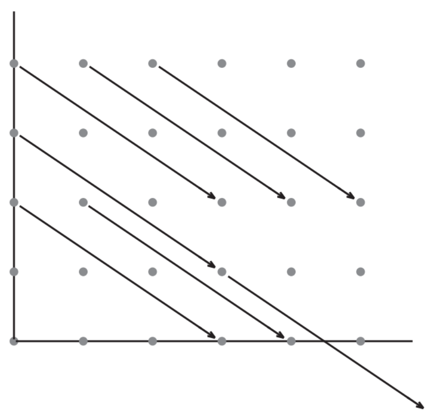

Figure 5.1: A page E_r in a spectral sequence. Arrows mapping of the page correspond to zero mappings. [McC01]

We can now say what it means for a spectral sequence to converge to a graded vector space.

Definition 5.46. A spectral sequence $\{E_r^{*,*}, d_r\}$ of vector spaces is said to converge to a graded vector space H^* if H^* has a filtration F^* and

$$E_\infty^{p,q} \cong F^p H^{p+q}/F^{p+1}H^{p_q} = E_0^{p,q}(H^*)$$

There are still some issues around uniqueness of the convergence that we shall not discuss. We also formulated our definitions in terms of graded vector spaces as oppose to more generally graded R-modules. The theory extends to these in a very similar fashen except there are now more issues with reconstructing our graded module H^* as for example there may be multiple extensions of a R-module A by another R-module e.g. these are not all split as in the vector space case.

We now give the spectral sequence that we shall use in order to comput the group cohomology of $\pi_1(K) = \mathbb{Z} \rtimes \mathbb{Z}$.

Lemma 5.47 (Hochschild-Lydon-Serre Spectral Sequence [HS53])**.** Let G be a group and $N \subseteq G$ a normal subgroup and let A be a G-module. Then there is a spectral sequence

$$E_2^{p,q} = H^p(G/N, H^q(N, A)) \Rightarrow H^{p+q}(G, A)$$

For the fundamental group of the Klein bottle, we have the short exact sequence

$$0 \to \mathbb{Z} \to \mathbb{Z} \rtimes \mathbb{Z} \to \mathbb{Z} \to 0$$

and so $\mathbb{Z}$ sits inside $\mathbb{Z} \rtimes \mathbb{Z}$ and $\dfrac{\mathbb{Z} \rtimes \mathbb{Z}}{\mathbb{Z} \times 0} \cong \mathbb{Z}$. The differential on the E_2 page is

$$d : E_2^{p,q} \to E_2^{p+2,q-1}$$

Using the fact that for any $\mathbb{Z}$-module G, the group cohomology groups $H^i(\mathbb{Z}, G) = 0$ for $i > 1$ it follows that the only entry of the spectral sequence we need to calculate is $H^1(\mathbb{Z}, H^1(\mathbb{Z}, A))$ where $A = \mathbb{Z}^2$ viewed as a $\mathbb{Z}$-module.

5.5.2 Calculating $H^1(\mathbb{Z}, H^1(\mathbb{Z}, A))$

Let $G = \mathbb{Z}\langle a \rangle$ and let $A = \mathbb{Z}^2$ be a G-module. A free resolution of $\mathbb{Z}[G]$ is given by

$$0 \to \mathbb{Z}[G] \xrightarrow{a-1} \mathbb{Z}[G] \xrightarrow{\epsilon} \mathbb{Z} \to 0$$

where ϵ is the augmentation map. Applying $\mathrm{Hom}_{\mathbb{Z}[G]}(-, A)$ to the deleted resolution yields

$$0 \to A \to A \to 0$$

and so the cohomology group is given by

$$H^1(G, A) = \frac{A}{(a-1)A}$$

Now in order to calculate the full cohomology group $H^1(\mathbb{Z}, H^1(\mathbb{Z}, A))$, there is the question of what the $\mathbb{Z}$-module structure on $H^1(\mathbb{Z}, A)$ should be.

Lemma 5.48. The action of $Z\langle b\rangle$ on $H^1(G, A)$ is given by

$$(b \cdot c)(a) = -(ba^{-1})c(a)$$

where $c \in \mathrm{Der}(G, A)$ and $G = \langle a\rangle$.

Proof. We have that $H^1(G, A) = \dfrac{\mathrm{Der}(G, A)}{\mathrm{PDer}(G, A)}$. Let $c \in \mathrm{Der}(G, A)$. Then

$$(b \cdot c)(a_1) = b \cdot c(b^{-1}a_1 b)$$

where this action is as in [HS53]. Now, using that c is a derivation

$$\begin{aligned} c(a^{-1}a) &= c(1) \\ &= 0 \\ &= c(a^{-1}) + a^{-1}c(a) \end{aligned}$$

and so $c(a^{-1}) = -a^{-1}c(a)$. This gives us that

$$\begin{aligned} b \cdot c(b^{-1}ab) &= b \cdot c(a^{-1}) \\ &= (-ba^{-1}) \cdot c(a) \end{aligned}$$

as required. □

This gives the following expression for the relevant cohomology group keeping in mind the respective actions of $\mathbb{Z}\langle a\rangle$ and $\mathbb{Z}\langle b\rangle$.

Corollary 5.49. The cohomology group $H^1(\mathbb{Z}\langle b\rangle, H^1(\mathbb{Z}\langle a\rangle, A))$ can be expressed as

$$\frac{\dfrac{A}{(a-1)A}}{(b-1)\dfrac{A}{(a-1)A}}$$

Now it remains to calculate this group for the holonomy representations of the integral affine structures on the Klein bottle. Firstly, recall that the integral affine structures on the Klein bottle are given by the following.

The possible integral integral affine structures on the Klein bottle are

1. $a = \left(I, \begin{pmatrix} 0 \\ x_2 \end{pmatrix}\right), \quad b = \left(\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} y_1 \\ 0 \end{pmatrix}\right)$, with $x_2, y_1 > 0$.
2. $a = \left(\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 \\ x_2 \end{pmatrix}\right), \quad b = \left(\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} y_1 \\ x_2 \end{pmatrix}\right)$, with $x_2, y_1 > 0$.
3. $a = \left(I, \begin{pmatrix} x_1 \\ -x_1 \end{pmatrix}\right), \quad b = \left(\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} y_1 \\ 0 \end{pmatrix}\right)$, with $x_1, y_1 > 0$.
4. $a = \left(\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 \\ x_2 \end{pmatrix}\right), \quad b = \left(\begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} y_1 \\ \frac{n-1}{n}x_2 \end{pmatrix}\right)$, with $y_1 > 0$ and $2y_1 \neq -\frac{n-1}{n}x_2$.

Theorem 5.50. The group cohomology $H^1(\mathbb{Z}, H^1(\mathbb{Z}, A))$ is given by

$$H^1(\mathbb{Z}, H^1(\mathbb{Z}, A)) = \begin{cases} \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}, & \text{for 1)} \\ \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}, & \text{for 2) with } n \equiv 0 \bmod 2 \\ \mathbb{Z}, & \text{otherwise} \end{cases}$$

Proof.
Case 1 (Family 1)**.** In this case the action of $\mathbb{Z}\langle a\rangle$ on $A = \mathbb{Z}^2$ is trivial and hence $(a-1)A = 0$. The action of b on $\dfrac{A}{(a-1)A}$ is given by the matrix action of $-(ba^{-1})$. Since a acts as the identity the action is given by

$$\begin{aligned} b \cdot \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} &= \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} \\ &= \begin{pmatrix} -x_1 \\ y_1 \end{pmatrix} \end{aligned}$$

and hence the action by $(b-1)$ is

$$(b-1) \cdot \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} -2x_1 \\ 0 \end{pmatrix}$$

This gives us that $(b-1) \cdot \dfrac{A}{(a-1)A} = 2\mathbb{Z} \oplus 0$. Taking the quotient of $A = \mathbb{Z}^2$ by this gives the required result.
Case 2 (Family 2)**.** In this case, a acts non-trivially. The action of $(a-1)$ is given by

$$(a-1) \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} ny \\ 0 \end{pmatrix}$$

It follows that

$$\frac{A}{(a-1)A} = \frac{\mathbb{Z}}{n\mathbb{Z}} \oplus \mathbb{Z}$$

We calculate $-ba^{-1}$ to be $\begin{pmatrix} -1 & n \\ 0 & 1 \end{pmatrix}$.

and so

$$\begin{aligned} \begin{pmatrix} -1 & n \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} &= \begin{pmatrix} ny_1 - x_1 \\ y_1 \end{pmatrix} \\ &= \begin{pmatrix} -x_1 \\ y_1 \end{pmatrix} \end{aligned}$$

where in the last equality we are reducing modulo n in the first component. Therfore the action of $(b-1)$ on $\begin{pmatrix} x_1 \\ y_1 \end{pmatrix} \in \dfrac{A}{(a-1)A}$ is given by

$$(b-1) \cdot \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} -2x_1 \\ 0 \end{pmatrix}$$

and so

$$\begin{aligned} H^2(\Lambda) = H^1(\mathbb{Z}, H^1(\mathbb{Z}, A)) &= \frac{\frac{\mathbb{Z}}{n\mathbb{Z}} \oplus \mathbb{Z}}{\langle \tilde{2} \rangle \oplus 0} \\ &= \frac{\mathbb{Z}}{\gcd(2, n)\mathbb{Z}} \oplus \mathbb{Z} \end{aligned}$$

giving the required result for the second family.

Case 3 (Family 3)**.** In this case, a again acts trivially and so it remains to calculate the action of b. We compute

$$-ba^{-1} = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$$

and so $(b-1) \cdot \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -y-x \\ -y-x \end{pmatrix}$ and so it follows that

$$\begin{aligned} H^1(\mathbb{Z}, H^1(\mathbb{Z}, A)) &= \frac{\mathbb{Z} \oplus \mathbb{Z}}{\langle (1,1) \rangle} \\ &= \mathbb{Z} \end{aligned}$$

Case 4 (Family 4)**.** In this case, the $\mathbb{Z}\langle a \rangle$ action on $\mathbb{Z}$ is the same as in Family 2. It remains to calculate the action of b. We calculate

$$-(ba^{-1}) = \begin{pmatrix} -1 & n-1 \\ 0 & 1 \end{pmatrix}$$

and so $(b-1) \cdot \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -2x+ny-y \\ 0 \end{pmatrix}$. Since this action is on $\frac{\mathbb{Z}}{n\mathbb{Z}} \oplus \mathbb{Z}$ we reduce mod n in the first component. This now gives that

$$\begin{aligned} H^1(\mathbb{Z}, H^1(\mathbb{Z}, A)) &= \frac{\frac{\mathbb{Z}}{n\mathbb{Z}} \oplus \mathbb{Z}}{\frac{\mathbb{Z}}{n\mathbb{Z}} \oplus 0} \\ &= \mathbb{Z} \end{aligned}$$

and this completes the calculations.

$\square$

Conclusion

TODO: Add Conclusion

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